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"Evaluating Brain Alignment as a Training Signal for Language Models"

Author's complete name: MohammadErfan Jabbari
1st Tutor complete name: Alejandro Lancho Serrano
2nd Tutor complete name: Pablo Martínez Olmos

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Evaluating Brain Alignment as a Training Signal for Language Models

MohammadErfan Jabbari

Abstract

Language-model representations can predict recorded functional magnetic resonance imaging (fMRI) responses to language, but that does not show that brain responses can improve model training. We tested whether adding recorded fMRI responses, or synthetic brain responses generated from text, to knowledge distillation gives a smaller student model an advantage that comes specifically from brain-response information. We first measured how well model representations predicted held-out fMRI responses, beyond nuisance features and beyond architecture-matched untrained models, using two assays: one over brain regions responding to isolated sentences, one over voxels responding to a spoken story. Both of these assays supported brain prediction under those controls. We then compared training on correctly paired responses against training on permuted responses, which carry the same values but lose their pairing with the sentence that produced them. For synthetic responses we asked three further questions: whether training made the synthetic target easier to recover from the student, whether the resulting changes remained after removing the prediction head used only during training, and whether the student improved on independently recorded fMRI responses beyond ordinary knowledge distillation (KD). We also compared the synthetic-response arm against a text-derived control, since only such a comparison can attribute a difference to brain-response content rather than to any similarly structured target. A separate assay tested one narrow pathway alone: whether a 50-component linear projection of the synthetic target improved held-out fMRI prediction beyond nuisance features and beyond a control that keeps the same target values but breaks their pairing with the sentences. For a GPT-2 student trained from scratch, the gap between teacher and student, which is the room an intervention could close, was confounded by language quality. At comparable language quality that gap remained unresolved for a pretrained GPT-2 student, and we did not measure it for the Qwen student. Recorded-response training did not reliably improve participant-level brain prediction over permuted-response training. Synthetic-response training did make the target easier to recover, and it did change the student retained after the head was removed, while staying inside the language-quality tolerance fixed in advance. The projected target, however, did not clear the improvement threshold set beforehand. The text-derived control was too poorly matched to show whether any difference was specific to brain-response information, and direct participant-level transfer over ordinary KD was not demonstrated. Mean scores on the practical endpoint, held-out language-modeling quality on text from outside the training corpus, favored recorded-response training. We did not report uncertainty for that paired comparison, however, and the brain-prediction improvement it depends on was unstable across random seeds, so brain-specific utility was not demonstrated. The routes we tested therefore did not establish a reliable brain-specific training ad-

vantage, and they do not show that brain-response-guided training is ineffective in general. The contribution is the separation of claims: for each route, the thesis says which of measurement, manipulation, direct transfer, attribution, and utility the evidence actually supports.

1 Introduction

Brain responses to language could help train a smaller language model (LM): when several internal representations predict text about equally well, brain responses could favor one of them over the others. In knowledge distillation (KD), a smaller student learns to reproduce the output distribution of a larger teacher, but comparable output behavior does not require the two models to organize information identically in their middle layers [1]. Under learning using privileged information (LUPI), a recorded functional magnetic resonance imaging (fMRI) response or a synthetic brain response generated from text, the output of a separate model trained to predict fMRI responses from text can serve as a training-only target: it may favor one model representation over another without changing the inputs available at deployment. Feature-distillation and LUPI research support this possibility, while also showing that an auxiliary target can produce a similar improvement for reasons unrelated to whether it contains information about brain responses [2, 3]. Brain-response supervision is thus best treated as a training signal that may shape model representations, not as evidence that the student has acquired a human cognitive mechanism.

We assess brain alignment by fitting an encoding model that predicts recorded fMRI responses from frozen LM representations, then evaluating its predictions on stimuli withheld from fitting. Successful prediction indicates that the representations contain information useful for predicting recorded fMRI responses to the same stimuli [4, 5]. The resulting score characterizes the complete assay rather than the LM alone: it depends on the participants and response aggregation, stimulus distribution, data split, nuisance features (simple stimulus properties such as position and length), response reliability, the class of readout that maps representations to responses, and the unit over which uncertainty is computed. In this manuscript, controlled brain predictivity refers to brain alignment that survives the nuisance, split, and control-model checks specified below. Section 3.2 defines the main held-out scores and summarizes the readout protocol; Appendix B specifies the nuisance features and preprocessing, and Appendix D formalizes the estimators and inference.

Because that score only measures prediction of recorded responses, it does not establish that training with brain responses can alter the representations the student keeps after temporary training components are removed. Even if such a change occurs, the score does not establish three further claims: (1) that the change is caused specifically by brain-response content rather than generic geometric properties of the target, or the extra optimization pressure of any auxiliary loss; (2) that it transfers to independently recorded fMRI responses; and (3) that it improves language performance. These distinctions matter because data splits that scatter locally related stimuli across fitting and evaluation, simple stimulus features, and analysis choices can produce or reorder apparent model–brain correspondences [6, 7]. An encoding model should therefore be treated as a measurement instrument between model representations and recorded fMRI responses, not as an account of how either the LM or the brain computes.

Prior work does not resolve whether brain responses can help train a smaller model. Encoding studies show that model representations can predict recorded brain responses and examine which linguistic properties and model layers support that prediction [4, 5]. Brain-response-guided optimization studies show that recorded brain responses or synthetic brain responses generated from text can alter model representations or task behavior in several language and speech settings [8–11]. Research on privileged information and feature distillation shows more generally how dense training-only targets may benefit a student [2, 3]. Compression studies also show that brain alignment after pruning or quantization can depend on the compression method and model scale [12]. However, these studies differ in their recording modality, temporal resolution, participant specificity, model size budget, evaluation endpoint, comparator, training seeds, and inference unit. They therefore do not establish whether training with brain responses gives a fixed smaller student an incremental advantage over a matched auxiliary target derived from text alone, when the compared students reach comparable LM quality.

We therefore evaluate each claim about brain-specific training against the specific evidence that claim requires. First, the model representations must show controlled brain predictivity, and the smaller model's brain predictivity must fall short of its teacher's by more than worse LM quality alone can explain. For synthetic brain responses generated from text, we then ask whether the exact training target contains measurable information about recorded fMRI responses, whether the student learns that target, and whether training produces changes that remain after removing the prediction head, an auxiliary output layer added only for training. Finally, training with synthetic brain responses must be compared with training that uses a similarly constructed text-derived auxiliary target. Because both conditions retain the same LM objective, this comparison tests whether any difference is specific to brain-response content. Tests must then determine whether those changes transfer to independently recorded fMRI responses and improve performance on an external task. A failure at one of these steps limits what the claims depending on it can establish; it leaves untouched the claims that do not depend on it.

In the systems we tested, trained LM representations contained information useful for predicting held-out recorded fMRI responses beyond the specified nuisance variables and untrained-model controls. The distilled student had lower controlled brain predictivity than its teacher, but worse LM quality remained a competing explanation for this gap. Training with recorded fMRI responses did not produce a reliable participant-level benefit in the tested regimes. For synthetic brain responses generated from text, the training target, after compression to 50 principal components, did not meet the rule fixed in advance for continuing: in the linear assay declared beforehand it no longer added held-out prediction of recorded fMRI responses beyond the nuisance model. The target was nevertheless learnable on the training corpus, and training changed the student's retained representations while preserving LM quality within the prespecified tolerance. A post-training readout on an independently recorded fMRI dataset (Tuckute) did not resolve whether the student retained more information about the target than a student trained by ordinary KD. The available text-derived auxiliary target was not sufficiently comparable to determine whether the observed differences arose specifically from brain-response content. The direct participant-level comparison between training with synthetic brain responses and ordinary KD found a near-zero difference in prediction of recorded fMRI responses. Returning to the recorded-response route: on the practical endpoint, the downstream task where a brain-specific advantage should pay off, the recorded-response arm scored higher on average than its permuted-response control. We did not report uncertainty for that paired comparison, and the controlled-predictivity prerequisite was unstable across random seeds, so brain-specific utility was not demonstrated.

This study contributes controlled measurement of brain predictivity alongside LM quality, and participant-level tests of training with recorded fMRI responses. It also contributes an evaluation framework that distinguishes supported results, claims not demonstrated, unresolved questions, and comparisons that cannot attribute a difference to brain-response content. The conclusion is restricted to what was actually tested here: the models, English stimuli, datasets, and participants, and the specific targets, objectives, controls, readouts, comparators, and endpoints used. It does not provide a successful brain-response-guided distillation method, but neither does it show that brain responses can never help train a smaller model; it separates which claims the evidence supports, which it does not demonstrate, and which remain unresolved.

2 Background and Related Work

Four adjacent literatures inform this thesis but answer different questions: brain-encoding studies [4, 5], brain-response-guided optimization [8–11], privileged-information and feature-distillation work [2, 3], and model-compression studies [12]. Predicting recorded brain responses does not establish that training with those responses changes a student model, the smaller model trained to imitate a larger teacher. A change in the student also does not establish that the change (1) arises specifically from brain-response content, (2) transfers to independently recorded fMRI responses, or (3) improves a practical endpoint, such as a downstream task, while the student's language-modeling quality stays comparable to the baseline's. This section connects these claims and defines the evidence criteria for a brain-specific training advantage, a benefit attributable to brain-response content rather than to any auxiliary target. Five words recur throughout the thesis for the five things that must be told apart: measurement is predicting recorded responses from a frozen model, manipulation is changing the student with a brain-response objective, attribution is tracing that change specifically to brain-response content, transfer is improvement on independently recorded responses, and utility is improvement on a practical endpoint.

2.1 What controlled brain predictivity establishes

Controlled brain predictivity tests whether representations extracted from a frozen LM contain information useful for predicting recorded fMRI responses to held-out stimuli. An encoding model is fitted to predict responses from those representations on one set of stimuli and evaluated on a separate set. This held-out evaluation tests whether prediction generalizes beyond the fitting stimuli rather than merely reflecting overfitting to them [4, 5]. Under a linear readout, a positive controlled score means that a linear combination of LM representation dimensions improves prediction beyond the controls described next.

Three controls are required to determine whether predictive performance is associated with learned LM representations rather than simpler alternatives. First, a nuisance-only model predicts recorded fMRI responses from stimulus properties such as position, length, and static lexical features. Comparing it with a full model that also includes the tested representations measures what those representations add. Second, an architecture-matched untrained LM retains the tokenizer, dimensionality, and architecture of the trained model but lacks its learned parameters. This comparison tests whether predictive performance depends on learned representations rather than the model's fixed structure. Third, contiguous or blocked splits keep locally related samples from being scattered across fitting and evaluation sets, reducing inflated scores caused by dependence or leakage between them [6]. Together, these controls restrict the interpretation of the observed effect to held-out predictive value added by learned LM representations under the declared assay, that is, the stated combination of stimuli, response definition, readout, and controls.

Even with these controls, controlled brain predictivity remains bounded by the declared assay. A linear readout tests whether recorded responses can be predicted from a linear combination of LM representation dimensions; it does not test every form of information that might be present. Participant-specific responses measure predictivity for an individual, whereas averaged responses measure predictivity of a group-level response summary. How reliably the response was measured, and how many repetitions were averaged to estimate it, set a ceiling on what any model can predict [13]. The stimulus distribution, how the response is defined and aggregated, and the selected brain regions bound the achievable score further. This thesis therefore states each of them for every reported score. The inference unit determines the scope of uncertainty: many voxels or repeated folds from one participant do not provide evidence about many independent participants. Controlled brain predictivity therefore supports a bounded claim about held-out prediction under the declared assay, not a shared biological mechanism, causal equivalence, or general cognitive competence [7]. Sections 3.1, 3.2, and 3.5 specify the response construction, estimator, controls, and inference unit used to instantiate this assay.

2.2 Why brain-response targets might help a student

Ordinary KD trains a student to reproduce the teacher's output distribution, but this objective does not uniquely determine how the student organizes information in its middle layers. Different bases, subspaces, and feature organizations can support similar token predictions. Two students can therefore show similar language behavior while retaining different information in their model representations. This leaves room for an auxiliary training target to favor one representation over another without necessarily improving or degrading the language objective. Feature-distillation research further shows that the value of such guidance depends on which information is emphasized and how teacher and student capacities are matched, rather than simply on copying a large dense vector [3].

This flexibility leaves room for brain-response targets to serve as privileged information, meaning data available during training but unnecessary at deployment [2]. The student retains the same LM objective, while an auxiliary objective favors representations from which the training target can be predicted. Recorded fMRI responses provide observed biological measurements, whereas synthetic brain responses generated from text carry only the structure learned by a fixed generator, a model trained beforehand to map text to brain responses. Because each synthetic response is determined by the text and generator, it cannot provide example-specific information that varies independently of them. It may nevertheless present generator-learned structure in a form that is easier for the student to learn or recover. This is a hypothesis about the intervention, motivated by privileged-information and feature-distillation results, not a consequence of the target's deterministic construction [2, 3].

Evidence that the auxiliary objective affected training does not establish a brain-specific training advantage. Reduced target loss may show only that components present during training learned to predict the auxiliary target; those components can be discarded afterwards, leaving the retained student unchanged. Target uptake instead requires a new prediction model, fitted after training, to recover the target from the frozen representations of the retained student, the model kept once the training-only components are removed, better than from the declared baseline. Retained-student movement separately requires parameter or representation differences from ordinary KD that remain after components used only during training are removed. These checks show, respectively, that target information became recoverable beyond the baseline and that the intervention changed the retained student beyond ordinary KD. They do not establish that the changes (1) arose specifically from brain-response content, (2) transferred to independently recorded fMRI responses, or (3) improved a practical endpoint. Sections 3.1, 3.3, 3.4, and 3.5 specify the target construction, training objectives, gradient paths, retained components, and comparison baselines used for these checks.

2.3 What prior brain-response-guided training leaves unresolved

Prior work assigns brain responses two distinct roles: post-training evaluation and training supervision. Post-training encoding studies use recorded fMRI responses to evaluate model representations after the model has been changed for another reason. Narrative-summarization tuning can increase brain-predictivity scores without using brain responses during training, while model scaling can increase naturalistic brain alignment even when matched-size instruction tuning does not under the studied assay [14, 15]. These findings show that brain predictivity can change because of the training objective or model scale alone; they do not estimate the value of brain responses as training targets. Brain-response-guided studies instead use recorded brain responses or model-generated brain-response predictions during training [8–11]. Their positive findings show that training with these responses can change model representations, brain-predictivity scores, or selected task outcomes. However, the claim supported by each result depends on its comparator, evaluation data, and inference unit. Table 1 compares selected designs without treating every positive brain-predictivity result as evidence for the same claim.

Table 1: Prior designs linking model interventions to brain predictivity.

Study	Intervention and relevant result	Why it does not settle this study's question
Aw and Toneva [14]; Gao et al. [15]	Post-training evaluation after summarization tuning, scaling, or instruction tuning shows that brain predictivity can change without brain-response training.	Uses no brain-response target, no smaller student, and no brain-response-versus-control intervention.
Bilgin et al. [16]	A cosine loss against participant-specific fMRI responses, together with a token loss, updates upper-layer adapters; held-out encoding, new-participant readouts, and a model-specific external task improve.	No shuffled or matched text-derived target, smaller student, or ordinary KD comparator.
Merlin, Moussa, and Toneva [10]	Participant-specific fMRI-response, stimulus-language, or joint objectives update adapters; brain-response training wins more linguistic-probe comparisons than stimulus tuning.	Target geometry and difficulty are not matched between the brain-response and stimulus-language objectives; no participant-held-out transfer or compression comparison.
Merlin and Toneva [17]	Complementary objectives add or remove brain-predictive information; a permuted-response control and matched LM loss accompany changes in linguistic probes.	No smaller student; the readout used to add or remove information may itself carry linguistic variance not specific to brain-response content.
Freteault et al. [18]	Individual or group fMRI responses fine-tune selected auditory-network layers; held-out brain predictivity and external auditory tasks improve.	No matched text-derived or shuffled-target arm; the model and modality differ from text distillation.
Pirlot et al. [19]	A monkey-V1-response regularizer trains a vision model; permuted targets reproduce the clean-accuracy gain, while real targets improve moderate-attack robustness.	The clean-task gain is generic regularization; only the robustness endpoint distinguishes brain-response content in that design.

Positive studies span language, speech, and vision; they do not estimate a single shared effect of brain-response training. Speech models adapted with recorded fMRI responses have improved brain-response prediction and performance on selected semantic or auditory tasks [8, 18, 20]. Some of those gains also appear for participants whose responses were not used in adaptation [20]. In one of these studies the brain-prediction gains were scored on stimuli drawn from the same distribution as the adaptation data, so they do not establish transfer across stimulus distributions [18]. Text-model interventions report changes in multilingual performance, linguistic probes, visual commonsense, or reasoning after training with brain-response targets [9, 10, 16, 21]. Training with temporally precise electrocorticography responses has improved prediction within the electrocorticography modality, while training with slower fMRI responses has transferred to prediction in that faster modality for new participants and stimuli after fitting the required readout [11, 22]. Together, these studies show that brain-response-guided objectives can alter model representations or selected task outcomes. However, they test different models, training targets, endpoints, and transfer settings. Some adapt a full encoder or adapters rather than a smaller student, some use participant-specific targets, and some require different amounts of brain-response training data per participant. Their positive results therefore do not establish one common effect or the incremental value of brain responses for a fixed smaller student.

Different controls rule out different explanations for a positive result. A target permutation (the permuted-response control) destroys the stimulus–target pairing while preserving the target values and their marginal distribution, testing whether the correct pairing matters. A task-only or stimulus-language condition tests whether additional training or a conventional objective is sufficient. Evaluation on a held-out participant or dataset tests whether the change transfers beyond the training setting, often after fitting a new readout. None of these controls replaces a text-derived auxiliary target matched to the brain-response target in dimensionality, covariance, baseline predictability, learnability, and optimization pressure; only such a matched target separates brain-response content from the generic benefit of any dense auxiliary target. The vision study is instructive: permuted targets reproduce the clean-accuracy improvement, while only the robustness endpoint distinguishes the real brain-response target [19]. Positive results should therefore be interpreted only within the scope of the claims their designs support, and a result should be attributed to brain-response content only to the extent that the design's controls rule out the competing explanations.

Privileged-information and feature-distillation studies provide a competing explanation for any benefit from training with brain responses. A dense training-only target may produce an improvement by emphasizing distinctions useful to the student or, as a so-far untested hypothesis, by making optimization easier, regardless of whether the target contains brain-response content. Its value depends on the information carried by the target, the student's capacity, and how the teacher and student representations are matched [2, 3]. An improvement from a brain-response target therefore requires comparison with a similarly learnable text-derived auxiliary target under the same LM objective before it can be attributed specifically to brain-response content.

Measurement-validity research identifies separate requirements for evaluating the student after training. Post-training brain predictivity must be measured with leakage-resistant splits and nuisance controls [6, 7]. It must also be measured at an inference unit appropriate to the claim: the number that enters a test is the number of independent units, not the number of observations pooled from them [23]. Reduced auxiliary loss during training cannot replace this fresh evaluation. Compression studies further show that pruning or quantization method and model scale can change brain predictivity without brain-response training [12]. An intervention comparison must therefore separate the effect of the brain-response objective from changes in LM quality and must evaluate the retained student with a fresh measurement rather than rely on the value of its own training loss.

The unresolved question posed before any new experiment is therefore narrower than whether training with brain responses can change a model. For a fixed smaller student trained by ordinary KD, do recorded fMRI responses or synthetic brain responses generated from text provide an incremental, brain-specific advantage over the appropriate control: permuted responses for recorded targets, and a matched text-derived target for synthetic targets at comparable LM quality? The next subsection states the evidence criteria for that advantage claim.

2.4 Evidence criteria for a brain-specific training advantage

Figure 1 organizes the evidence by what each claim depends on, rather than as one fixed sequence every claim must pass through. Controlled brain predictivity defines the biological endpoint shared by both routes. Route-specific headroom describes whether an intervention has room to improve that endpoint, but it does not determine the validity of a later paired intervention contrast. The recorded- and synthetic-response routes then require different evidence for manipulation, attribution, transfer, and utility claims.

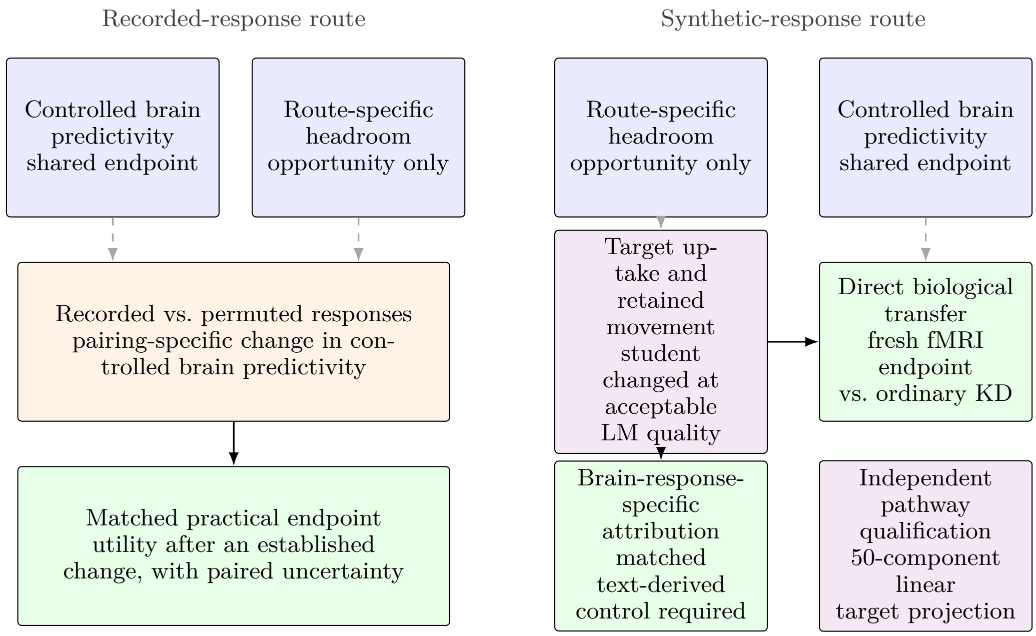


Figure 1: Claim dependencies for recorded- and synthetic-response training. Dashed arrows link shared endpoint or opportunity context; solid arrows mark claim dependencies. The target-projection assay independently qualifies only the declared linear pathway.

No single comparator supports every claim in Figure 1. An architecture-matched untrained LM tests whether controlled brain predictivity depends on learned parameters. An ordinary KD arm isolates the incremental effect of adding a brain-response objective to the same student-training regime, while comparable LM quality limits generic quality differences. The route-appropriate attribution control depends on the training target. In the recorded-response route, the attribution control compares correctly paired with permuted responses, testing whether the stimulus–response pairing matters while preserving the response values and their marginal distribution. In the synthetic-response route, the attribution control compares the synthetic brain-response target with a text-derived auxiliary target matched on properties that we fix or calibrate in advance, before interpreting the target-arm outcomes. These properties include geometry, scale, baseline predictability, remaining headroom (how much the target's prediction can still improve), and initial optimization pressure (the early gradient magnitude the target exerts). Target uptake and retained-student movement, defined above, can explain how two target interventions differed, but they are outcomes of those interventions rather than conditions the comparator must satisfy. For either route, brain-response-specific attribution requires the brain-response-target arm to outperform its control on the outcome being claimed.

A freshly fitted readout on independently recorded fMRI responses tests biological transfer; successful recovery of the optimized training target cannot substitute for this endpoint. The direct transfer comparison, the synthetic-response arm against ordinary KD, is interpretable without the text-derived auxiliary control. The auxiliary control is additionally required only when the claim attributes a difference to brain-response content.

The 50-component linear target-projection assay tests whether a declared 50-component linear projection of the training target adds controlled predictivity; it qualifies that linear pathway alone. It does not gate any manipulation, transfer, or attribution claim, so Figure 1 draws it as an isolated box with no arriving or departing dependency arrows. In particular, a failure of this assay does not overturn a direct-transfer result that its own comparison identifies. A practical-utility claim requires improvement on its prespecified endpoint; when the claim is that improved controlled brain predictivity pays off practically, it additionally requires that the predictivity improvement itself be reliably established first.

Each result therefore licenses only the claim that depends on it. A failed manipulation check (target uptake or retained-student movement) limits the interpretation of a null intervention result, but it does not invalidate a positive independently measured endpoint contrast. Failure in one branch does not erase evidence supporting another branch or an earlier, narrower claim. An unresolved comparison must remain unresolved rather than be converted into a null result, and failure under one readout, cohort, or target must not be generalized beyond that scope. This dependency interpretation prevents one type of evidence from substituting for another while preserving valid route-specific conclusions. Section 3 maps each claim to its data, training arms, estimand, comparator, endpoint, and inference unit; Section 4 reports the corresponding verdicts.

3 Methods

Section 2.4 defines the evidence criteria for converting controlled brain predictivity into a brain-specific training claim. This section instantiates those criteria: for each claim it names the data, the intervention or assay, the comparator, the endpoint, and the inference unit. An estimand is the specific quantity or effect that an analysis is designed to estimate. Table 2 maps each claim role to its design tuple (comparison, endpoint, and inference unit). A role may appear in more than one row when different datasets, response constructions, or population units define different estimands. This section specifies the study design. Section 4 reports what the evidence settled for each role.

Table 2: Claim roles and their primary designs.

Claim	Scientific role	Primary comparison and endpoint	Inference unit
Shared	Controlled regional predictivity assay	Trained representations versus nuisance and architecture-matched untrained LM; held-out ordinary and unique R^2	Fixed sentence set; five contiguous folds
Shared	Controlled naturalistic voxelwise predictivity assay	Trained versus untrained representations beyond nuisance; held-out-story voxelwise unique R^2	Story blocks within one deeply sampled participant
Context	Distillation-headroom analysis	Declared-layer brain predictivity before and after ordinary KD, interpreted with held-out LM quality	Model condition; training seed for stochastic arms
Linear path	50-component target-projection assay	Synthetic target versus nuisance and frozen row twin; held-out recorded-response unique R^2	Participant on a fixed sentence set
Synthetic	Synthetic-target recovery assay	Synthetic-response versus ordinary KD arms; saved block-permuted variants are sensitivity checks	Training seed
Synthetic	Retained-student movement audit	Synthetic-response versus ordinary KD arms; retained parameter or representation change at comparable LM quality	Training seed; fixed target sample for geometry
Recorded	Recorded-response intervention, participant-averaged estimand	Correctly paired versus permuted response targets; fresh averaged-response unique R^2	Shared stimulus fold; seeds are technical repetitions
Recorded	Recorded-response intervention, participant-specific estimand	Correctly paired versus permuted response targets; fresh participant-response unique R^2	Participant after fold and seed aggregation
Attribution	Comparator-adequacy audit and attribution assessment	Synthetic-response versus text-derived auxiliary-control arms; target construction, KD-baseline recoverability and headroom, pre-update calibration, and downstream arm contrast	Seed for trained models; fixed target sample for matching checks
Synthetic	Saved-student target-retention analysis	Synthetic-response versus ordinary KD students; held-out synthetic-target recovery	Training seed
Linear path	Composed-path diagnostic	Aligned versus frozen row-twin targets and synthetic-response versus ordinary KD students; held-out recorded-response unique R^2 through the composed path	Participant
Transfer	Saved-student biological-transfer assay	Synthetic-response versus ordinary KD and text-derived auxiliary-control arms; fresh participant-response unique R^2	Participant
Utility	Practical-utility evaluation	Brain-response-target arm versus the quality-matched control declared for that route; held-out out-of-domain perplexity endpoint	Technical seed; fixed corpora and procedure

Appendix A maps these scientific roles to the experiments that own their final evidence and records the interpretations that were superseded during review.

3.1 Study Design, Datasets, and Data Partitioning

Two datasets provide the recorded fMRI responses used in this study. The Tuckute condition-B dataset [24] contains 1,000 isolated English sentences, responses from 10 participants, and five fixed left-hemisphere language regions of interest (ROIs). The dataset supports the controlled regional predictivity assay and the distillation-headroom analysis. The same sentences later support the recorded-response route and the Tuckute endpoints of the target-projection and saved-student biological-transfer assays. The LeBel corpus [25] provides continuous story stimuli for the controlled naturalistic voxelwise predictivity assay. That assay uses participant UTS03 and 20 stories divided into five held-out-story folds. A separate repeated story determines which voxels meet the reliability criterion. Holding out complete stories prevents temporally dependent samples from the same story from crossing the evaluation boundary. The resulting estimand concerns held-out-story prediction within UTS03 and does not support population inference across people.

The Tuckute responses can be constructed at two response-aggregation levels. Participant averaging combines the five Tuckute training participants declared in Appendix B into one sentence-by-ROI matrix. These five are the participants the original Tuckute encoding fit used, so the averaged target follows the published construction rather than a subset selected here. This matrix describes the shared stimulus-locked response, but it cannot support participant-level inference. Participant-specific construction begins from the 10 recorded participants. One participant is excluded because that participant's required five-ROI response matrix is incomplete, leaving nine valid participants. Each remaining participant retains a separate response matrix. Contrasts can therefore be aggregated within participant before inference across the nine tested participants. Spatial resolution is a separate distinction. An ROI coordinate summarizes responses within a predefined language region, whereas a voxel coordinate provides a finer spatial measurement. Section 3.5 pairs these constructions with their estimands and inference units.

Each training and evaluation role draws on a declared text and response source. The recorded-response intervention trains on Tuckute sentences and their recorded responses. The synthetic-response intervention instead trains on 95,999 WikiText-103 sentences [26]. A separate 1,999-sentence WikiText split supports language-quality evaluation and synthetic-target recovery. For the target-projection assay, the fixed synthetic brain-response target is generated solely from the Tuckute sentence text, by a published text-to-brain response model introduced below. Recorded fMRI responses enter only the assay's held-out evaluation. Saved students, the students kept from each trained arm after the training-only components are discarded, then remain frozen; for each evaluation we fit a fresh readout to test biological transfer to those responses. Pereira sentences and LeBel story transcripts provide out-of-domain language-quality endpoints for the practical-utility evaluation [25, 27]. Recorded LeBel fMRI responses are confined to the naturalistic assay.

The data partitions preserve these separate roles. The principal Tuckute roles use five contiguous outer partitions. Each partition holds out 200 sentences and leaves the other 800 as its training complement. For the controlled regional predictivity assay, distillation-headroom analysis, target-projection assay, and saved-student biological-transfer assay, this boundary separates fresh-readout fitting from evaluation. The recorded-response intervention uses the outer boundary differently. It optimizes the student on the 800-sentence training portion, which excludes the held-out 200 sentences. Within that held-out block, five inner folds separate fresh-readout fitting from readout evaluation. The synthetic-response route uses its two WikiText partitions for student training and target-recovery evaluation. Within that route, recorded Tuckute responses enter only the later target-projection and saved-student biological-transfer assays. Fold-dependent transformations are fitted on the corresponding training portion. The target-projection assay has one declared exception: it reuses the target mean and scale that were fixed once during target construction, before any fold split existed. Its subsequent target principal component analysis (PCA) and all readout transformations remain fold-local. Appendices B and D report the exact inclusion, preprocessing, partition, and nesting definitions.

Figure 2 condenses these assignments into three routes.

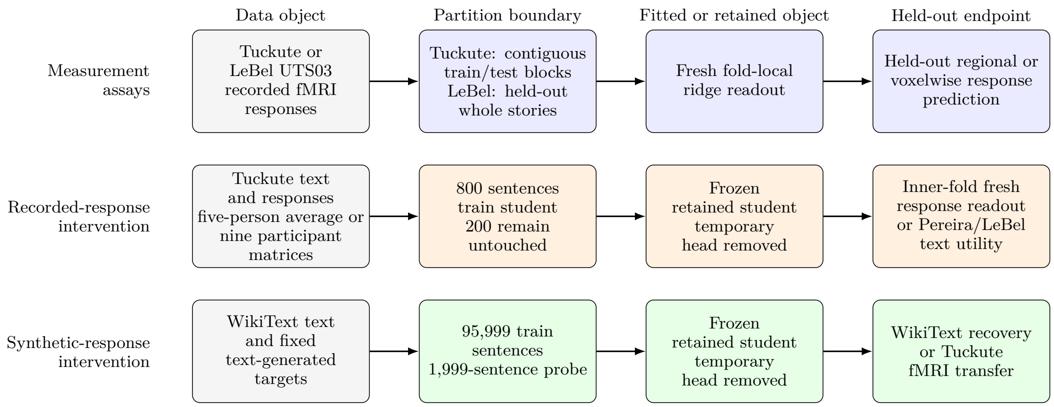


Figure 2: Data and partition routes used by the designs in Section 3. All learned fold-dependent transforms are fitted on training data; held-out responses are used only by the declared evaluation.

The measurement route fits readouts without changing an LM; the two intervention routes first produce a retained student and then evaluate it on held-out data. The arrows denote data use, training, or evaluation rather than causal influence.

3.2 Measurement Assays and Opportunity Diagnostics

Both controlled brain predictivity assays address the same question: does a contextual LM representation improve held-out prediction of recorded fMRI responses beyond specified nuisance features? Each assay first fits a nuisance-only readout containing its declared low-level and static features. The full readout adds a contextual representation block. The contextual and static blocks are each reduced to the same PCA rank, so their widths match. Let Y_j denote response coordinate j, N the nuisance block, X the contextual block, and $\mathcal{T}$ one held-out fold. For any readout design M, ordinary held-out R^2 is

$$R_j^2(M) = 1 - \frac{\sum_{i\in\mathcal{T}}(y_{ij} - \widehat{y}_{ij}^{M})^2}{\sum_{i\in\mathcal{T}}(y_{ij} - \bar{y}_{\mathcal{T}j})^2}. \tag{1}$$

Equation 1 expresses held-out performance as the fraction of response variation explained beyond a prediction that always uses the held-out mean. A value of one denotes perfect prediction, zero matches that mean baseline, and a negative value means that the fitted readout predicts worse than the baseline. The nuisance-only and full scores are $R_j^2(N)$ and $R_j^2([N,X])$. Their cross-validated semipartial, or unique, score is

$$\Delta R_j^2(X) = R_j^2([N,X]) - R_j^2(N). \tag{2}$$

Equation 2 measures the change in held-out predictive value when the contextual block is added to the declared nuisance design. A positive value therefore means that the contextual LM representation improves held-out response prediction beyond those nuisance features.

Figure 3 separates this nuisance comparison from the two additional controls. One control uses an untrained LM with the same architecture and rank. In the naturalistic assay, this control uses the matched nuisance construction and tests whether learned parameters matter. The regional assay as executed here recomputed the static nuisance block for each model, so its trained-minus-untrained gap does not isolate learned parameters against a fixed nuisance block. The frozen row twin, a row-permuted copy of the fixed synthetic brain-response target with the permutation fixed once and reused, instead tests whether correct sentence–target pairing matters. Repeating the contextual comparison across the teacher and students supplies the scores used by the distillation-headroom analysis.

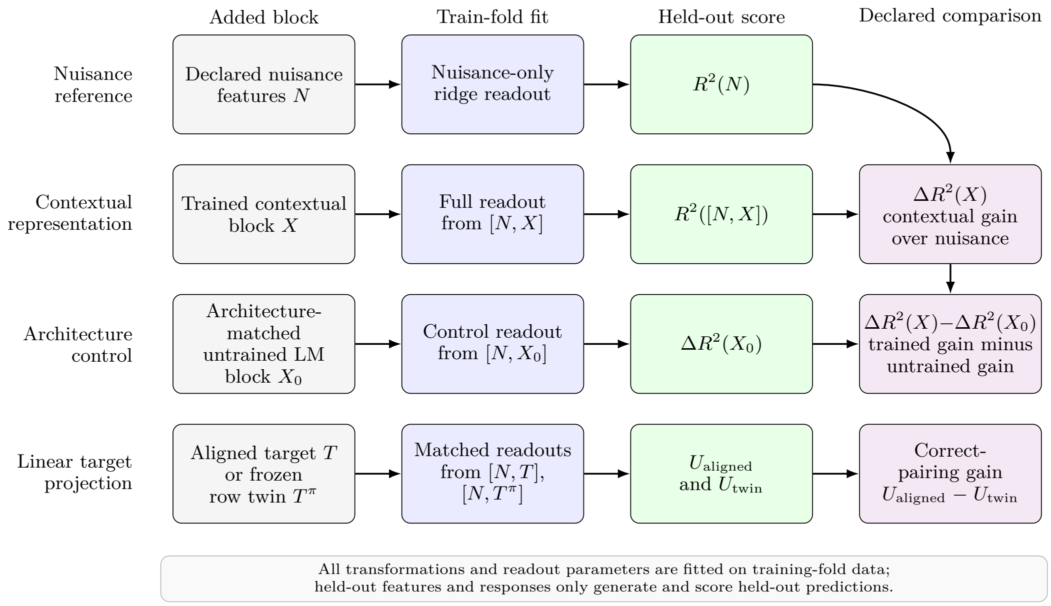


Figure 3: Fold-local readout comparisons used by the measurement assays.

The test-fold mean $\bar{y}_{\mathcal{T}j}$ supplies the reference prediction in Equation 1. Both $R_j^2(M)$ and $\Delta R_j^2(X)$ can be negative. For Tuckute, these readouts use contiguous sentence folds, and released reliability estimates provide descriptive context. For LeBel, they hold out complete stories, and a separate repeated story determines voxel eligibility.

The controlled assays define the biological measurement context, but they do not show how much predictivity a particular teacher–student route leaves available to recover. The distillation-headroom analysis addresses that separate opportunity question by applying the same Tuckute assay at the declared layers. Its reference models are a pretrained GPT-2 Medium (`gpt2-medium`) teacher, an untrained GPT-2 Small (`gpt2`) architecture-matched to the students, and an off-the-shelf pretrained GPT-2 Small (`gpt2`). The analysis also evaluates two GPT-2 Small students after ordinary KD. One student begins from random initialization. The other begins from pretrained weights. Let $\overline{\Delta R^2}(m, s)$ denote the declared-ROI and fold average of Equation 2 for model m and technical seed s. For teacher m_T, the seed-specific headroom is

$$\mathcal{H}(m, s) = \overline{\Delta R^2}(m_T) - \overline{\Delta R^2}(m, s). \tag{3}$$

Equation 3 measures the teacher predictivity that the tested student has not recovered. Positive headroom means that the teacher retains a predictivity advantage, while zero headroom means that their averaged scores are equal under this assay. The seed index is omitted for deterministic reference models. For Figure 7a, let A_0, A_S, and A_T denote the corresponding untrained-model floor, student, and teacher scores. The floor-anchored retention is

$$\rho' = \frac{A_S - A_0}{A_T - A_0}. \tag{4}$$

Equation 4 reports the fraction of the teacher-to-untrained-floor interval retained by the student. Values of zero and one place the student at the floor and teacher anchors, respectively. Each brain-predictivity score is reported alongside held-out LM quality because quality may explain part of the difference. Perplexity provides this comparison among the models above, which share a tokenizer. A separate cross-family analysis uses tokenizer-comparable bits per byte. That cross-family analysis relates $\overline{\Delta R^2}(m)$ directly to model quality; it does not define teacher-relative headroom across unrelated model families. The analysis measures route-specific headroom under the tested conditions and provides opportunity context rather than a universal intervention criterion. It does not identify model size, architecture, initialization, training history, distillation, or language quality as the cause of that headroom.

The 50-component linear target-projection measurability assay, called the target-projection assay from here on, examines the fixed synthetic brain-response target that TRIBE [28], a published text-to-brain response model, generates from text. It asks whether the declared fold-local PCA representation of that target predicts recorded Tuckute responses on the same 1,000 sentences. For each participant, the assay fits three otherwise matched response readouts. The first contains nuisance features only. The second adds the aligned synthetic target. The third adds its frozen row-twin control. This control applies one deterministic permutation to complete target rows within each outer block and has no fixed points. It preserves the exact target-row multiset within that block but breaks sentence–target pairing. Let T denote the aligned target block and T^{π} its frozen row twin. Both use the same train-fold target-PCA basis and the same nuisance and readout procedure. Write $R^2_{u,f,j}(M)$ for Equation 1 applied to participant u, fold f, and response coordinate j. For the J declared response coordinates, define

$$
\begin{aligned}
U_{\text{aligned}}(u,f) &= \frac{1}{J}\sum_{j=1}^{J}\left[R^2_{u,f,j}([N,T]) - R^2_{u,f,j}(N)\right], \\
U_{\text{twin}}(u,f) &= \frac{1}{J}\sum_{j=1}^{J}\left[R^2_{u,f,j}([N,T^{\pi}]) - R^2_{u,f,j}(N)\right].
\end{aligned}
\tag{5}
$$

These two quantities apply the same nuisance subtraction to the correctly paired target and its row-twin control. Their difference can therefore test whether correct sentence–target pairing adds predictive value beyond the information retained by the target rows after permutation.

Let $A(u)$ denote participant-level measurability beyond nuisance. Let $Q(u)$ denote the additional predictive value of correct sentence–target pairing. The two participant-level estimands are

$$
\begin{aligned}
A(u) &= \frac{1}{K}\sum_{f=1}^{K} U_{\text{aligned}}(u,f), \\
Q(u) &= \frac{1}{K}\sum_{f=1}^{K}\left[U_{\text{aligned}}(u,f) - U_{\text{twin}}(u,f)\right], \qquad K = 5.
\end{aligned}
\tag{6}
$$

Equation 6 first averages the aligned target's unique predictivity across folds and then averages the aligned-minus-twin pairing contrast across those same folds. Computing both quantities separately for each participant makes the participant, rather than a fold or response coordinate, the biological inference unit. The nine pairs $\{A(u), Q(u)\}$ enter participant-level inference. The assay therefore tests participant-level predictive measurability on this fixed stimulus set. It does not establish causal mediation, brain-response-specific attribution, or stimulus-population generality.

All assays in this subsection use the same general held-out readout protocol. A fresh ridge readout is fitted to frozen features for every evaluated model or target. Ordinary R^2 compares prediction error with variation in the held-out responses. Negative values are retained when predictions are worse than the held-out-mean baseline. Equations 1 and 2 apply across the prerequisite assays. All learned preprocessing, response centering, regularization selection, and readout fitting follow the training-partition rule in Section 3.1. Appendix B specifies the exact inputs, preprocessing, and declared standardization exception. Appendix D specifies the formulas, aggregation, and inference procedures.

3.3 Recorded-Response Intervention

Within each outer fold and technical seed, the recorded-response arm and permuted-response control start from the same student state and use the same sentences. They also share the ordinary KD objective, optimizer schedule, loss weight, and training budget. Only the pairing between recorded-response rows and their sentences differs. Both arms distil a Qwen2.5-1.5B teacher into a Qwen2.5-0.5B student using low-rank adaptation (LoRA) [29].

The teacher remains frozen. Using the partitions defined in Section 3.1, each run trains on 800 Tuckute sentences and leaves the same contiguous block of 200 sentences untouched. Five outer folds and three technical seeds repeat this paired design.

Ordinary KD provides the shared language-preservation objective. Let $\mathcal{L}_{\mathrm{KD}}$ denote the temperature-scaled, teacher-to-student Kullback–Leibler (KL) divergence over all non-padding tokens. Let $\mathcal{L}_{\mathrm{R}}^{(a)}$ denote the auxiliary response loss for arm a. The complete training objective is

$$\mathcal{L}_{\mathrm{train}}^{(a)} = \mathcal{L}_{\mathrm{KD}} + \lambda_{\mathrm{R}} \mathcal{L}_{\mathrm{R}}^{(a)}, \qquad a \in \{\mathrm{recorded}, \mathrm{permuted}\}. \tag{7}$$

Equation 7 states that both paired arms are anchored by the same teacher-defined language objective while receiving an additional response-based training signal. The arm identity changes the response pairing, not the language objective or the strength assigned to the response loss. For a minibatch $\mathcal{B}$, let $\bar{h}_i^{(12)}$ be the masked-mean layer-12 student state, W a temporary linear map, $d = 5$ the number of response coordinates, and $y_i^{(a)}$ the target for sentence i. The auxiliary loss is

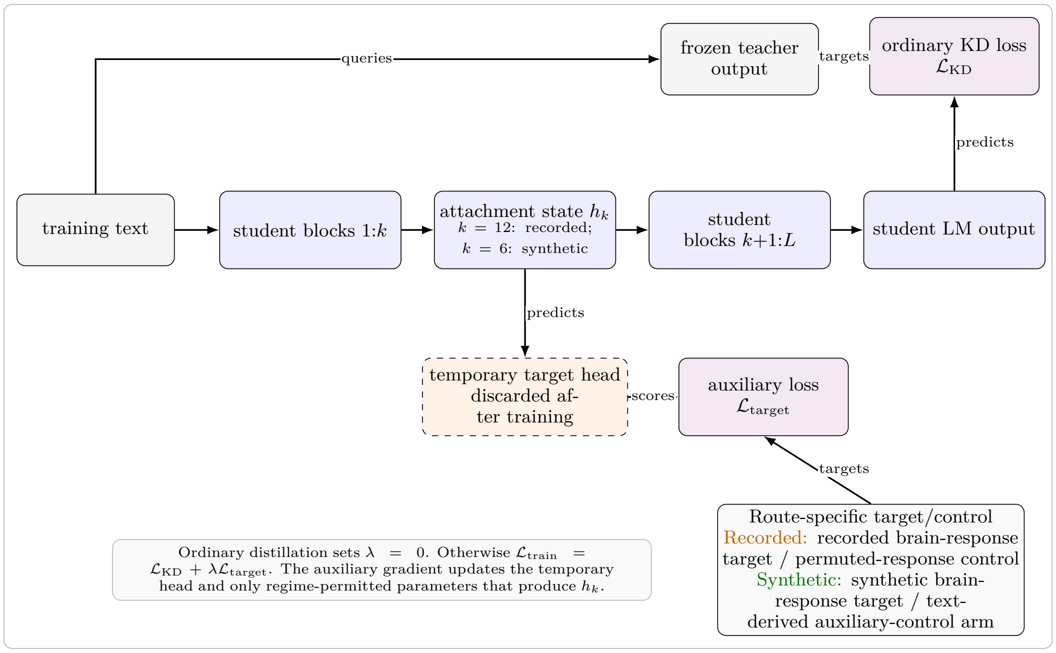
$$\mathcal{L}_{\mathrm{R}}^{(a)} = \frac{1}{|\mathcal{B}|d} \sum_{i \in \mathcal{B}} \left\| W \bar{h}_i^{(12)} - y_i^{(a)} \right\|_2^2. \tag{8}$$

Equation 8 measures the distance between the prediction that the temporary head computes from the student's layer-12 representation and the assigned response target. Minimizing it can change the retained student only when its gradient reaches trainable parameters that produce that representation. The primary intervention uses $\lambda_{\mathrm{R}} = 10$. The ordinary KD reference sets $\lambda_{\mathrm{R}} = 0$, but it is a secondary reference because its language quality need not match that of the response-target arms. Appendix C specifies the exact temperature, masking reduction, optimizer, schedule, and parameter states.

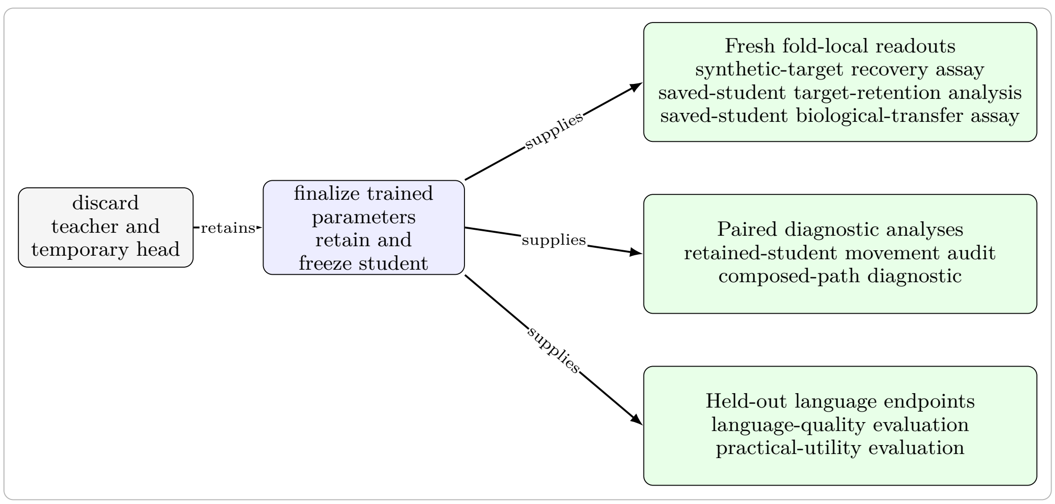
The recorded-response arm uses the response vector paired with each sentence. The control applies the same loss to response rows permuted in contiguous sentence blocks. This permutation preserves the response values, their marginal distribution, and their within-block structure while disrupting the sentence–response correspondence. Because a random block permutation can leave some blocks in their original positions, it is a matched permutation rather than an exact derangement (a permutation with no fixed points). The participant-specific assay repeats this control with five permutation draws and averages their evaluated scores when forming the paired contrast. Appendix B gives the exact permutation construction.

Participant-averaged and participant-specific responses instantiate the same paired intervention but answer different questions. In the participant-averaged regime, each fold–seed run uses the fixed participant-averaged response construction (averaged cell by cell) defined in Section 3.1. Its contrast describes the effect of correct pairing for that shared stimulus-locked target. The participant-specific regime instead trains separate paired arms for each participant with a complete five-ROI response matrix, nine of the ten released. Its contrast supports inference about variation across the tested participants. Averaging responses before training is not equivalent to averaging participant effects after training. The participant-averaged contrast therefore cannot be interpreted as the cohort-mean intervention effect.

The auxiliary branch attaches to the masked-mean layer-12 student state through the temporary head W. In the primary LoRA regime, the pretrained base model and tied input/output embedding remain frozen, while W and the adapters are optimized. The auxiliary gradient can reach W and trainable adapters that produce the layer-12 state. It cannot reach later adapters through this branch, although those adapters can still receive gradients from $\mathcal{L}_{\mathrm{KD}}$. After training, the temporary head and teacher are discarded, the adapters are merged into the retained student, and that student is frozen for evaluation. Figure 4 separates the shared training core from frozen-student evaluation. Figure 4a shows where the ordinary KD and route-specific auxiliary objectives attach. Figure 4b shows that every post-training endpoint uses the retained student rather than the temporary target head.



(a) Shared training core.



(b) Frozen-student evaluation.

Figure 4: Shared intervention training and frozen-student evaluation paths.

After each run, evaluation uses only the untouched 200-sentence block. A fresh ridge readout is fitted and tested within that block using five contiguous inner subfolds. The endpoint of each run is the ROI-averaged unique R^2 defined by Equation 2. Let $U^{(a)}_{ufs}$ denote this score for participant u, outer fold f, technical seed s, and arm a. With $P = 5$ permutation draws indexed by p, the paired cell contrast and participant aggregate are

$$\begin{aligned} d_{ufs} &= U^{(\text{recorded})}_{ufs} - \frac{1}{P}\sum_{p=1}^{P} U^{(\text{permuted})}_{ufsp}, \\ e_u &= \operatorname{median}_{f,s} d_{ufs}. \end{aligned} \tag{9}$$

Equation 9 first compares the recorded-response student with the average of its five matched permuted-response controls within each fold–seed cell. The median over fold–seed cells was predeclared before the participant outcome run because the earlier participant-averaged analysis showed that one unusually large fold could dominate an arithmetic mean. It therefore summarizes technical repetitions within each participant without allowing one fold or seed to define that participant's effect. The participant-level point estimate is

$$\widehat{\delta}_{\text{participant}} = \frac{1}{9}\sum_{u=1}^{9} e_u. \tag{10}$$

Equation 10 averages the nine participant effects to estimate the intervention contrast in the tested cohort. The nine values e_u, rather than the underlying seeds or folds, are the biological inference values. A separate aggregation across the five shared outer folds checks whether the contrast is concentrated in particular stimuli. The participant-averaged analysis instead aggregates seeds within each shared fold and remains descriptive of its fixed response target. A held-out WikiText probe checks language-quality balance between arms. Appendix D specifies the intervals, tests, shared-fold checks, and sensitivity analyses.

The frozen auxiliary-head, increased-LoRA-capacity, contrastive-objective, naturalistic voxelwise, and full-fine-tuning studies retain the paired recorded-versus-permuted comparison while changing the target, target loss, language anchor, trainable subset, or evaluation scope. Appendix C defines these variants, and Section 4 reports their study-specific outcomes. None replaces the primary nine-participant contrast.

3.4 Synthetic-Response Intervention and Downstream Assays

The synthetic-response intervention asks whether fixed synthetic brain-response targets generated from text change a retained student beyond ordinary KD. Across six technical seeds, a frozen GPT-2 Medium teacher supervises a GPT-2 Small student in the synthetic-response, ordinary KD, and text-derived auxiliary-control arms. These arms share the initialized student, training text and order, tokenizer, ordinary KD objective, optimizer schedule, and training budget. The synthetic-response arm adds a fixed text-only TRIBE target, whereas the text-derived auxiliary-control arm mean-pools frozen teacher states and maps them through a fixed Gaussian projection. Both auxiliary targets are standardized to 20,484 coordinates and predicted from the student's layer-6 representation with a mean-squared-error loss. Training updates the student and a temporary prediction head; the head is then discarded and the retained student is frozen. Saved block-permuted variants provide sensitivity checks rather than arms that could identify the claim on their own. Figure 4a locates the synthetic target and text-derived auxiliary control within the shared training core. Figure 4b separates target recovery, retained-student movement, biological transfer, composed predictive overlap, and language evaluation after the temporary head has been removed. Appendices B and C specify the target construction and exact training procedure.

Two manipulation checks ask whether the intervention affected the retained student. First, the synthetic-target recovery assay fits the scaler and ridge readout on the 95,999-example WikiText training partition, selects the ridge penalty from $\{1, 10, 100, 1000, 10000\}$ by training-partition cross-validation, and evaluates the frozen student once on the untouched 1,999-example partition. It then averages target-coordinate R^2 with equal weight across coordinates. For the synthetic-response arm and the text-derived auxiliary-control arm separately, let Δ_s^{rec} be the seed-specific recovery difference between that arm's target-trained student and its seed-matched ordinary KD student. Target uptake, the recoverability of the training target from the retained student, passes only when

$$\min_s \Delta_s^{\text{rec}} > 0 \qquad \text{and} \qquad L_{0.95}\left(\overline{\Delta}^{\text{rec}}\right) > 0, \tag{11}$$

where $L_{0.95}$ is the lower endpoint of the two-sided t interval computed across technical seeds. Equation 11 requires the target-trained student to improve recovery in every seed and requires the mean improvement to remain positive after technical variation is included.

Second, the movement audit compares each trained student with its ordinary KD counterpart through parameter displacement, centered relative-Frobenius distance, and layer-6 centered kernel alignment (CKA) [30]. The retention rule below uses the Frobenius and CKA distances; global parameter displacement is reported separately. Let $D_{F,s}$ and $D_{\text{CKA},s}$ denote the centered relative-Frobenius and linear-CKA distances from the seed-matched ordinary KD student, and let η_{CKA} denote duplicate-extraction noise, the CKA variation between repeated extractions of the same saved student. Retained-student movement passes only when every seed satisfies

$$D_{F,s} > 10^{-4} \qquad \text{and} \qquad D_{\text{CKA},s} > \max\bigl(10^{-6}, 10\eta_{\text{CKA}}\bigr)\,. \tag{12}$$

Equation 12 requires movement to exceed both a fixed numerical floor and the measured extraction-noise floor in every saved seed. These measurements establish retained change, but they do not identify which property of the auxiliary target caused it. The six seeds quantify technical variation rather than biological uncertainty. This WikiText recovery assay is distinct from the saved-student target-retention analysis (Appendix D), which scores retention in response space with nuisance control and variance weighting.

Brain-response-specific attribution additionally requires an adequate text-derived comparator. The comparator audit groups its checks by the source of a possible alternative explanation. Training parity covers the shared optimization conditions and the initial auxiliary-loss and gradient scales, which constitute the pre-update calibration checks. Target parity covers scale, covariance, effective rank, row order, and nuisance predictability. Baseline learnability compares ordinary KD recovery and remaining headroom. These target-construction, baseline, and pre-update calibration properties are identification conditions because they are fixed or measured independently of the target-arm outcomes. Parameter and representation movement are post-training diagnostics that may explain how the interventions differed; they are not comparator-matching conditions. Passing every required pre-outcome or baseline axis is necessary but not sufficient for attributing an effect to brain-response content. Failure on any such axis makes the contrast non-identifying; if no required axis fails but a required measurement is missing, attribution remains unresolved. The available block-permuted targets are also non-identifying, for a reason in how they were built. The construction splits the sentences into ten contiguous blocks, permutes the block order, and pairs each destination block with its source block only up to the shorter of the two lengths. Because the blocks come out unequal in length, a destination row past that cut keeps its own target, and a source row past that cut is never used, so the result retains some correctly aligned rows and can duplicate or omit a row rather than being an exact permutation. The frozen row-twin control therefore supplies an exact-permutation sensitivity check that alters only the target. Because no student was trained against that control, it cannot replace the text-derived auxiliary-control arm.

The saved-student biological-transfer assay then asks whether the synthetic-response intervention improves prediction of independently recorded responses. Let $U^{(a)}_{ufs}$ be the nuisance-controlled, ROI-averaged unique R^2 for participant u, contiguous fold f, technical seed s, and training arm a. The direct biological-transfer endpoint uses the ordinary KD arm as its reference. With $F = 5$ folds and $S = 6$ seeds, define

$$\begin{aligned} d^{\text{direct}}_{ufs} &= U^{(\text{synthetic})}_{ufs} - U^{(\text{KD})}_{ufs}, \\ e^{\text{direct}}_{u} &= \frac{1}{FS}\sum_{f=1}^{F}\sum_{s=1}^{S} d^{\text{direct}}_{ufs}, \\ \widehat{\delta}_{\text{direct}} &= \frac{1}{9}\sum_{u=1}^{9} e^{\text{direct}}_{u}. \end{aligned} \tag{13}$$

Equation 13 first compares paired synthetic-response and ordinary KD students within each fold–seed cell, then averages the fold–seed cells within each participant. The final mean therefore describes biological transfer across the nine tested participants, with participants rather than folds or seeds as the biological inference units. All saved students remain frozen while, in each of the five contiguous folds, fresh ridge readouts are fitted on the 800 training sentences and evaluated on the held-out 200 sentences. We fixed Layer 7 as the primary biological-transfer layer before participant responses were scored and did not allow best-layer selection. We retained Layer 6, where the auxiliary target attached during training, as a prespecified mechanistic sensitivity analysis. This design prevents outcome-driven layer selection while keeping the attachment layer available as a direct check. The direct contrast answers whether the synthetic intervention improves recorded-response predictivity relative to ordinary KD, without attributing any difference to brain-response content.

The attribution endpoint was fixed in the study design before participant scoring and uses the same cell scores with a different comparator. The arm label textctrl denotes the text-derived auxiliary-control arm. Its cell contrast is

$$\begin{aligned} d_{ufs}^{\text{control}} &= \left(U_{ufs}^{(\text{synthetic})} - U_{ufs}^{(\text{KD})}\right) - \left(U_{ufs}^{(\text{textctrl})} - U_{ufs}^{(\text{KD})}\right) \\ &= U_{ufs}^{(\text{synthetic})} - U_{ufs}^{(\text{textctrl})}. \end{aligned} \tag{14}$$

Equation 14 expresses the difference between two gains over ordinary KD and shows why it reduces to the synthetic-response arm minus the text-derived auxiliary-control arm. The reduction holds algebraically, and the byte-identical ordinary KD baselines within each seed ensure that both gains are measured against the same baseline. This contrast was predeclared as the primary attribution comparison, but its interpretation remains conditional on comparator adequacy.

The same saved students support the composed-path diagnostic. Let $A_{H\to T|N}$ denote the representation-specific coefficient contribution in the train-fold model that predicts the synthetic target from nuisance features N and a frozen student representation H. Let $A_{T\to Y|N}$ denote the target-specific coefficient contribution in the train-fold model that predicts recorded responses Y from N and the target T, and let $\widehat{Y}_N$ be the nuisance-only response prediction. The composed prediction is

$$\widehat{Y}_{\text{comp}} = \widehat{Y}_N + A_{T\to Y|N}\left(A_{H\to T|N}(H_{\text{test}})\right). \tag{15}$$

Equation 15 asks how much held-out recorded-response predictivity can pass through the specific synthetic-target subspace recovered from a frozen student. Both conditional coefficient contributions and all fold-dependent preprocessing transforms are estimated from training-fold data only. The externally fixed moments from the synthetic-target construction first standardize the generated target. Before response mapping, the predicted target contribution $A_{H\to T|N}(H_{\text{test}})$ is converted with the train-fold target scale retained from the target-to-response model. Comparing aligned and frozen row-twin targets tests whether sentence-level target pairing matters, while comparing synthetic-response and ordinary KD students tests whether training changed access to that path. This composed score measures predictive overlap; it does not establish causal mediation. The direct saved-student endpoint remains the primary biological-transfer endpoint.

Language-quality checks qualify the synthetic-response contrasts but do not themselves establish practical benefit. Perplexity and bits per byte test whether the compared students remain within the declared quality margin. Passing the 0.005 bits-per-byte margin only excludes discrepancies larger than that margin on this metric; it does not prove the absence of language-quality confounding. A separate practical-utility evaluation belongs to the recorded-response route. It compares approximately quality-matched recorded-response and permuted-response students through the ratio of out-of-domain to in-domain perplexity on Pereira sentences and LeBel story text. For corpus d, arm a, and technical seed s, define

$$R_{d,a,s} = \frac{\text{PPL}_{d,a,s}^{\text{OOD}}}{\text{PPL}_{a,s}^{\text{ID}}}, \qquad \Delta R_{d,s} = R_{d,\text{recorded},s} - R_{d,\text{permuted},s}. \tag{16}$$

The ratio measures out-of-domain perplexity relative to the same student's in-domain perplexity. A negative $\Delta R_{d,s}$ favors the recorded-response arm. The prerequisite contrast in controlled brain predictivity is defined at the technical-seed level as

$$A_s = U_s^{\text{recorded}} - U_s^{\text{permuted}}, \tag{17}$$

where U_s is held-out controlled brain predictivity. This contrast tests whether the recorded-response arm produced the representational change whose downstream value the utility comparison is intended to assess. Ordinary KD and text-derived auxiliary-control arms remain descriptive references when their language quality is unmatched. The ratio contrast estimates endpoint behavior on each fixed corpus. A brain-specific utility interpretation additionally requires two things: a reliable brain-response-specific change in the prerequisite contrast, and an uncertainty estimate for the paired endpoint contrast at the technical-seed unit. An minimum detectable effect (MDE) summarizes the effect size that an assay could reliably resolve under its stated variance model; it is not a pass criterion, hypothesis test, or equivalence margin. The seeds of this practical-utility analysis quantify technical variation, not independent biological evidence. Section 3.5 states the formal aggregation and decision rules.

3.5 Cross-Cutting Estimands and Decision Rules

Table 2 binds each scientific role to its estimand, comparator, endpoint, and inference unit; the focal and comparator conditions are first evaluated on the same endpoint within the smallest matched analysis cell, such as a single fold–seed cell. The interpretation of a numerical difference changes when its comparator, endpoint, or inference unit changes, even if the same scoring function is used.

Figure 5 shows the common order from a matched analysis cell to the declared inference unit. Its three tracks separate biological inference across participants, fixed-stimulus inference across shared folds or stories, and technical inference across seeds. The eligibility check introduced below qualifies interpretation but neither changes the inference unit nor supplies endpoint evidence.

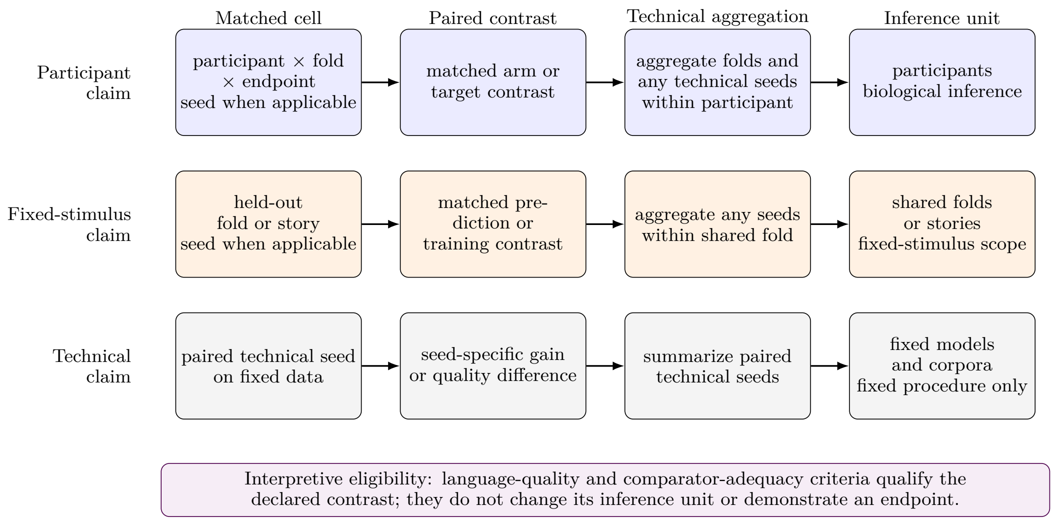


Figure 5: Aggregation paths for participant, fixed-stimulus, and technical claims.

A training contrast becomes eligible for interpretation only under its declared LM-quality rule. The quality metric follows the declared assay. The scaled synthetic-target family comprises all target arms of the synthetic-response route trained on the 95,999-sentence corpus; these arms and the recorded-response probes use perplexity under a shared tokenizer and scoring protocol. Cross-tokenizer or cross-family comparisons require bits per byte, and the saved-student transfer assay uses its predeclared bits-per-byte criterion even though its models share a tokenizer. Let $\mathrm{PPL}_{a,s}$ be the held-out perplexity of target arm a in seed s, and let b denote the arm against which a is compared. Let $L_{a,s}$ and $B_{a,s}$ be its total negative log-likelihood in bits and the number of decoded UTF-8 bytes in the same scored span. Let $\epsilon_{\mathrm{ppl}} = 0.05$ and $\epsilon_{\mathrm{bpb}} = 0.005$ be the declared relative-perplexity and bits-per-byte tolerances, respectively. The two principal quality criteria are

$$\begin{aligned} q_{\mathrm{ppl}}(a,s) &= \frac{|\mathrm{PPL}_{a,s} - \mathrm{PPL}_{\mathrm{KD},s}|}{\mathrm{PPL}_{\mathrm{KD},s}} \le \epsilon_{\mathrm{ppl}}, \\ q_{\mathrm{bpb}}(a,b,s) &= \left| \frac{L_{a,s}}{B_{a,s}} - \frac{L_{b,s}}{B_{b,s}} \right| \le \epsilon_{\mathrm{bpb}}. \end{aligned} \tag{18}$$

The first rule requires each scaled synthetic-target arm to remain within 5% of its seed-matched ordinary KD reference on the fixed WikiText quality probe. The second requires the saved-student comparisons to remain within 0.005 bits per byte in every seed. Passing a rule makes the contrast eligible under that metric and tolerance; it does not prove identical language behavior or remove possible differences inside the accepted margin. The paired recorded-response arm and permuted-response control are evaluated on the same fixed WikiText probe under a shared scoring implementation, with comparable perplexity by construction of the paired design. Recorded-response comparisons with ordinary KD additionally require perplexity matching across checkpoints or training curves. Without the required match, the corresponding biological contrast remains descriptive.

Aggregation follows the declared inference unit. Technical repetitions are combined before uncertainty is calculated across participants or fixed stimulus blocks; when technical seeds are the inference unit, each seed enters the uncertainty calculation directly. Figure 5 summarizes these routes; Appendix D gives the exact nesting, estimators, tests, and multiplicity rules.

Detectable-effect bounds and continuation bounds, the thresholds fixed in advance for deciding whether a follow-up proceeds, answer narrower questions than statistical intervals. An MDE, defined above, does not imply that smaller effects are zero or biologically irrelevant. We exclude an earlier detectable-effect calculation for the naturalistic assay because it used unmatched absolute fold variability rather than the intended paired contrast. The participant analyses calculate detectable-effect bounds from between-participant variation, whereas the practical-utility analysis uses variation across technical seeds. The +0.002 unique-R^2 threshold was reused from the biological-transfer design and fixed in advance as the rule for deciding whether the participant assay warrants a follow-up. It is not a universal biological-importance threshold and cannot be transferred to another response scale, nuisance model, layer, or population.

Evidence status follows the claim dependencies in Figure 1. A passed manipulation check establishes target uptake or retained-student movement, but it does not establish brain-response-specific attribution, biological transfer, or practical utility. Failure of a required manipulation or quality criterion limits only a claim that depends on that condition; it does not invalidate a positive endpoint contrast whose own comparison is identifying. Comparator adequacy requires every required axis to pass: failure on any one of them makes the attribution contrast non-identifying. An unmeasured required axis leaves adequacy unresolved only when no measured required axis has already failed. An unmeasured stage supplies no evidence and is not a null result. The label not demonstrated is reserved for an identifying, adequately measured contrast that does not pass its decision rule. We apply these statuses in Section 4 by claim role rather than experiment number or execution date.

4 Results

Each subsection reports one decisive observation, its uncertainty at the declared inference unit, the resulting evidence status, and the narrow claim that the status licenses. The subsections follow the claim dependencies in Figure 1 and the scientific roles defined in Section 3. Table 3 summarizes these role-specific outcomes using supported, not demonstrated, unresolved, unmeasured, inconclusive, and non-identifying comparator.

Table 3: Role-specific evidence status and decisive observations.

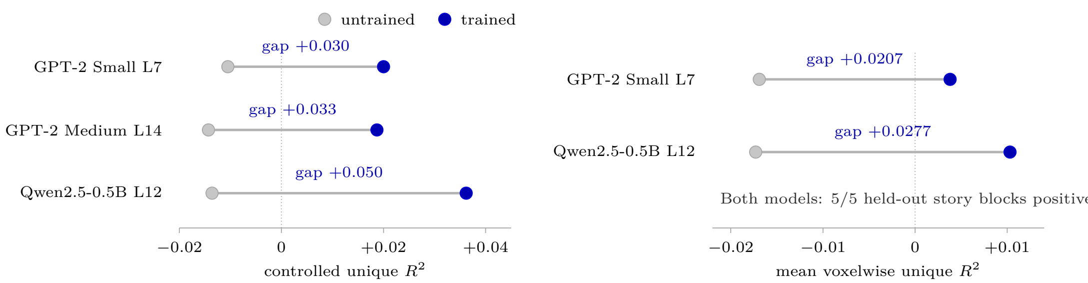
Scientific role	Status	Decisive observation
Controlled regional predictivity assay	Supported; attribution of the gap to learned parameters unresolved	Trained Qwen2.5-0.5B controlled unique R^2: $+0.036 \pm 0.010$. The +0.050 trained-minus-untrained gap was computed with model-specific static nuisance blocks.
Naturalistic assay	Supported	GPT-2 Small and Qwen2.5-0.5B gaps over matched untrained controls: +0.0207 and +0.0277.
Distillation-headroom analysis	GPT-2 warm route unresolved; Qwen route unmeasured	The cold-start GPT-2 Small gap is +0.0181, but initialization and LM quality differ. The warm-start comparison provides context for this route but does not establish a teacher–student gap at comparable language quality.
Recorded-response intervention, participant-averaged estimand	Not demonstrated	Paired gain: +0.0081 (five-fold t interval [−0.0030, +0.0193]); one fold dominates the mean.
Recorded-response intervention, participant-specific estimand	Not demonstrated	Nine-participant mean: +0.00010 (participant t interval [−0.00037, +0.00058]).
Target-projection assay	Not demonstrated; row-pairing specificity unresolved	Aligned-target increment: −0.004494 (Bonferroni family-95% interval [−0.010695, +0.001707]); row-twin contrast unresolved.
Synthetic-target recovery assay	Supported	WikiText recovery gain over seed-matched KD: +0.078298 (two-sided 95% t interval across technical seeds [+0.077198, +0.079398]).
Retained-student movement audit	Supported	All seeds pass; mean relative-Frobenius movement from KD: 0.2151, at acceptable LM quality.
Saved-student target-retention analysis	Unresolved	Fresh held-out Tuckute retention increment: −0.000855 (Bonferroni family-95% interval [−0.002246, +0.000537]).
Comparator-adequacy audit and attribution assessment	Non-identifying comparator	In the corrected audit, the text-derived auxiliary-control arm fails 16/37 identification checks; post-training movement remains diagnostic and does not determine comparator validity.
Saved-student biological-transfer assay	Not demonstrated	Participant-level synthetic-response-minus-KD mean: −0.000014 (95% interval [−0.000739, +0.000711]).
Composed-path diagnostic	Unresolved	Synthetic-response-minus-KD increment: +0.000031 (Bonferroni family-95% interval [−0.000080, +0.000142]).
Practical-utility evaluation	Endpoint inconclusive; brain-specific utility not demonstrated	Pereira and LeBel means are −0.019 and −0.068, both favoring the recorded-response arm; no paired endpoint interval or equivalence analysis was retained.

Status key. Supported: the declared criterion passed. Not demonstrated: the tested contrast did not support the claim. Unresolved: the analysis ran, but a confound or a missing control prevents separating the competing explanations. Unmeasured: the route-matched comparison was not run. Inconclusive: the analysis ran and its comparison is interpretable, but the retained evidence supports no conclusion in either direction. Non-identifying comparator: the available control cannot attribute an effect to brain-response content.

4.1 Controlled brain predictivity under regional and naturalistic assays

Trained representations exceeded the declared nuisance baseline in the controlled regional predictivity assay. For the primary Qwen2.5-0.5B layer-12 comparison, controlled unique R^2 was $+0.036 \pm 0.010$, where the $\pm$ term is the descriptive standard deviation across five contiguous stimulus folds. The previously recorded trained-minus-untrained difference was +0.050, but we recomputed the static nuisance block separately for each model. That gap therefore does not cleanly isolate learned parameters against one fixed nuisance design. Because we averaged the response target across five participants before readout fitting, the folds describe stimulus-block variation and do not support participant-level inference. The assay therefore supports controlled regional predictivity for this fixed participant-averaged target.

Trained representations also exceeded their matched untrained controls in the naturalistic assay when complete stories were held out. For participant UTS03, the mean differences were +0.0207 for GPT-2 Small and +0.0277 for Qwen2.5-0.5B, and all five held-out-story blocks had positive differences. The voxels are spatially dependent, and the story folds share training stories, so neither the voxel set nor the story folds provide independent biological replication. This assay therefore supports a within-participant controlled brain predictivity result conditional on UTS03 and the fixed naturalistic protocol.



(a) Controlled regional predictivity. (b) Controlled naturalistic voxelwise predictivity.

Figure 6: Assay-specific trained–untrained contrasts; scales and inference units remain separate.

Figure 6 places the two trained-versus-untrained comparisons in separate panels. Together, the assays show that trained LM representations contain linearly accessible information useful for predicting held-out recorded fMRI responses beyond the specified nuisance features. In the naturalistic assay the matched trained–untrained gap is itself supported; learned-parameter attribution remains unresolved for the regional trained-minus-untrained gap. Their numerical estimates are not pooled or compared in magnitude because the response construction, target resolution, nuisance set, and inference unit differ.

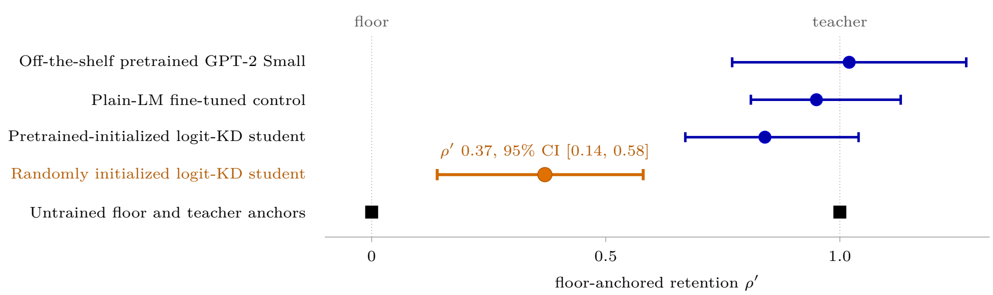
The assays do not identify a shared computation between the LMs and the brain or establish population-general controlled brain predictivity. They also do not test whether using brain responses during training changes a retained student or improves language behavior. Section 4.2 next asks what route-specific headroom is actually established.

4.2 Observed distillation headroom and its association with language quality

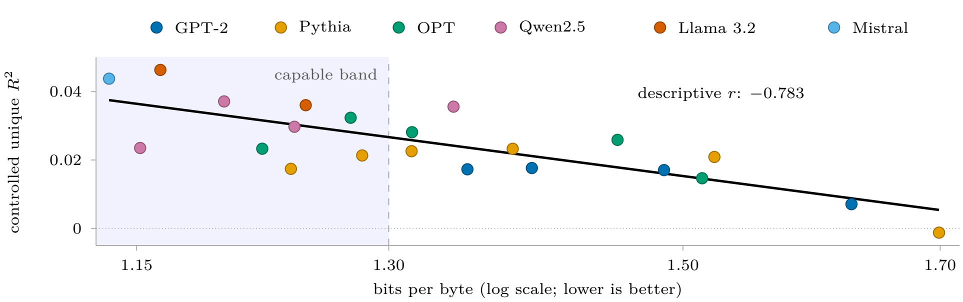
The from-scratch GPT-2 comparison contains an observed teacher-relative gap, but it does not establish quality-aware headroom (a teacher–student gap measured at comparable language quality) for the intervention routes. Figure 7a places the GPT-2 teacher, floor, and student conditions on the floor-anchored retention scale defined in Equation 4. At the fixed Tuckute layers, the randomly initialized GPT-2 Small logit-KD student is +0.0181 below the GPT-2 Medium teacher in controlled brain predictivity (bootstrap no-drop p =0.000 over the cold arm's seeds). Relative to the architecture-matched untrained GPT-2 Small floor, the student retains 0.37 of the teacher-to-floor interval. The recorded 95% bootstrap interval over folds and seeds is [0.14, 0.58]. The tested cold-start route therefore leaves an observed gap within this assay, although the comparison does not identify its source.

Language quality is a material competing explanation. The randomly initialized student remains substantially weaker as an LM, with perplexity 477, compared with 74 for the teacher. The evaluated conditions also differ in initialization and training history, not only in objective. Their ordered predictivity estimates establish a descriptive teacher–student gap, but they do not show that compression or KD caused the gap. The pretrained GPT-2 Small condition is closer to the teacher in initialization and language quality, which makes it the relevant reference for the synthetic-response route, but the completed analysis did not establish a formal quality-aware headroom result for that warm route. The recorded-response intervention uses Qwen2.5-1.5B and Qwen2.5-0.5B, for which we did not measure route-matched teacher–student headroom.

The broader model comparison shows why later interventions must control language quality. Across 22 decoder-only LMs from six families, controlled brain predictivity is negatively associated with log bits per byte, where higher values indicate poorer language quality. Pearson r is -0.783, with a family-cluster 95% confidence interval (CI) of $[-0.911, -0.500]$. Figure 7b shows this full-range association and marks the capable-model band, defined as the $n = 10$ models with bits per byte below 1.30. Within that band, Pearson r is -0.501, with a Fisher 95% CI of $[-0.859, +0.188]$. The point estimate remains moderately negative, but its interval includes zero. Model family, scale, training data, and language quality remain entangled, so the association identifies language quality as a design constraint rather than a causal explanation.



(a) Floor-anchored retention; the cold-start gap is quality-confounded and quality-aware headroom for the warm route remains unresolved.



(b) Controlled brain predictivity across 22 decoder-only LMs.

Figure 7: Observed headroom and its association with language quality.

Together, the results support an observed, assay-conditional gap between the GPT-2 teacher and randomly initialized student. Quality-aware headroom remains unresolved for the warm GPT-2 route and unmeasured for the Qwen route. They do not establish that the headroom is brain-specific or recoverable through a brain-response objective. A valid intervention comparison must therefore test the brain-response objective against a same-procedure control while maintaining comparable LM quality. Section 4.3 applies this requirement to the recorded-response intervention.

4.3 Recorded-response intervention at valid inference units

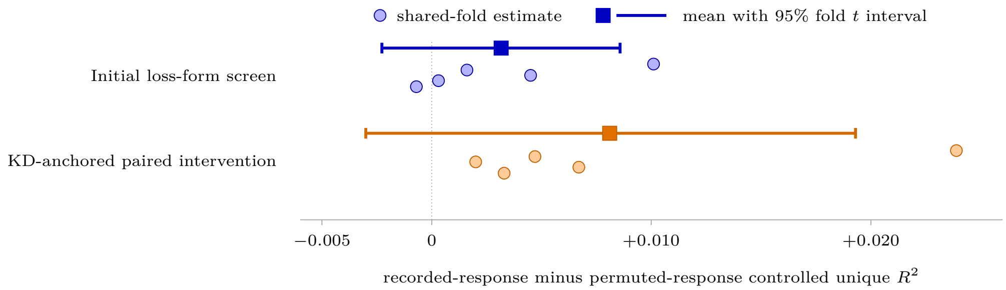
Participant-averaged recorded responses produce positive recorded-response-minus-permuted-response point estimates, but neither comparison establishes a reliable pairing advantage at the shared-fold inference unit. In the initial loss-form screen (a comparison of the candidate training losses), the mean contrast is +0.00316, with a five-fold t interval of [−0.00227, +0.00858]. In the KD-anchored paired intervention, the mean contrast is +0.0081, with a five-fold t interval of [−0.0030, +0.0193]. Both intervals include zero, and both estimates are sensitive to a dominant fold. The point estimates are positive, but a pairing advantage is not demonstrated.

These averaged-response comparisons answer a narrower question than participant-level inference. Averaging responses before training creates one shared stimulus-locked target, so the fold-level contrast asks whether correct stimulus–response pairing helps for that fixed target. It does not estimate the mean intervention effect across participants. The recorded-response arm and permuted-response control have perplexities 55.4 and 55.3, respectively, showing no gross quality imbalance without establishing a formal equivalence margin.

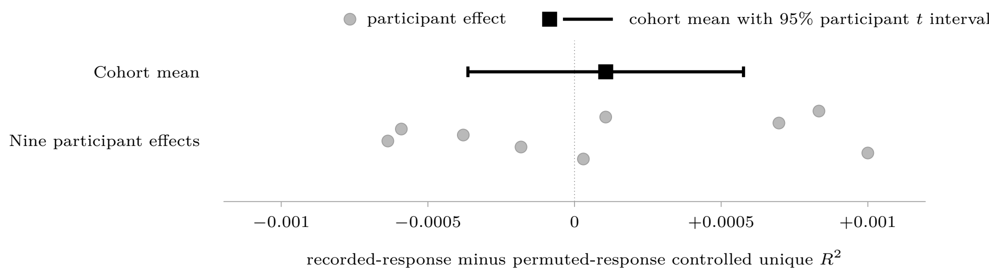
The participant-count diagnostic probes how the contrast varies with the number of participants. In the original nested sequence each subset contained the previous one, so participant count was confounded with participant identity; with random subsets instead, the diagnostic shows no stable monotone relationship. With the student frozen, averaging nevertheless increases controlled brain predictivity by reducing response noise and emphasizing the shared response component; individual responses still contain measurable signal. Averaging can therefore strengthen measurement of the shared response without creating biological replication.

The participant-specific intervention does not demonstrate a participant-general advantage at the biological inference unit, the participant. After the technical seed and fold repetitions are aggregated within each participant as defined in Section 3.3, the mean recorded-response-minus-permuted-response effect across nine participants is +0.00010, with participant t interval [−0.00037, +0.00058]. The shared-fold summary is also centered near zero at +0.00026, with five-fold t interval [−0.0004, +0.0009].

The minimum detectable effect at 80% power, computed from the observed variance, is approximately 0.0006–0.0013 for the participant analysis. This range is far below the earlier participant-averaged point estimate of +0.0081. If an individual-level effect of that earlier magnitude existed on this scale, the participant analysis would therefore have detected it. Because the two results use different response constructions, scales, and estimands, that transfer of effect size is not guaranteed. The participant result neither proves an exactly zero effect nor extends beyond the tested target, model, cohort, objective, and readout. Figure 8 therefore shows them in separate panels rather than pooling them.



(a) Positive averaged-target trends remain uncertain at the shared-fold unit.



(b) Participant-specific effects center near zero at the biological inference unit.

Figure 8: Recorded-response contrasts at their valid fold and participant inference units.

The tested variants address specific alternative explanations, but none provides evidence that changes the participant-general conclusion. Table 4 separates participant contrasts from studies in which the intended manipulation was not demonstrated, or in which the biological sample supported only an existence probe. An existence probe is a low-power per-subject check of whether the intervention effect appears at all, without population scope. Objectives and schedules are documented in Appendix C, secondary-variant dispositions in Appendix E, and full numerical tables in Appendix F.

Table 4: Recorded-response robustness checks and their scoped conclusions.

Variant	Strongest valid observation	Scoped conclusion
Higher-capacity LoRA	Participant mean +0.00043, with t interval [−0.0005, +0.0013].	Additional adapter capacity at the fixed loss weight does not rescue the effect and does not produce a stronger manipulation.
Contrastive objective	Participant mean −0.00019, with t interval [−0.0008, +0.0005].	Changing the loss does not rescue the effect at five-ROI granularity.
Naturalistic voxelwise LoRA	The recorded-response arm does not produce the required improvement over its starting model and permuted control.	The intended manipulation is not demonstrated; this is not a naturalistic population null.
Full fine-tuning	Lower-intensity training shows no reliable paired gain in a three-participant analysis; higher-intensity training causes forgetting of previously learned language-model behavior.	The tested full-parameter route does not rescue the effect, and no population-general inference is supported.

Across the tested recorded-response interventions, the evidence does not demonstrate a participant-general advantage of correct stimulus–response pairing. The participant-general advantage is not demonstrated for the tested targets, models, objectives, cohorts, and readouts; this is not a universal null. The synthetic-response route is a separate intervention route. Section 4.4 first evaluates the declared 50-component linear pathway from the synthetic target to recorded responses.

4.4 Measurability of the 50-component linear target projection

This assay asks whether the 50-component PCA representation of the TRIBE target, the synthetic brain response generated from text, adds held-out linear predictivity of recorded fMRI responses beyond the nuisance model. Section 3.2 defines $A(u)$ as this unique increment for participant u. It defines $Q(u)$ as the additional value of correct sentence–target pairing over the frozen row-twin control. Both quantities are aggregated across folds within each participant, making the participant the biological inference unit.

The aligned-target increment does not pass the continuation rule carried over from the biological-transfer design. Let $\bar{A}$ denote the mean of $A(u)$ across the $n = 9$ participants, let $U^{\text{family}}_{95\%}(\bar{A})$ denote the upper endpoint of its Bonferroni family-95% interval, and let τ_{cont} denote that continuation threshold. The result is

$$\begin{aligned} \bar{A} &= -0.004494, \\ \mathrm{CI}^{\text{family}}_{95\%}(\bar{A}) &= [-0.010695,\ +0.001707], \\ U^{\text{family}}_{95\%}(\bar{A}) &= +0.001707 < \tau_{\text{cont}} = +0.002. \end{aligned} \tag{19}$$

Equation 19 shows that even the Bonferroni family-95% upper endpoint remains below the continuation threshold. Only 3 of 9 participant effects are positive, and every leave-one-out mean, dropping any single participant or stimulus block, remains negative. The assay therefore justifies the decision not to continue this route under the fixed conditions of the assay. That continuation threshold marks no universal boundary for biological importance.

Correct sentence–target pairing remains unresolved. The frozen row twin preserves the target rows and readout procedure while breaking their pairing with sentences. The participant mean of $Q(u)$, the aligned-minus-row-twin contrast, is +0.001758. Its Bonferroni family-95% interval is [−0.003575, +0.007091]. 5 of 9 participant contrasts are positive, but the interval spans zero. The assay therefore does not establish that correct pairing adds predictive value over this frozen row-twin control.

Figure 9 places the two participant-level contrasts on a common scale. The red guide applies only to the aligned-target increment and marks the continuation threshold. Colored squares and horizontal bars show participant means and Bonferroni family-95% intervals.

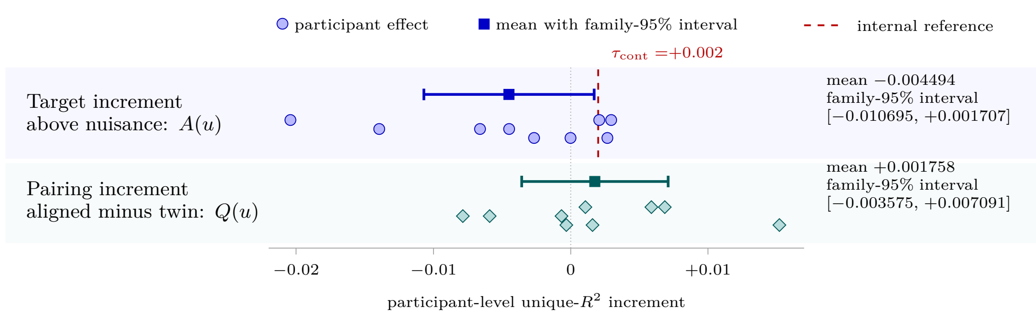


Figure 9: Linear target-projection measurability and row-pairing specificity across participants.

On the fixed sentence set, the target-projection assay does not meet that continuation rule, and row-pairing specificity remains unresolved. Training used the full 20,484-coordinate target, so not clearing the continuation rule says nothing about full-dimensional or nonlinear access to that target. The assay also leaves open total predictivity, predictivity redundant with the nuisance model, and effects under another target transform, cohort, or endpoint; it does not causally explain the later student results. The result limits only the declared linear target-projection claim. The next subsection asks the independent question of whether training produced target uptake and retained-student movement, and whether the available comparator attributes any arm difference to brain-response content.

4.5 Synthetic-response manipulation checks and attribution status

Training changed the saved student despite the noncontinuation result for the separate target-projection assay. The first manipulation check asks whether a fresh readout recovers more of the synthetic target from the synthetic-response student than from its seed-matched ordinary KD student. Let $R^2_{s,a}(T)$ denote held-out WikiText target recovery for technical seed s, training arm a, and target T. Let S denote the number of saved technical seeds ($n = 6$). The mean uptake contrast is

$$\widehat{\Delta}^{\text{Wiki}}_{\text{uptake}} = \frac{1}{S}\sum_{s=1}^{S}\left[R^2_{s,\text{synthetic}}(T) - R^2_{s,\text{KD}}(T)\right]. \tag{20}$$

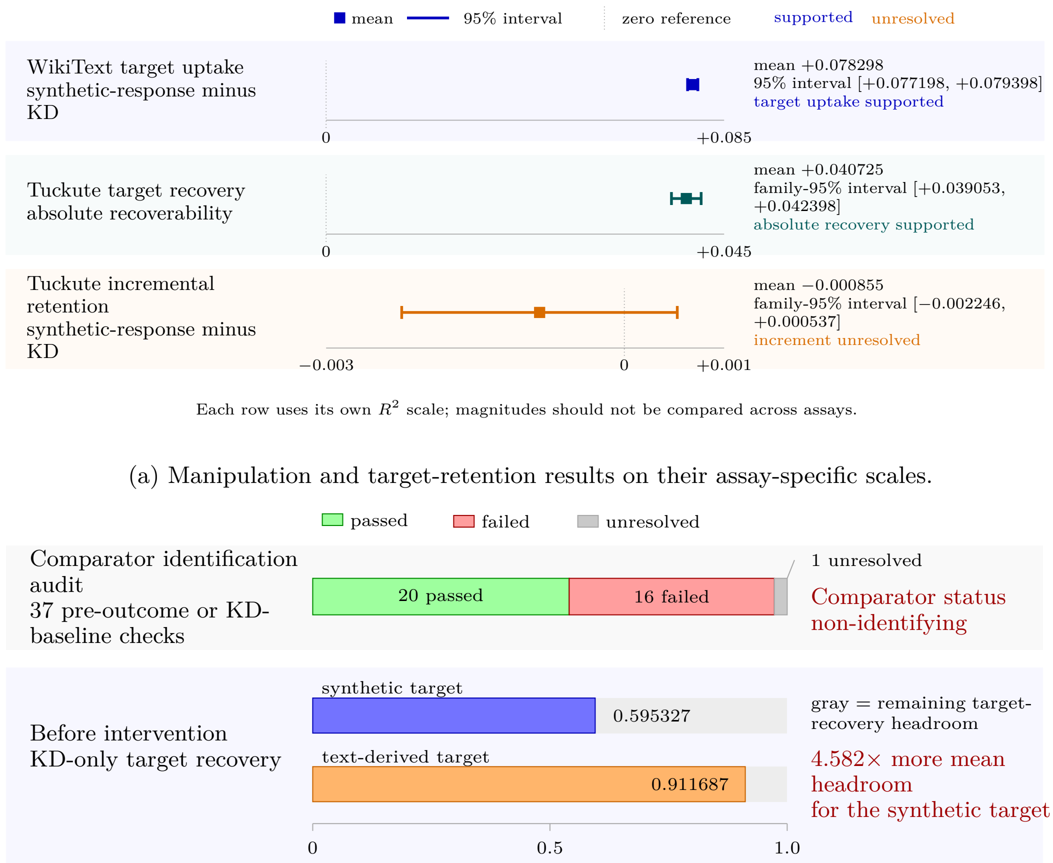
Equation 20 measures the increase in linear target recoverability after synthetic-response training relative to ordinary KD. Its estimate is +0.078298, with two-sided 95% seed-axis t interval [+0.077198, +0.079398]. The saved procedure therefore passes the WikiText target-uptake check. These seeds quantify technical stability, not biological or model-population uncertainty.

The retained student also moved while measured language quality remained within the declared margin. Mean movement from the ordinary KD student, measured by layer-6 centered relative Frobenius distance, is 0.2151. Every saved technical seed passes the movement check. Across the direct saved-arm comparisons, the largest absolute language-quality difference is 0.002579 bits per byte, below the frozen 0.005 margin. These results show that the intervention changed the retained student without a detected language-quality discrepancy larger than the declared margin. They do not show that brain-response content caused the change or improved language quality.

A fresh readout on the Tuckute sentences gives a narrower result. Absolute target extractability from the synthetic-response students is +0.040725, with Bonferroni family-95% interval [+0.039053, +0.042398]. The incremental synthetic-response-minus-ordinary KD estimate is instead −0.000855, with Bonferroni family-95% interval [−0.002246, +0.000537]. Thus, the target is linearly recoverable from the trained students on these sentences, but whether synthetic-response training increased its recoverability beyond ordinary KD remains unresolved. The $n = 6$ seeds again describe stability of the saved training procedure rather than biological replication.

Differences between the arms in these manipulation results cannot be attributed to brain-response content, because the available text-derived auxiliary-control arm is not adequately matched. In the corrected audit, the text-derived auxiliary-control arm fails 16/37 identification checks. Those checks test conditions that were fixed or measured before any outcome was seen, and each required check is a necessary condition for attribution, so a single failure is enough to leave the comparator unable to identify an effect of brain-response content. Before intervention, ordinary KD representations recover mean target R^2 of 0.595327 versus 0.911687 for the synthetic and text-derived targets, respectively. This leaves 4.582 times as much remaining room for target recovery for the synthetic target. The text-derived auxiliary-control arm also fails the required identification margins for covariance geometry, effective rank, and low-level nuisance predictability. Initial auxiliary-loss and gradient comparability remain unresolved because those quantities were not retained. Separately, 14/18 post-training diagnostic margins depart from their original engineering ranges. Parameter displacement, centered relative-Frobenius movement, and CKA describe how the interventions affected the students; they do not determine comparator validity. The pre-outcome and baseline mismatches can affect apparent learnability and retained-student movement without any difference in brain-response content. The audit does not establish which mismatch caused the observed cross-target difference.

Figure 10 separates these manipulation results from the attribution audit.



(a) Manipulation and target-retention results on their assay-specific scales.

(b) Comparator-audit outcomes and baseline target-recovery mismatch.

Figure 10: Synthetic-response manipulation and attribution evidence.

A block-permuted arm of the synthetic-response intervention was also saved, but it cannot resolve this attribution problem. Its construction leaves some rows aligned and duplicates or omits other rows, so it remains a sensitivity check rather than an exact-null trained comparator.

The intervention therefore produced target uptake and retained-student movement, but the available comparator remained non-identifying for the attribution claim. The next subsection tests biological transfer while preserving that limit.

4.6 Saved-student biological-transfer assays

The preceding manipulation checks show that synthetic-response training changed the saved student, but they do not show that the change improves prediction of recorded brain responses. The direct biological-transfer assay therefore compares each synthetic-response student with its seed-matched ordinary KD student. Folds and technical seeds are averaged within each participant, and the participant is the biological inference unit ($n = 9$). The participant-level result is

$$\begin{aligned} \widehat{\Delta}_{\text{direct}} &= -0.000014, \\ \mathrm{CI}_{95\%} &= [-0.000739, +0.000711]. \end{aligned} \tag{21}$$

Here, $\widehat{\Delta}_{\text{direct}}$ is the participant-mean change in held-out unique R^2 produced by synthetic-response training relative to ordinary KD. The participant t interval spans zero, so the direct assay does not support participant-general biological transfer in the tested cohort. The one-sided 95% participant upper bound +0.000653 lies below the internal continuation threshold +0.002, so a participant-mean effect of at least that size is excluded. It does not establish that the effect is exactly zero, and its scope remains limited to the fixed target, student, layer, nuisance design, sentences, participants, and ROI endpoint.

The synthetic-response-versus-text-derived auxiliary-control contrast is numerically positive, but it does not overturn the direct result:

$$\begin{aligned}\widehat{\Delta}_{\text{synthetic}-\text{text}} &= -0.000014 - (-0.000150) = +0.000136, \\ \mathrm{CI}_{95\%} &= [-0.000505,\ +0.000777].\end{aligned} \tag{22}$$

This decomposition shows that the positive relative contrast arises because the text-derived auxiliary-control arm is more negative than ordinary KD, not because the synthetic-response arm has a positive direct effect. The participant t interval also spans zero. Section 4.5 shows that the text-derived auxiliary-control arm is not adequately matched as a comparator. The contrast therefore cannot identify an effect of brain-response content.

The composed-path diagnostic asks a different question. It measures whether the synthetic-target component that a fresh readout recovers from the frozen student also predicts recorded responses. Figure 11b presents its three participant-level estimands:

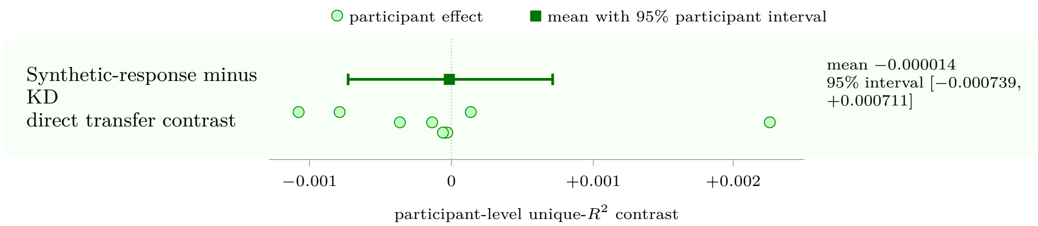
$$\widehat{O} = +0.001242, \qquad \mathrm{CI}_{95\%}^{\text{family}}(\widehat{O}) = [-0.000400,\ +0.002885], \tag{23}$$

$$\widehat{D} = +0.000031, \qquad \mathrm{CI}_{95\%}^{\text{family}}(\widehat{D}) = [-0.000080,\ +0.000142], \tag{24}$$

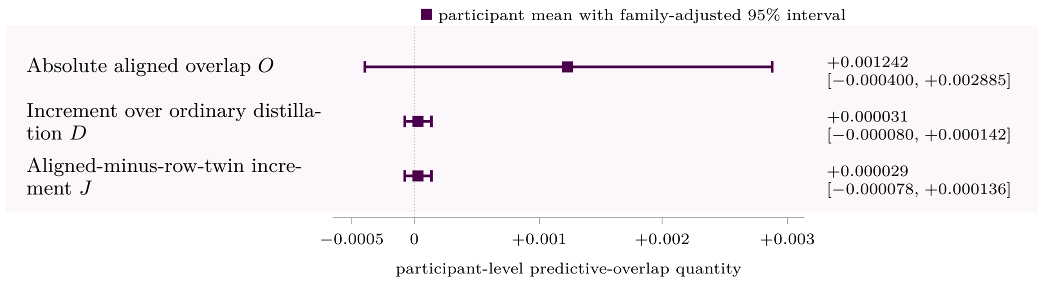
$$\widehat{J} = +0.000029, \qquad \mathrm{CI}_{95\%}^{\text{family}}(\widehat{J}) = [-0.000078,\ +0.000136]. \tag{25}$$

Here, O measures absolute aligned predictive overlap, D measures its increment over seed-matched ordinary KD, and J asks whether that increment is larger for correctly aligned targets than for the frozen row twin. Every Bonferroni family-95% interval spans zero. The composed-path diagnostic therefore leaves predictive overlap, incremental overlap over ordinary KD, and any aligned-over-row-twin increment unresolved under participant inference. These quantities are predictive-path diagnostics, not causal-mediation estimates or replacements for the direct biological-transfer endpoint.

Figure 11 summarizes the direct assay and the composed-path diagnostic after their estimands have been defined. Panel a shows the nine direct participant effects and their participant-level interval; the zero line makes the uncertain direction visible without treating folds or technical seeds as additional biological observations. Panel b shows the Bonferroni family-95% intervals for O, D, and J.



(a) Direct synthetic-response-versus-ordinary KD effects across participants.



(b) Bonferroni family-95% composed-path diagnostic intervals.

Figure 11: Direct biological transfer and the composed-path diagnostic.

Neither assay demonstrates participant-general biological transfer, and the text-derived comparison remains non-identifying.

4.7 Practical-utility evaluation

This final subsection returns to the recorded-response route, which contains the only completed practical-utility evaluation. It compares the averaged recorded-response arm with its approximately quality-matched permuted-response control across $n = 8$ technical training seeds. Equation 16 defines the out-of-domain to in-domain perplexity ratio and the recorded-response-minus-permuted-response contrast; negative values of the contrast favor the recorded-response arm. Averaging that contrast across seeds gives

$$\overline{\Delta R}_{\text{Pereira}} = -0.019, \qquad \overline{\Delta R}_{\text{LeBel}} = -0.068. \tag{26}$$

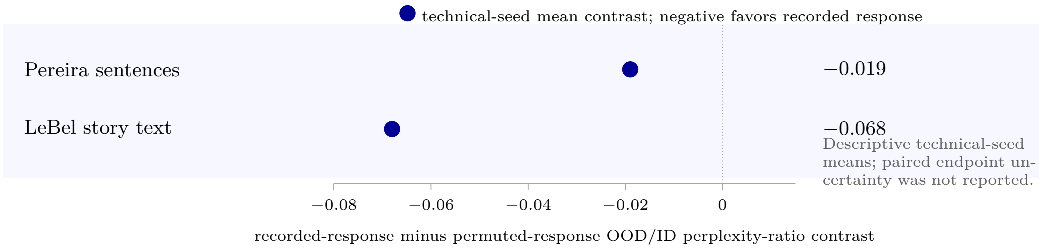
These are the mean contrasts for Pereira sentences [27] and LeBel story text [25], respectively. Both point estimates favor the recorded-response arm. The completed analysis retained no paired technical-seed uncertainty interval or formal equivalence test, and an MDE cannot substitute for either. The endpoint contrasts are therefore inconclusive.

Interpreting this downstream result requires a reliable difference in controlled brain predictivity between the two arms, because the endpoint means can be credited to brain-response training only if that training reliably changed brain predictivity. Equation 17 defines this prerequisite contrast in controlled brain predictivity as A_s. Its seed-level summary is

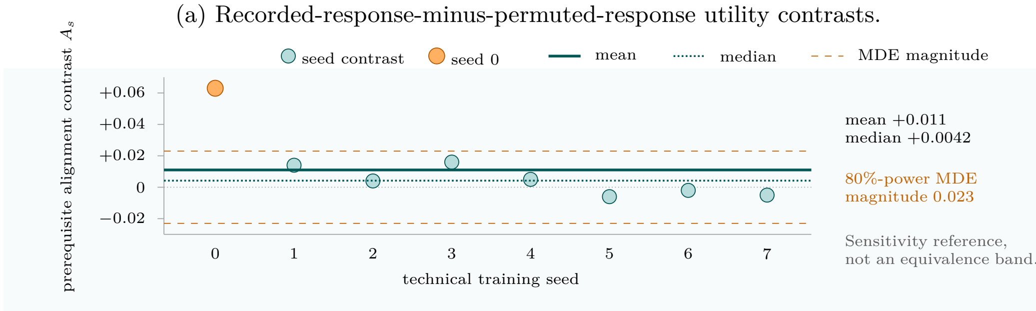
$$\text{mean}_s(A_s) = +0.011, \qquad \text{median}_s(A_s) = +0.0042, \qquad \text{MDE}_{80\%} = 0.023. \tag{27}$$

The MDE in Equation 27 is a sensitivity reference for the prerequisite contrast, not an acceptance threshold or equivalence margin. Only 5/8 contrasts are positive. Seed 0 contributes +0.063, whereas the remaining seeds average approximately +0.004. The mixed signs and seed-0 influence leave the controlled-predictivity prerequisite unstable across technical seeds. The completed evidence therefore does not show that the endpoint means measure the benefit of a reliably induced, brain-response-specific change in the student.

Figure 12 shows the two limits on the practical-utility conclusion. Panel a shows the two completed utility contrasts, and Panel b shows the seed variation and measured sensitivity of the prerequisite contrast in controlled brain predictivity.



(a) Recorded-response-minus-permuted-response utility contrasts.



(b) Technical-seed prerequisite contrasts with an observed-MDE sensitivity reference.

Figure 12: Descriptive endpoint contrasts and their prerequisite contrast in controlled brain predictivity.

The completed analysis does not demonstrate a reliable brain-response-specific practical advantage for these corpora, this student–teacher pair, and this recorded-response procedure at approximately matched LM quality. The result also does not establish the absence of an endpoint advantage: both means favor the recorded-response arm, but their paired uncertainty was not retained. Section 5 places this result within the claim-dependency framework. Supporting reproductions, stopped routes, and prospective designs remain in Appendix E.

5 Discussion

The experiments achieved two outcomes, controlled brain predictivity and a student that training measurably changed, but not a third: a brain-specific training benefit. Controlled brain predictivity establishes that trained LM representations contain information useful for predicting recorded fMRI responses. Two supported results establish that the synthetic-response objective altered the saved student: post-training target uptake, meaning the target became easier to recover from the student, and retained-student movement, meaning the student still differed from its baseline once the prediction head was removed. These results do not by themselves establish brain-response-specific attribution, biological transfer, or practical utility. Support for the recorded-response and synthetic-response routes stops at different stages, in both cases before a participant-general training benefit is established.

5.1 Where support is lost across the two intervention routes

Controlled brain predictivity settles the measurement stage of the recorded-response route: under the declared controls, model representations do predict recorded responses. We did not measure the Qwen teacher–student headroom for that route. Headroom describes how much room an intervention has to improve on, not whether the paired comparison is valid, so this gap limits the opportunity we can claim rather than the support for the contrast. The recorded-response intervention itself does not demonstrate a participant-general pairing advantage. Training with participant-averaged responses produced positive trends for a fixed shared target. Those comparisons still did not demonstrate a reliable pairing advantage at the unit they actually support, which is a held-out fold of the one shared averaged target rather than an individual participant. Averaging can strengthen measurement of shared response structure, but it changes the estimand and does not create participant replicates. When responses and technical repetitions were aggregated within participants, correct stimulus–response pairing did not demonstrate a participant-general advantage. The tested variants narrow only specific alternative explanations. Greater adapter rank did not induce stronger movement, and the contrastive, naturalistic, and full-parameter analyses bound their conclusions to the loss, data, and optimization regimes they tested. They do not establish that every recorded target or training procedure would fail. The practical-endpoint means are higher for the recorded-response arm, but we did not report their paired uncertainty and the controlled-predictivity prerequisite was unstable across technical seeds. The endpoint is therefore unresolved, and the analysis does not identify the observed endpoint contrast as the payoff of a reliably induced brain-response-specific change.

The synthetic-response evidence separates into parallel claims rather than one failure sequence. The unique linear predictivity of the 50-component PCA target representation did not meet the continuation rule, while its advantage over the frozen row-twin control, the same target rows with their sentence pairing broken, remained unresolved. Post-training target uptake and retained-student movement show that training altered the saved student while measured LM quality remained within the declared margin. These manipulation results answer whether training changed the student. They do not change the earlier measurability finding or establish that brain-response content caused the retained change. The text-derived auxiliary-control arm is non-identifying, meaning it cannot isolate an effect of brain-response content, and neither the direct participant-level assay nor the composed-path diagnostic demonstrates biological transfer. A supported manipulation can therefore coexist with biological transfer that is not demonstrated. Neither the measurability result nor the comparator audit identifies the cause of the transfer outcome.

Together, the controls narrow the explanation without identifying it. The nuisance and untrained-network controls make it unlikely that no controlled signal was present at all, and the manipulation and quality checks make it unlikely that training either failed to alter the student or badly damaged its language quality. Four explanations remain compatible with the evidence: the target may lack the needed information or geometry; the measurement modality, temporal resolution, or readout may be inadequate; the optimization, model, or data scale may be too small; or the signal may need nonlinear access, or may differ from participant to participant. These possibilities are hypotheses suggested by where support was lost, not causes established by the experiments. The evidence therefore identifies which claim roles are supported, unresolved, non-identifying, or not demonstrated within the targets, models, objectives, participants, controls, and endpoints tested here. It does not provide a causal explanation or establish a universal null for brain-response-guided training. Because this localization is conditional on the tested designs, it can coexist with positive findings from materially different interventions and estimands.

5.2 Compatibility with positive prior studies

The route-specific account of where support stopped does not directly contradict positive brain-response-guided studies. As Section 2.3 documents, prior work differs in recording modality and temporal resolution, participant-specific or pooled supervision, model and training budget, and whether the endpoint is target recovery, biological prediction, or task performance. Those choices change both the intervention and the effect being estimated. The present results are conditional on a fixed smaller student, the declared fMRI targets, and the tested controls and inference units. Because the designs are not equivalent, the results here cannot directly adjudicate those studies. It does not show that any one design difference caused the outcomes to diverge.

What a positive contrast establishes also depends on the reference condition it is measured against (Section 2.4). Improvement over a pretrained or language-only model does not isolate the incremental contribution of brain-response supervision in the way that comparison with the ordinary KD arm at comparable LM quality does. For recorded-response training, a permuted-response control tests whether correct stimulus–response pairing matters. For synthetic-response training, attribution to brain-response content additionally requires a suitably matched text-derived auxiliary target. Training seeds can establish procedural stability, but they cannot replace participant-level uncertainty when the claim concerns people. Endpoint choice sets another boundary: recovery of the optimized target establishes uptake, prediction of independently recorded responses tests biological transfer, and improvement on an external task addresses utility. A positive result for one of these outcomes need not satisfy the others.

By applying these distinctions within one controlled framework, we show why positive findings at different stages can be valid without supporting the same training claim. Our contribution is to keep measurement and manipulation distinct from attribution, transfer, and utility, and to specify the comparator, quality condition, endpoint, and inference unit needed for each claim. Whether materially different prior results would pass those additional stages remains an open empirical question.

5.3 Design implications for future brain-response-guided training

The two routes lost support at different points, so they call for different design changes. For recorded-response training, response construction and inference must match the intended claim. A target averaged across participants estimates the shared response. Participant-general claims instead require population inference that keeps the participant as the biological unit. Here we aggregated folds, seeds, and other technical repetitions within each participant before comparing across participants. The two estimands are complementary, not interchangeable.

For synthetic-response training, the point at which support stopped is a reason to move several checks before expensive optimization. The training target, after compression to 50 principal components, should first be evaluated on held-out data from the intended recorded responses. That assay should use the planned nuisance model and readout, a frozen row-twin control, the inference unit required by the claim, and a prespecified assay-specific continuation rule. The frozen control tests sentence–target pairing; content attribution requires a separately trained text-derived auxiliary-control arm. Before training, that auxiliary target should be checked for comparable scale, covariance geometry, nuisance predictability, baseline recoverability, remaining headroom, and held-out learnability. Initial auxiliary losses and gradient scales should also be retained so that optimization pressure can be audited directly. Passing these checks would establish a better-controlled opportunity under the declared assay, not attribution or transfer. Not meeting the 50-component PCA linear criterion would still not rule out every nonlinear intervention.

Once target uptake and retained-student movement are supported at acceptable LM quality, evaluation must move beyond the optimized target. Attribution requires comparison with the text-derived auxiliary-control arm after its pre-outcome target properties, KD-baseline recovery and headroom, and initial optimization scale have been checked. Post-training learnability and movement help interpret how the interventions differed, but they are not among the conditions a comparator must satisfy to identify a cause. Biological transfer then requires an independent recorded-response endpoint, with technical repetitions aggregated within each participant before population inference. These outcomes must remain separate because a reliable manipulation can coexist with biological transfer that is not demonstrated. Utility is route-specific. It can be read as the payoff of improved controlled brain predictivity only once that prerequisite is reliably established and the practical endpoint is reported with paired uncertainty. The completed utility evaluation belongs to the recorded-response route and does not estimate synthetic-route utility.

The stage status should determine the next use of data and compute. A supported stage permits the next test but does not license its claim. A stage at which the claim is not demonstrated justifies further work only when additional participants, a more informative endpoint, or a redesigned assay could change the decision. An unresolved stage calls for a targeted test that can distinguish the remaining alternatives. A non-identifying comparator requires redesign before attribution can proceed. When no feasible follow-up can change the finding at the relevant stage, further dependent target-fitting or endpoint evidence has little interpretive value. The route should then stop, or redesign its target, control, or assay. This claim-specific continue–redesign–stop logic preserves informative results while avoiding tests that cannot repair a condition required by the intended claim. These recommendations follow from the diagnostic pattern observed here, not from an experimentally proven universal procedure. They apply within the populations, measurements, and intervention spaces set out in Section 6.

6 Limitations

Section 5 identified where the evidence stopped supporting the recorded-response and synthetic-response routes. The reach of those conclusions is bounded by the populations and measurements covered, by what each analysis can identify, and by the range of interventions and utility endpoints searched.

6.1 Population, data, and measurement scope

The biological evidence is limited to fMRI responses to English-language stimuli from the studied cohorts. The sentence-level assays summarize responses within fixed left-hemisphere language ROIs. The naturalistic voxelwise predictivity assay instead uses voxels selected for response reliability in one participant who was scanned extensively while listening to stories. Some recorded-response targets average responses across participants to emphasize a shared stimulus-locked component; the participant-level analyses instead retain one within-participant summary effect per person. The conclusions therefore concern these isolated-sentence and spoken-story response summaries, not language processing across the whole brain or population at large.

The study does not cover recording modalities with finer temporal resolution, populations that differ in language background or in clinical and developmental status, non-English or nonlinguistic stimuli, or broader reading, listening, and interactive contexts. It also does not test whether nonlinear mappings could recover information that the declared linear ridge readouts miss or at different temporal scales. These untested settings may change what brain information is measured and how that information can enter training, so the route-specific conclusions do not extend to them.

6.2 Identification and inference limitations

What each result identifies depends on the nuisance set, comparator, readout, and inference unit. Unique predictivity rules out, as alternative explanations, only the nuisance features that were actually included. Where both arms pass through byte-identical nuisance features, the trained-versus-untrained contrast shows that training produced the predictive access; it does not show a shared biological or computational mechanism. In the regional assay the untrained arm's static-embedding nuisance was built from its own model, so that contrast leaves unresolved whether the predictivity gain comes from the learned parameters. The text-derived auxiliary-control arm differs from the synthetic target in baseline predictability, remaining headroom, covariance geometry, effective rank, and nuisance predictability. Its cross-target contrast therefore cannot identify the contribution of brain-response content. The target-projection result is restricted to the declared PCA assay, and specificity against the frozen row-twin control remains unresolved. To keep these boundaries explicit, the thesis separates three inference tracks: biological, fixed-stimulus, and technical. On the biological track, participant-general claims aggregate technical repetitions within each participant before comparing across participants. On the fixed-stimulus track, analyses condition their inference on the declared sentence or story set. On the technical track, seeds, folds, stories, voxels, and ROIs count as repetitions of a procedure and never stand in for participants. Even after this aggregation, the fixed or shared stimulus sets do not support generalization to all language stimuli. Appendix D formalizes these identification and dependence boundaries.

The study contains positive checks for separate parts of the argument: the measurement assays support controlled brain predictivity, and the synthetic-response intervention satisfies the target-uptake and retained-student-movement criteria. It does not contain a positive control, that is, a brain-response intervention already expected to improve participant-level biological transfer through the complete evaluation path. The positive component checks therefore show that the readouts detect existing signal and that training changes the retained student. They do not establish that the full intervention-to-transfer path is sensitive enough to reveal a genuine brain-specific training advantage. This limitation leaves open whether a smaller or differently structured benefit would pass the complete path, subject to the interval and minimum-detectable-effect bounds reported in Section 4.6.

Within those design bounds, the participant-level and exact-target intervals reported in Sections 4.3, 4.4, and 4.6 answer different sensitivity questions under different decision rules. The recorded-response pairing analysis reports a participant-level interval and, separately, a minimum detectable effect as a sensitivity quantity. Neither is interpreted as excluding effects at least as large as a declared continuation threshold. By contrast, the participant upper bound from the biological-transfer assay and the exact-target interval from the target-projection assay are read against their declared continuation thresholds. Each disfavors effects at least as large as its own threshold, for direct biological transfer and for target measurability respectively. They do not establish that the corresponding effects are exactly zero. The direct synthetic-response versus ordinary-KD transfer contrast is interpretable independently of the auxiliary comparator. Brain-response-specific attribution, however, is not identified, because the available comparator cannot isolate an effect of brain-response content. As noted above, specificity against the frozen row-twin control in the 50-component linear assay remains unresolved. Smaller average contrasts across participants, heterogeneous participant effects, and effects in other cohorts remain compatible with the evidence.

6.3 Intervention and utility scope

The intervention search aimed to diagnose where each route failed, not to cover the design space. The central synthetic-response route tested a GPT-2 Small student paired with a GPT-2 Medium teacher, representing one teacher–student scale pairing within one architecture family (Section 3.4). The recorded-response route instead distils a Qwen2.5-1.5B teacher into a Qwen2.5-0.5B student (Section 3.3). Qwen was selected for that route because it was the strongest aligner in the regional assay of Section 4.1 and the better-powered substrate. The two routes therefore run in different model families. Each route is evaluated only against its own control at comparable language quality, under the eligibility rule in Section 3.5, so no result compares one route with the other and no reported contrast depends on the family difference. Across both routes, the study did not jointly explore broader student architectures, objective families, loss weights, schedules, data volumes, target constructions, or participant-personalization strategies. The executed adapter-capacity, contrastive, voxelwise, and full-parameter variants inform only their tested parameterizations and data regimes. The naturalistic and full-parameter variants were probes of mechanism, or three-participant existence checks, not participant-general tests, so they do not broaden population inference. Section 4.3 and Appendix E report the individual variant designs and dispositions. Different model scales, targets, objectives, architectures, and optimization procedures remain open.

The practical-utility evaluation in Section 4.7 rests on a narrow set of endpoints. Perplexity served both as a language-quality control and as the primary out-of-domain utility measure; the study therefore did not evaluate task accuracy, calibration, robustness, sample efficiency, or deployment-specific value. The completed endpoint analysis reports descriptive means across eight technical seeds but no paired endpoint interval or equivalence test, so the direct endpoint advantage remains unresolved. The recorded-response route also did not establish the technically stable change in controlled brain predictivity that would be required to interpret that endpoint contrast as brain-specific utility. No synthetic-response utility test was run. Sections 4.4 and 4.6 report that the target-projection assay did not meet its continuation rule and that participant-level biological transfer was not demonstrated. The supporting studies summarized in Appendix E tested narrow design levers, stopped before intervention, or stopped because the planned measurement was unusable. They therefore do not broaden the set of completed endpoints that identify brain-specific utility. No completed route supports a brain-specific practical-utility claim under the tested procedures. Whether an intervention with a successfully identified biological effect would also help at untested practical endpoints remains open.

7 Conclusion

For the systems tested using fMRI responses to English-language stimuli, neither intervention route established a participant-general training benefit attributable to brain-response content. The recorded-response route supported controlled brain predictivity, but did not demonstrate, across the tested objectives and variants, that correctly paired responses beat permuted responses for participants in general. WikiText assays in the synthetic-response route supported target uptake (the target became easier to recover from the student) and retained-student movement (the retained student still differed from its baseline), both at acceptable measured language quality. However, the target-projection assay did not clear the improvement threshold set beforehand, and the text-derived auxiliary-control arm was too poorly matched to attribute any difference to brain-response content. Direct participant-level biological transfer over ordinary KD was also not demonstrated. The utility endpoint for recorded-response training remained inconclusive, and the evidence collected did not support a brain-specific practical-utility claim. The overall answer is therefore inconclusive: the tested routes do not establish brain alignment as a usable training signal, but they also do not rule out a benefit from materially different targets, interventions, or populations. The supported contribution is narrower than the governing question: controlled brain predictivity in the declared assays, a demonstrated ability to change the student through training, and an evidence framework that keeps those findings distinct from attribution, biological transfer, and utility.

A controlled brain predictivity score alone does not establish a usable training signal. Within the evidence criteria used in this thesis, such a claim depends on route-specific requirements, some of which must be met in order and others of which can be assessed on their own. Route-specific headroom should be measured before optimization to describe the opportunity for improvement. For synthetic targets, the target-projection assay tests only the declared linear pathway. After optimization, the following outcomes must be evaluated separately: target uptake and retained-student movement at acceptable measured language quality; direct participant-level transfer over ordinary KD; brain-response-specific attribution against an adequate route-appropriate control; and any claimed utility endpoint. For recorded-response training, a permuted-response control tests whether correct stimulus–response pairing matters. For synthetic-response training, an adequately matched text-derived auxiliary-control arm tests whether any observed benefit is specific to brain-response content. Seeds are technical repetitions, folds are data partitions, and voxels and ROIs index brain-response measurements. None can substitute for participants when the claim concerns people. Keeping these claim dependencies separate shows what an intervention achieved and what evidence remains necessary before controlled brain predictivity can be treated as a usable training signal.

A Evidence Records and Provenance

The E001–E030 identifier range covers the analyses and dispositions used in this thesis. Each existing record states what was executed, what its retained artifacts support, and which interpretation survived review. The main text states the resulting thesis claims. This appendix connects those claims to their final evidence owners, classifies every record identifier, identifies earlier readings that were superseded, and reports the retained path–hash pairs used for reproduction. Engineering-only, unused, unexecuted, and precondition-stopped records support only their stated disposition.

A.1 Evidence-record dispositions and claim ownership

Table 5 classifies every identifier once. The categories separate completed scientific records from restricted studies, stopped routes, design work, and unused identifiers.

Table 5: Final dispositions of evidence identifiers E001–E030.

Disposition	Records	Permitted use
Final executed scientific record	E008, E011, E016, E025, E030	Supports its settled scientific conclusion.
Executed; interpretation narrowed by a recorded correction	E002, E009, E026	Supports the raw measurements, which are unchanged, and only the narrowed learned-parameter, utility, or comparator-identification reading that replaced the original one.
Executed; conclusion corrected or scope-limited	E003, E004, E005, E006, E010, E014, E015, E021	Supports only the corrected or scope-limited conclusion.
Executed with a restricted diagnostic scope	E013, E017, E019, E020, E022	Supports only the stated local or diagnostic conclusion.
Precondition-stopped	E024	Supports the stopping decision; the intervention was not tested.
Deferred, analysis-only, or unauthorized	E012 deferred; E023 analysis-only; E027 unauthorized	Supports only its stated design, its unmet activation criterion, or the decision not to run it; no intervention result was produced.
Engineering or design record only	E001, E028, E029	Documents pipeline work or an outcome-blind design; supports no scientific result.
Unused identifier	E007, E018	Supports no scientific claim.

Table 6 follows the claim roles used in Sections 3 and 4. Each row names one scientific role, the record that owns its final conclusion, and the Results subsection where the evidence is interpreted.

Table 6: Scientific roles, final evidence owners, and settled conclusions.

Scientific role	Evidence owner(s)	Settled conclusion	Results owner
Controlled regional predictivity assay	E002	Controlled Tuckute ROI predictivity is supported under contiguous folds, nuisance subtraction, and an architecture-matched untrained control.	4.1

Continued on next page

Scientific role	Evidence owner(s)	Settled conclusion	Results owner
Controlled naturalistic voxelwise predictivity assay	E006	Trained representations outperform matched untrained representations for UTS03 and five held-out story blocks. The assay does not support claims about individual voxels or a population of participants.	4.1
Distillation-headroom analysis	E003; E015 supports the quality analysis	The cold GPT-2 route shows an observed quality-confounded gap; quality-aware warm-route headroom is unresolved and Qwen-route headroom is unmeasured.	4.2
Recorded-response intervention, participant-averaged estimand	E004 and E005; E010 and E014 bound the averaging interpretation	Fold-unit reanalyses of E004 and E005 supersede their earlier excludes-zero readings. E010 does not identify a participant-count response. E014 shows that averaging reduces response noise and emphasizes shared response while changing the estimand.	4.3
Recorded-response intervention, participant-specific estimand	E008; E011, E013, and E017 bound tested variants	No participant-general pairing advantage is demonstrated across the tested objectives and variants.	4.3
50-component linear target-projection assay	E030	The fixed 50-component PCA target does not clear the improvement threshold of +0.002 set for it in advance. Aligned-minus-row-twin specificity remains unresolved.	4.4
Synthetic-target recovery assay	E016	WikiText assays support target uptake.	4.5
Retained-student movement audit	E025; E016 supports the earlier assay	Retained-student movement is supported at acceptable measured LM quality.	4.5
Comparator-adequacy audit and attribution assessment	E026; E016 supplies the original comparator	The saved text-derived auxiliary-control arm is non-identifying because pre-outcome target and KD-baseline comparability checks do not hold; post-training movement remains diagnostic.	4.5
Saved-student target-retention analysis	E030	The exact target is recoverable from the synthetic-response students on Tuckute, while incremental retention beyond ordinary KD remains unresolved.	4.5
Composed-path diagnostic	E030	Within the declared linear pathway, predictive overlap, incremental overlap, and the aligned-over-row-twin increment remain unresolved across participants. These diagnostics do not constrain the independent direct-transfer contrast.	4.6
Saved-student biological-transfer assay	E025; E026 bounds attribution	Participant-level biological transfer is not demonstrated. The direct synthetic-response-minus-KD contrast is near zero, while the text-derived comparison is non-identifying.	4.6
Practical-utility evaluation	E009	Mean out-of-domain ratios favor the recorded-response arm, but paired endpoint uncertainty was not reported; the prerequisite contrast is unstable across technical seeds, so brain-specific utility is not demonstrated.	4.7

A.2 Superseded interpretations

Some records were technically complete before their final interpretation was settled. Table 7 preserves the corrections that change how those records may be cited.

Table 7: Earlier interpretations and the verdicts that supersede them.

Records	Earlier interpretation	Final permitted reading
E003	KD uniquely destroys or preserves alignment.	The cold GPT-2 gap is quality-confounded; quality-aware warm-route headroom remains unresolved.
E004, E005, E008	Seed–fold cells provide independent uncertainty, and the averaged contrast excludes zero.	Fold-unit inference supersedes the cell bootstrap; E008 owns participant inference.
E006	A voxel bootstrap supports population inference or a minimum detectable effect.	Those claims are withdrawn; the retained result concerns UTS03 and five held-out story blocks.
E009	A minimum detectable effect on the practical endpoint tests the null.	The minimum detectable effect is a sensitivity quantity only; no point estimate changed and the paired endpoint uncertainty was never reported.
E010, E014	Nested participant subsets establish a monotone participant-count response.	Corrected random subsets do not identify a count response; E014 owns the response-reliability and estimand effects of averaging.
E015	A narrow-family best-layer correlation establishes a general scaling relation.	The full-range quality association survives; the capable-band relation and architecture residual remain uncertain.
E021	The apparent advantage is cognition-specific.	A phase-randomized shape twin matches the apparent advantage; the final finding is shape-specific.
E002	The 0.491 correlation ceiling can normalize unique R^2.	Correlation and variance-scale R^2 cannot be divided; the raw contrasts and the conclusion remain unchanged.
E013	The initial damaging variant establishes a stronger intervention result.	Only the anchored and contrastive tested branches inform the conclusion; the larger expansion was not built.
E016, E025, E026	The defective permutation or the text-derived comparator matched only in dimension identifies brain-response content.	E025 owns the direct participant contrasts; E026 establishes that the auxiliary comparator is non-identifying.
E017	The aggressive forgetting configuration represents the full-tuning result.	The gentle matched-quality probe owns the scoped conclusion.
E019	The local probe reproduces or refutes the external study.	It is a local lever diagnostic.
E020	The controlled residual closes the stimulus-predictable ceiling question.	Reference reliability fails, so the ceiling remains bounded but unresolved.
E030	Failure of the 50-component linear target projection explains the E025 result.	E030 limits the declared linear pathway without explaining or invalidating the independent direct-transfer result.

A.3 Retained artifacts the results depend on

Table 8 lists the verified path–hash pairs that a reported result depends on and that an owning evidence record explicitly binds. It excludes smoke outputs, intermediate caches, prospective-design artifacts, and hashes not bound to exact filenames. The table reproduces the bindings for reader verification; the owning evidence record remains the authority for each artifact. Unbound E025 and E030 hashes and the E028 and E029 design artifacts are omitted. Appendices C, D, and F report the corresponding training settings, estimators and inference rules, and complete numerical and sensitivity tables.

Table 8: Retained paths and verified SHA-256 digests.

Record	Retained path and SHA-256
E002	`outputs/E002_tuckute_feasibility.json` sha-256: `ce45eb1786514a168e81248c426c472077a44245d8863ee5d0bbf6e4aacab44d`
E003	`outputs/E003_cold.json` sha-256: `a30dceb63832a2fce838907cc9c782b4f6e8297c2a0136f94b5aacc9056edd6e`
	`outputs/E003_warm.json` sha-256: `6e75770387b0a84866060049d8fb8d3aecda898a95b2ce159ceffc226a5bf387`
	`outputs/E003_perplexity.json` sha-256: `2385e67ec61b7c64276f01e7b8e717e1d54cb1c1b23023715922e685cab5f97b`
	`outputs/E003_dissociation.json` sha-256: `b7658aa0d1acf92bd62897aadec663aeda7a662abfa58890ca63eab6db5e5d38`
E004	`outputs/E004_brain_lever_Qwen.json` sha-256: `5f5582a651ecd86c25691fe24b6827d49a258c33e95127f630c522563d8dc6d4`
E005	`outputs/E005_Qwen.json` sha-256: `e9b7be5b739370ed224de1bd8c6d4fa34d33881db308d3508e1edba66ddfcf6e`
	`outputs/E005_lambda_sweep_avg_Qwen_analysis.json` sha-256: `85013e6ff961aa88bd40e9e16dd19aaab3c89656f0dac8cd74c64560469f7938`
E006	`outputs/E006_lebel_gpt2.json` sha-256: `ea424b7d2f8a32dd289e8ef8066c8609749f4edc1b4cdd132373a78bd6b6830b`
	`outputs/E006_lebel_Qwen.json` sha-256: `09b77b8801e937d8232ce7e7f16435e490f0b9300c78ce4ede23d00bf07d2581`
E008	`outputs/E008_g0.json` sha-256: `fa2d18629e3818412a493539438d211c60cb8052de13286dd632983642d056cd`
	`outputs/E008_g1.json` sha-256: `0a60a95009f8bdb2c9aca1ba8f17f9e88f2f1be584a7db13bd35f6c9b51e6b5a`
	`outputs/E008_g2.json` sha-256: `e51b58d2f4b19fa624ed7b46dd0e4bfe35d5fb5b57104d78aa06a269830aaecf`
	`outputs/E008_g3.json` sha-256: `70903174700a20b5eaf4b44f685a6e14317daff3f35c14b28d1d4ac645ac5803`
E009	`outputs/E009_a3_powered.json` sha-256: `1820c097a1a30d3a23bb0956221fbfb5b33c6dc67557e108ac6b79dc56c46afb`
E010	`outputs/E010_averaging_doseresponse.json` sha-256: `79312a649c3021df1fdc94aa46f3fcaf6c9b57fa30e59bd4fcd2e28db87ba5d0`
	`outputs/E010b_random_subsets.json` sha-256: `276d19bbf013f4f90d9ccbbd02151926a67fdc5af08dc5787902420d754f1d17`
E011	`outputs/E011_g0.json` sha-256: `299ca57c7733932d9272a44d06fe3cb6976ec535a887b5ad27072002426ea9e2`
	`outputs/E011_g1.json` sha-256: `f3963b67cc01e0fae3807de354b613d4e6a3ca6ff8c6e55e1eed6b7c8b719bd3`
	`outputs/E011_g2.json` sha-256: `136449382dc627d5007df84b248b3c63d8b9bd4e2d1164c7399afbb9b867b4ce`
	`outputs/E011_g3.json` sha-256: `4da6639f3943c931beab6b7189fd6a03248161c846180a225e0e9c17af5b0745`
E013	`outputs/E013b_g0.json` sha-256: `a98346f08ad41103d1cc7b9c28715dd810927c3fa09c63bf4d1db858ece19976`
	`outputs/E013b_g1.json` sha-256: `e472ee523b6af62701472bca554f79eed64f319944549349b01d6b54d3b43346`
	`outputs/E013b_g2.json` sha-256: `f9df4143b2b1cce4e871778f30a0b4f2bb48f561f4847693be0b5a4bd2c03361`
	`outputs/E013b_g3.json` sha-256: `ae741c2c4e948356841e45b9053686823071365f8cfb9e362018b66213126cad`
	`outputs/E013v2_check.json` sha-256: `a7ea8adbda9312c55192015f1b704948d13774fc2105937ac6b58d159c6e99de`
	`outputs/E013v2_lam3.json` sha-256: `35869107e166aade0097ef2e337d133a2ae98b27d7f494c1dc3a7b4b09c6a926`
	`outputs/E013v2_lam6.json` sha-256: `dcb0ab4a31f21da71cf2aea87bdfdada8a1eb530d2955fa8923eccde57e3421a`
E014	`outputs/E014_averaging_encoding_seeded.json` sha-256: `f16fbff6c9a86b65ac1ae6cc51a9ee9f9a1a82c1e3ef3aa4f0f17a7b980a5835`
E015	`outputs/E015_expand/E015_expand_merged.json` sha-256: `0aa4f149c1ae471f3b008ff77707570e2df9c4e3a5df7e61c189aa87f438abad`
	`outputs/E015_expand/panel_followup.json` sha-256: `6c5d9738e08028636a5bbbf21dfad1acde2da5756edeb3c628dd3f8664a6eefa`

Continued on next page

Record	Retained path and SHA-256
E016	`outputs/E016_tribe/phase3/phase3_combined_tribe_gpt2_n95999_s0-5_lam0.1.json` sha-256: `9fd5a657bde1d95cfad93a85a3e6513514c33b45d21d7245ef0a0336f7ee9d98`
	`outputs/E016_tribe/phase3/phase3_combined_textfeat_gpt2_n95999_s0-5_lam0.1.json` sha-256: `793af7beaab54aff1267e44ef16bdffa5b4f5854d8fc99bcfc99a2bbec0d382d`
	`outputs/E016_tribe/phase3/phase3_combined_tribe_vs_textfeat_s0-5_comparison.interpret_audit.json` sha-256: `2716234438f2b0a0519fab9d16bceebaffea2e1c6fc93e9cd6f90a6819df2f5e`
	`outputs/E016_tribe/phase3/phase3_rerun_tribe_gpt2_n95999_s0-2_lam0.1_save.tuckute_alignment.json` sha-256: `f7ea1607f48bb6b91c9e9f4087ac691aca527ac4c46ddfecb758fd90554827f7`
	`outputs/E016_tribe/phase3/phase3_extra_tribe_gpt2_n95999_s3-5_lam0.1.tuckute_alignment.json` sha-256: `cf5e7805bc5bd5fec4b6bd5df0d652b1b28368741c874d686c19ffa0c37096ac`

	`outputs/E016_tribe/phase3/phase3_combined_textfeat_gpt2_n95999_s0-5_lam0.1.tuckute_alignment.`
	`json`
	sha-256: `fa94779f2b0a06c748ef9016b9494720d82c9b345ea9fb577bc4bde6ee898024`

	`outputs/E016_tribe/phase3/phase3_combined_tribe_vs_textfeat_s0-5_tuckute_alignment.interpret_`
	`recompute.json`
	sha-256: `9fe1a4253bc39b1b75ce5d25184332825706b95518a493532af3a276d10f10eb`

E017	`outputs/E017_matched_ppl_control/fullft_gentle_3subj.json` sha-256: `8f426e7b012f8461208bfb6a0e3e0bf3981eacf2feef91fb95f40638e5ca4e38`
E019	`outputs/E019_negi/results.json` sha-256: `983805956f37d8e17d81d3172113d46bdfe174c6ccce581ce5d1ae040d165758`
	`outputs/E019_negi/probe_strong.json` sha-256: `2fed0402fcce0067a81d81330289829c6fd1d6c5128dfb30cccda14f9787664b`
	`outputs/E019_negi/eval_posctrl.json` sha-256: `c7a9a5f5c07b3802842c73dbde891f6e9185751f3898522dd13c8347c45c214f`
E020	`outputs/E020_eys/eys_story11_results.json` sha-256: `d16d6f94f2f8f389db3178262bb399c93a8f973321684e1246bf1009f063acab`
	`outputs/E020_eys/eys_diagnose.json` sha-256: `e8fab13e0fed3840a94fb9605d2ca4868ddc6e78d6cc975cede6f7f03b8579b0`
	`outputs/E020_eys/eys_diagnose2.json` sha-256: `84d268dc4a6211e7775cc5bbeca84e5d64eb52fbd8af963799011014eacb1879`
E021	`outputs/e021/arms_v5_results.json` sha-256: `95c05b8d259d82fb83c84d54c595176198a9ba9cfe50a70941c2cddd24d69b5b`
	`outputs/e021/arms_v5_shard_all.json` sha-256: `b53bf1c88de36b397a9be753587e0e3846fde107c648eb496250850a1b7314e1`
E022	`outputs/e022/pilot_results.json` sha-256: `a95a1e23be864322fce4f26ad215927b8fb74739ae35156c0da0091a0eccd24c`
E024	`outputs/e024/zuco_nr_reliability.json` sha-256: `34f412c1d40177510e2369296a223c00267da1aa3af940514ece47c3e5dfc4e2`
	`outputs/e024/positive_control.json` sha-256: `acea6d09191cbd1e31062dd4c3ed1963de5ff95387f87f4a8d26b836edaa8e1a`
	`outputs/e024/gaze_richness_probe_NR.json` sha-256: `b8eb0f32bd16e7eda77f957f23ad329c881ce2f550e9e5f4de0b48bef4ae606e`
	`outputs/e024/gaze_richness_probe_TSR.json` sha-256: `5d23870ad33f24a52044983446b347e51a9b6ee437c1e3659eaa633e8cc694a8`
E025	`outputs/E025/extraction.json` sha-256: `b49c2795c55961cae161084afa1d3409dd662b893f9a40fa6a4e9373bc2c84c5`
	`outputs/E025/analysis.json` sha-256: `9a43410bfafced8130d286faa8c011c6e2097707979d476b5ce01d9ec44f106c`
E026	`outputs/E026/e026_audit.json` sha-256: `dfac7498d89d488483f129ebd30f36013d98778a8304f0d3d97d9e0a8c1932ad`
	`outputs/E026/e026_independent_result_audit.json` sha-256: `b2dca54c87efeca60c78a0de8287b4571991dcf3e653ce17dacc60583d2fda9f`
E030	`outputs/E030/tribe_tuckute_condition-b_n1000_full.npz` sha-256: `a1fb8667e5c27a7e9bde35cb5f3a4f24622ad13c3b79e91a884c119ef3b65233`
	`outputs/E030/raw_scores.json` sha-256: `cebea58cd7f7d82908b26a25714a1d256ab674a7b08d9a18d94835b0f7fcb6e4`
	`outputs/E030/analysis.json` sha-256: `43a4f3b01641072345149e651cdd0b1ae0c903bb74366db2006f31b89d8dd818`

B Data, Responses, Targets, and Controls

This appendix specifies the data and target constructions behind the assays in Sections 3 and 4. It moves from recorded-response inclusion and aggregation, through nuisance features and fold-local preprocessing, to synthetic and text-derived targets and their controls.

B.1 Data inclusion and response construction

The recorded-response analyses use two complementary datasets. The Tuckute resource provides isolated sentences and regional responses from multiple participants, whereas the LeBel resource provides continuous stories and voxelwise responses from one deeply sampled participant. Table 9 records the exact substrate and held-out partitions used by the two assay families.

Table 9: Included recorded-response material and held-out story folds.

Source or partition	Included material
Tuckute condition B	1,000 English sentences ordered by `item_id`; five left-hemisphere language ROIs: `AntTemp`, `IFG`, `IFGorb`, `MFG`, and `PostTemp`
LeBel fold 1	`adollshouse`, `adventuresinsayingyes`, `afatherscover`, `againstthewind`
LeBel fold 2	`alternateithicatom`, `avatar`, `backsideofthestorm`, `becomingindian`
LeBel fold 3	`beneaththemushroomcloud`, `birthofanation`, `bluehope`, `breakingupintheageofgoogle`
LeBel fold 4	`buck`, `catfishingstrangerstofindmyself`, `cautioneating`, `christmas1940`
LeBel fold 5	`cocoonoflove`, `comingofageondeathrow`, `exorcism`, `eyespy`

The Tuckute resource was acquired during silent sentence reading with 3-T fMRI, a 2.0-s repetition time (TR), and 2-mm isotropic voxels [24]. This work inherits the released `response_target`; it does not reconstruct voxel time series or refit the functional localizers. In the released target, voxel responses were z-scored within scanning session and then averaged within each ROI. The remaining operations select condition B, order rows by `item_id`, and construct either an averaged or participant-specific response matrix.

The participant-averaged construction combines, cell by cell, the five participants the Tuckute release designates as its training set. Their unique identifiers (UIDs) are 848, 853, 865, 875, and 876. A missing contributor is skipped, but the resulting sentence-by-ROI matrix must be complete before analysis. This response is used by the controlled regional assay and the participant-averaged recorded-response intervention. It estimates predictivity for the shared response of this fixed group because averaging occurs before readout fitting and before held-out R^2 is calculated.

Participant-specific construction retains one complete sentence-by-ROI matrix per person. UID 853 is excluded because 60 entries are missing and the five-ROI grid is incomplete. The inclusion rules differ because cell-wise averaging can use an available entry, whereas participant-level inference requires a complete sentence-by-ROI matrix for each person. UID 853 has 60 missing sentence–ROI cells: those 60 participant-averaged cells use four contributors, while the other 4,940 use all five. This construction distinction does not assume that the missing entries are ignorable for every estimand. Ten participants have condition-B coverage; the nine that remain after this exclusion are UIDs 797, 837, 841, 848, 856, 865, 875, 876, and 880. Within this set, $\{848, 865, 875, 876\}$ form the inherited training-participant subgroup and $\{797, 837, 841, 856, 880\}$ form the held-out-participant subgroup. Appendix D specifies the fold and seed aggregation; the nine retained participants are the biological inference units. Thus participant averaging changes both the signal-to-noise ratio and the estimand; it does not create additional independent participants.

The naturalistic assay uses participant UTS03 from the LeBel story-listening resource [25]. The responses were acquired at a nominal TR of 2.0 s. The present implementation places the stimulus features and released response matrices on the 2.0045-s effective sampling grid recorded in the released dataset metadata. Each released HDF5 `data` matrix contains one row per scanner sample and 95,556 cortical-voxel columns. This work begins from those matrices and does not rerun reconstruction, motion correction, or cortical registration.

The 20 assay stories follow their recorded order and are divided into the five contiguous four-story folds in Table 9. Complete stories are held out, which prevents temporally dependent samples from the same story from crossing the evaluation boundary. The separate story `wheretheressmoke` is excluded from model fitting and scoring and is used only for voxel reliability selection.

Reliability is estimated from the ten released repeats of `wheretheressmoke`. With random seed 0, each of 50 random splits assigns five repeats to each half. Responses are averaged within each half, correlated voxelwise across the 291 scanner samples, and corrected with the Spearman–Brown formula. The mean corrected correlation across the 50 splits is the voxel's reliability estimate. Voxels with reliability strictly greater than 0.5 are retained, leaving 11,442 voxels. Because the repeated story is independent of the 20 assay stories, this screen does not select voxels from an evaluation outcome. The resulting estimand remains held-out-story prediction within UTS03; the retained voxels are spatial measurements, not independent biological replicates.

B.2 Stimulus, nuisance, and fold-local preprocessing

Each assay separates the contextual model feature from lower-level or static stimulus features that could explain recorded responses without the contextual representation under study. Table 10 summarizes these blocks before the fold-specific transformations are applied.

Table 10: Contextual and nuisance feature blocks by assay.

Assay	Contextual block	Low-level nuisance block	Static or additional block
Tuckute	Mean-pooled activations of the declared LM layer over non-padding token positions	Sentence token length and item position, the zero-based index divided by one less than the sentence count	Mean noncontextual input embedding; released imageability ratings in the primary design, which reuses the released response matrices unchanged
LeBel	Final-subtoken hidden state for each word under its available left context	Word rate, phoneme rate, word duration, word length, and log word frequency	the released `english1000` lexical-semantic basis (985 dimensions)

In the controlled regional assay, the original implementation recomputed the static embedding block for each trained or untrained LM. The trained score therefore remains controlled relative to its implemented nuisance set, but the trained-minus-untrained gap does not isolate learned parameters against a byte-identical static nuisance block. The saved-student biological-transfer assay instead fixes the static block to GPT-2 Medium so that every student arm is evaluated against the same nuisance representation. The 50-component linear target-projection assay adds the released imageability covariate in its primary design; Appendix F reports the design without imageability as a continuity sensitivity. GPT-2 XL surprisal and probabilistic context-free-grammar surprisal provide a secondary stress test because both contain model-derived contextual information that overlaps the construct being measured.

The Tuckute analyses use five contiguous outer folds of 200 sentences. For each fold, stimulus-domain scaling and contextual and static PCA are fitted on the other 800 sentences and applied unchanged to the held-out block. Contextual and static feature blocks are each reduced to rank 50, equalizing their nominal dimensions before readout fitting, while the scalar nuisances remain unprojected. The synthetic brain-response target has one fixed exception: its coordinate standardization is learned once from the external WikiText training corpus, as described below. Any target PCA used in a Tuckute assay is still fitted only on the corresponding 800 outer-training rows.

The LeBel preprocessing begins with the time-aligned word and phone annotations in the TextGrid files. Word duration is the successive word-onset difference clipped to 0–3 s, with 0.3 s assigned to the terminal word. Contextual word features are extracted in overlapping

256-token windows with stride 128; each word receives the hidden state of its final subtoken under the maximum available left context. Word length, log word frequency, word duration, and `english1000` features are Lanczos-resampled with window 3 to the 2.0045-s grid. Word and phoneme rates are direct counts in TR-centered histogram windows.

After these stream-specific operations, the first 10 and final 5 samples of each story are removed with the fixed slice `[10:-5]`. The contextual, low-level, and static feature streams then receive within-story z-scoring and finite-impulse-response copies at delays one through four TRs. The recorded blood-oxygen-level-dependent (BOLD) responses are z-scored voxelwise within story but receive no finite-impulse-response expansion. For a held-out story, this transformation uses that story's own response mean and variance before scoring. This is a shared transductive normalization across model conditions, so the estimand is prediction of within-story standardized responses rather than deployment to a story whose response moments are unavailable. Feature and response rows are then truncated to their common aligned length. Within each outer fold, the contextual and static blocks receive train-story PCA to rank 100, while the low-dimensional nuisances remain unprojected. A predictor scaler fitted on the training stories then z-scores the assembled design. The same folds, nuisance arrays, temporal transforms, and response rows are used for trained and architecture-matched untrained LMs. Appendix D specifies response centering, ridge selection, fresh-readout fitting, aggregation, and statistical inference.

B.3 Synthetic brain responses and text-derived auxiliary targets

Target construction. The synthetic-response route uses a fixed WikiText-103 corpus that is disjoint from the Tuckute sentences. The corpus builder reads the training split of `Salesforce/wikitext`, configuration `wikitext-103-raw-v1`, in dataset order. It removes blank lines and document headers, splits each remaining line after sentence-final punctuation, retains sentences of 4–45 whitespace-delimited words, and removes case-insensitive duplicates and exact case-insensitive matches to Tuckute sentences. The first retained sentences form the fixed training file and the following sentences form the untouched held-out file. The distillation-headroom corpus build recorded 96,000 training sentences and 2,000 held-out sentences. When the target builder later reads those files it skips any line that is empty once surrounding whitespace is removed, which discarded one line from each file. The synthetic-target caches therefore contain 95,999 training sentences and 1,999 held-out sentences; the two pairs of counts describe the same split before and after that step rather than one drifting split.

TRIBE v2 maps each sentence to synthetic brain responses on a standardized cortical surface with 20,484 fsaverage5 locations [28]. The target builder loads `facebook/tribev2`, sets `data.text_feature.model_name` to `unsloth/Llama-3.2-3B`, and sets `data.features_to_use` to `[text]` in the runtime configuration. The Llama-3.2-3B component is TRIBE's internal text encoder; it is distinct from the frozen GPT-2 Medium distillation teacher. The published TRIBE description does not list the Tuckute sentence set among its training datasets. This supports dataset separation at the level of reported provenance, but it is not a categorical proof that no overlapping text or derivative data entered any upstream component.[28] For this thesis, the sentence-level target is the average of TRIBE's one-second output samples for that sentence. This operation creates one training vector per sentence; it is not a claim of physical alignment to the acquisition timing of the Tuckute fMRI responses. Its `synthetic-text` configuration assigns each token a 0.35-s event followed by a 0.05-s gap, lengthens the final event of any timeline shorter than 1.0 s until the timeline reaches 1.0 s, and uses a 1.0-s TR. Each sentence has a separate timeline, and its retained one-second predictions are averaged coordinatewise to form one 20,484-dimensional vector. The same construction is applied to Tuckute item IDs 1–1,000, with `item_id=item_index+1`.

Let Y_{ij} be synthetic response coordinate j for training sentence i. Each coordinate is standardized using the training-corpus mean $\mu_{j,\text{train}}$ and standard deviation $\sigma_{j,\text{train}}$:

$$\widetilde{Y}_{ij} = \frac{Y_{ij} - \mu_{j,\text{train}}}{\sigma_{j,\text{train}} + 10^{-6}}.$$

This operation centers and scales each cortical coordinate using only the synthetic-response training corpus. The same stored moments are applied to held-out sentences, and the small constant prevents division by a near-zero scale. The moments and standardized targets are computed and stored in float32. The stored training target retains all 20,484 standardized coordinates. Appendix C specifies how the training objective uses them; no target PCA is applied before that use. When a later assay requires a lower-dimensional target, its PCA basis is learned only from the relevant outer-training rows.

The text-derived auxiliary target preserves the target width and standardization procedure while replacing the TRIBE construction with frozen LM features. Let $h_i \in \mathbb{R}^{1024}$ be the mean-pooled layer-12 representation of sentence i extracted from the same frozen GPT-2 Medium model used as the distillation teacher, with at most 64 token positions. Let $R \in \mathbb{R}^{1024\times 20484}$ be drawn once with `numpy.random.default_rng(0)`, with entries $R_{ab} \sim \mathcal{N}(0,1)$, and then frozen. The target for sentence i is

$$T_i = \frac{h_i R}{\sqrt{1024}}.$$

The projection gives the auxiliary target the same 20,484-coordinate width as the synthetic brain-response target. The factor $1/\sqrt{1024}$ keeps its scale from growing mechanically with the 1,024 input dimensions. Each projected column is standardized by the same formula, using its own mean and standard deviation computed on the WikiText training corpus.

Pairing controls. The controls differ in which relationship they disrupt. The following constructions make the preserved and disrupted relationships explicit.

The recorded-response intervention constructs its permutation only from the current 800-sentence training complement. The participant-averaged route uses one draw per fold and seed, whereas the participant-specific route fits five independently permuted controls per fold and seed and averages their scores; that replicate count is a fixed choice and is unrelated to the five ROIs. The control randomly reorders ten contiguous response blocks within the 800-sentence training complement. It disrupts sentence–response pairing while preserving the response-value multiset and within-block order, but it is not constrained to move every block.

The saved synthetic block permutation and the frozen row twin serve different purposes. The earlier saved block permutation split the rows into ten contiguous blocks, permuted the block order, and copied each source block into its destination block only up to the shorter of the two lengths. Because the blocks are unequal, a destination row past that cut keeps its own target and a source row past that cut is never used, so the result can retain correctly paired rows and can duplicate or omit a row. It is therefore only a sensitivity. The retained manifest does not record an exact fixed-row count, so the thesis does not quantify or treat this control as an identifying comparator. A derangement is a permutation with no fixed points, so every stimulus is paired with another stimulus's target. The frozen row-twin control instead uses Sattolo's algorithm to construct one derangement inside each 200-item outer block before target values or participant scores are read. Its deterministic seed is derived from the block identifier. For each outer fold, aligned TRIBE rows define one train-only 50-component PCA basis, and the same basis is applied to the twin. This comparison isolates row identity at matched target-row content and linear target capacity. It concerns one frozen row-twin control and therefore does not establish invariance across possible derangements or identify a causal brain mechanism.

Table 11 summarizes the construction and attribution role of each target or control.

The text-derived auxiliary target matches the synthetic target on row order, output width, standardization procedure, and corpus. The two targets do not share a source representation, and matching width and standardization does not establish matched covariance, effective rank, recoverability, or optimization difficulty. Those differences limit the attribution question the comparison can answer; Section 4.5 and Appendix F report the measured audit. With the data, targets, and controls fixed, Appendix C specifies how the training objectives use them.

Table 11: Synthetic targets and pairing controls used in the thesis.

Construction	Matched or preserved property	Attribution role
Synthetic brain-response target	Sentence order, TRIBE geometry, standardization, and width	Synthetic-response supervision under study
Text-derived auxiliary target	Sentence order, width, standardization, and corpus	Frozen GPT-2 Medium projection controlling for auxiliary supervision derived from text
Recorded-response permutation	Response values, marginal distribution, and within-block order	Randomly reordered sentence–response pairing; fixed blocks remain possible
Saved block-permutation sensitivity	Synthetic target values and approximate block structure	Legacy pairing sensitivity; fixed, duplicated, or omitted rows remain possible
Frozen row-twin control	Exact within-block row multiset and train-only 50-component PCA capacity	Row identity under one deterministic derangement with no fixed points

C Training Interventions and Reproduction

This appendix specifies the optimized objectives, the parameters reached by each gradient, and the student state available after training. Appendix B owns the construction of the data, targets, and controls; Appendix D owns fresh readouts and statistical inference; and Appendix A owns verified artifact paths and hashes. Section 4 and Appendices E–F own the resulting observations and verdicts. Temporary target heads never enter a later evaluation. A student checkpoint is retained only when the run explicitly saves one; otherwise evaluation uses the frozen in-memory student and only the result artifact is kept.

C.1 Ordinary KD and language-quality rules

Let $z_T(x)_t$ and $z_S(x)_t$ be teacher and student logits at a non-padding causal-language-model position t. At temperature τ, their softened token distributions are

$$p_T^\tau(\cdot \mid x,t) = \mathrm{softmax}(z_T(x)_t/\tau)\,, \qquad q_S^\tau(\cdot \mid x,t) = \mathrm{softmax}(z_S(x)_t/\tau)\,.$$

The implemented output-distillation loss is

$$\mathcal{L}_{\mathrm{KD}} = \tau^2 \frac{\sum_{x,t} m_{xt} D_{\mathrm{KL}}(p_T^\tau(\cdot \mid x,t) \,\|\, q_S^\tau(\cdot \mid x,t))}{\sum_{x,t} m_{xt}}, \tag{28}$$

where m_{xt} is one at non-padding positions and zero otherwise. Equation 28 averages teacher-to-student token-level KL divergence over valid positions. The factor τ^2 compensates for the gradient-scale change introduced by temperature softening. Every executed output-distillation route uses $\tau = 2$, gradient-norm clipping at 1, and AdamW at its library default decay rates and weight decay, with a constant learning rate and no warmup. Two route types recur below: an unanchored route, which trains the student without a teacher and keeps language quality with its own next-token objective, and a teacher-anchored route, which adds output KD from a frozen teacher.

The language objective is shared within a paired intervention, but the student initialization and the quality rule that applies differ across families. Table 12 records those differences.

Table 12: Language objectives and quality rules by training family.

Family	Teacher, student, and initialization	Language objective and corpus	Quality rule
Distillation-headroom analysis	GPT-2 Medium teacher; GPT-2 Small student initialized either from pretrained weights or randomly	Output KD on 96,000 WikiText sentences; 2,000 fixed held-out sentences	Held-out perplexity is a convergence measure; the warm and random-start routes differ in initialization and training duration
Recorded-response intervention (sentence)	Qwen2.5-0.5B student; Qwen2.5-1.5B teacher when the route is teacher-anchored	Masked next-token cross-entropy in the unanchored screen; output KD in anchored routes; fixed WikiText quality probe	The recorded-response arm and permuted-response control share the scoring implementation; ordinary KD comparisons require perplexity within the tolerance stated in Appendix D
Recorded-response intervention (naturalistic)	Qwen2.5-0.5B student; frozen Qwen2.5-1.5B teacher in anchored routes	Unanchored or output-KD-anchored language path, according to the executed grid	The full-parameter route holds the student to its predeclared maximum allowed loss of language quality on the fixed probe
Synthetic-response intervention	GPT-2 Medium teacher; pretrained GPT-2 Small student shared within seed	Output KD on 95,999 WikiText sentences; 1,999 fixed held-out sentences	Each target arm must satisfy Equation 18; saved-student comparisons also apply the within-seed bits-per-byte tolerance of Appendix D

The headroom family uses seeds 0–2, batch size 16, and maximum length 64. The pretrained student is trained for one epoch at 5×10^{-5}, whereas the random-start student is trained for two epochs at 3×10^{-4}. These are separate optimization regimes, not an initialization-only contrast. The recorded- and synthetic-response settings are specified with their target paths below.

C.2 Recorded-response intervention

For a recorded response y_i, the common sentence-level target path is

$$x_i \longrightarrow h_{12}(x_i) \longrightarrow W h_{12}(x_i) \longrightarrow \mathcal{L}_{\text{target}}.$$

Here h_j denotes entry j in the model's returned hidden-state sequence, whose entry 0 is the embedding output. The thesis shorthand "hidden-state index 12" therefore names the exact attachment returned by the implementation. The vector $h_{12}(x_i)$ is mean-pooled over non-padding token positions, and W is a temporary linear map from the pooled hidden state to the five-ROI response vector of Appendix B, unless the selected loss is readout-free.

The common joint objective is

$$\mathcal{L} = \lambda_{\text{lang}}\mathcal{L}_{\text{lang}} + \lambda_{\text{target}}\mathcal{L}_{\text{target}}, \qquad \lambda_{\text{lang}} = 1. \tag{29}$$

The first term preserves the declared language objective, while the second exposes the retained student path to the recorded response. For teacher-anchored routes, $\mathcal{L}_{\text{lang}} = \mathcal{L}_{\text{KD}}$; the unanchored screen instead uses masked next-token cross-entropy.

For predictions $\widehat{Y}$ and targets Y with minibatch size B and target width D, the squared-error target loss is

$$\mathrm{MSE}(\widehat{Y}, Y) = \frac{1}{BD} \sum_{b=1}^{B} \sum_{j=1}^{D} \left(\widehat{Y}_{bj} - Y_{bj} \right)^2.$$

This loss averages equally over minibatch rows and target coordinates. Consequently, every reported target weight is interpreted under this normalization.

The common sentence-level regime places rank-16 LoRA adapters with scaling 32 and dropout 0.05 in the attention and feed-forward projections of the student family: `c_attn`, `c_proj`, and `c_fc` for GPT-2, and `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj` for Qwen2.5 [29]. The pretrained base, input embeddings, and tied output matrix are frozen; the adapters and a co-trained W are optimizer parameters. The target gradient reaches W and the adapters that produce h_{12}, while adapters after the attachment state receive only the language-loss gradient. After training, the adapters are merged into the student and the temporary target map is discarded. The merged student is frozen before a fresh evaluation readout is fitted as specified in Appendix D.

Table 13 states only the implementation change relative to this common path.

Table 13: Recorded-response intervention variants.

Variant	Change from the common target path	Executed grid	Retained object
Co-trained mean squared error (MSE)	A new W is trained jointly with the producing adapters; correctly paired and permuted responses use the same path	Rank 16; seeds 0–2; 3 epochs; 2×10^{-4}; batch 16; target weight 10. Participant-averaged training uses one permutation draw; participant-specific training uses five	Merged student; W discarded
Frozen readout	W_0 is ridge-fitted on untuned outer-training features with $\alpha = 100$ and remains fixed during intervention training	Same grid and paired-permutation construction as the co-trained MSE row	Merged student; W_0 discarded
Contrastive	Symmetric InfoNCE compares normalized Wh_{12} with the paired five-ROI response using in-batch negatives [31]	Rank 16; seeds 0–1; 3 epochs; target weight 1; temperature 0.07; three permutation draws	Merged student; W discarded
Increased capacity	The co-trained MSE path is unchanged; adapter rank increases from 16 to 64	Seeds 0–1; 6 epochs; target weight 10; five permutation draws; otherwise as in the participant-specific grid	Merged student; W discarded
Naturalistic voxelwise	A participant-specific head maps pooled 25-word segments to reliable voxels after a 4-s response delay	UTS01–03; 14 tuning and 6 evaluation stories. The unanchored grid ran all three participants at seeds 0–1, two permutation draws, and target weight 10. The output-KD-anchored grid swept target weights 1, 3, and 6 on UTS03 over the 2,801 voxels whose repeat reliability exceeds 0.7	Merged student; voxel head discarded

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Table 13 (continued)

Variant	Change from the common target path	Executed grid	Retained object
Full parameter	The voxelwise MSE and output-KD anchor update all student parameters except the tied input/output embedding; no LoRA is used	UTS01–03; seeds 0–2; 2 epochs; 10^{-5}; batch 16; target weight 10	Student used for immediate frozen evaluation; voxel head discarded; no reusable student checkpoint retained

The unanchored sentence screen also explored alternative target losses, holding the co-trained MSE grid fixed and changing only $\mathcal{L}_{\text{target}}$. Cosine and squared-Pearson losses use a co-trained target map. Linear CKA is readout-free: it compares the minibatch geometry of h_{12} and the response matrix directly, so no temporary W participates in that branch [30]. These are implementation variants of the recorded-response intervention; their outcomes remain in Section 4.3 and Appendix E.

C.3 Synthetic-response intervention

The synthetic-response intervention uses the same joint objective in Equation 29, but the attachment and trainable path differ. For GPT-2 input x, the retained student path is

$$x \longrightarrow e_{\text{tok}}(x) + e_{\text{pos}}(x) \longrightarrow h_6(x) \longrightarrow h_{7:12}(x) \longrightarrow q_S^{\tau}(\cdot \mid x),$$

and the temporary target branch is

$$\text{meanpool}(h_6(x)) \longrightarrow W_{\text{tmp}}\, \text{meanpool}(h_6(x)) \longrightarrow \mathcal{L}_{\text{target}}.$$

As above, h_6 names hidden-state index 6 in the returned sequence, not an informal count of visible blocks. The output-KD gradient reaches the entire student path. The target gradient reaches W_{tmp}, token and positional embeddings, normalization components, and student blocks that produce h_6; blocks after the attachment receive only the output-KD gradient. Because GPT-2 ties its token embedding and output matrix, that retained matrix receives the target gradient through its input role and the output-KD gradient through both roles.

The teacher is frozen, whereas every student parameter and W_{tmp} is optimized. All target arms map the 768-dimensional mean-pooled attachment state to all 20,484 coordinates using the same mean MSE. Appendix B owns the construction and standardization of the synthetic brain-response target, the text-derived auxiliary target, and the saved block permutation. The common optimization grid uses 95,999 training sentences, one epoch, which is 24,000 optimizer steps at batch size 4,, learning rate 5×10^{-5}, batch size 4, and maximum length 64.

Table 14: Synthetic-response training arms and retained states.

Arm	Additional supervision	Shared optimization settings	Saved state
Ordinary KD arm	None; $\lambda_{\text{target}} = 0$ and no temporary target map	Pretrained GPT-2 Small initialization, frozen GPT-2 Medium teacher, corpus, output-KD loss, and seeds 0–5	Student checkpoint only when artifact saving is enabled; otherwise result artifact
Synthetic-response arm	Correctly paired synthetic brain-response target through W_{tmp}; $\lambda_{\text{target}} = 0.1$	Same initialization, teacher, corpus, schedule, and seeds 0–5 as the matched ordinary KD arm	Temporary map always discarded; student saved only under the configured artifact path
Saved block-permutation sensitivity	Saved block-permuted synthetic target and saved block-permuted text-derived target, each through the same W_{tmp} and mean MSE	Same initialization, teacher, corpus, schedule, target weight, and seeds 0–5	Temporary map discarded; result artifact retained; student saving follows the run configuration
Text-derived auxiliary-control arm	Text-derived auxiliary target through the same target-map shape and mean MSE; $\lambda_{\text{target}} = 0.1$	Same pretrained student, teacher, corpus, and schedule; executed for seeds 0–5	Temporary map discarded; seed-specific students saved when requested

The text-derived auxiliary-control arm was first executed for seeds 0–2 and later extended prospectively to seeds 3–5. The combined six-seed analyses use seeds 0–5; earlier three-seed tables retain only the initial run for provenance.

At the final optimizer step, the temporary target map is always discarded. When no checkpoint is requested, synthetic-target recovery uses the frozen in-memory student and only the resulting measurements are stored. When artifact saving is enabled, later assays reload the retained causal-language-model checkpoint and still fit a fresh readout; the temporary training map is never reused. Thus training can shape the retained student through the target gradient, but the training head itself cannot compute a later target-recovery or biological-transfer score.

The frozen row-twin control is not a training arm. It is constructed only for the later 50-component linear target-projection assay described in Appendix B. Comparator adequacy and all arm outcomes are reported in Section 4.5 and Appendix F.

C.4 Replay scope and provenance handoff

Replay requires four linked objects: the locked software environment, the executed entry point and arguments, the declared seed and final-state rule, and the retained result or student artifact. Table 15 provides the compact index. It is not a command-complete substitute for the owning result configurations or the verified path–hash ledger in Appendix A.

Table 15: Training entry points and retained replay information.

Family	Executed entry point	Final-state and retained-output rule	Provenance handoff
Distillation-headroom analysis	`scripts/run_kd_alignment.py`	Final warm or random-start student evaluated in memory; result bundle records configuration and quality measurements	Verified retained bundles are indexed in Appendix A
Recorded-response intervention (sentence)	`scripts/run_brain_lever.py`; target losses in `scripts/brain_loss.py`	Final merged student evaluated after each fold–seed cell; temporary target map discarded; no intermediate quality frontier retained	Primary and participant result shards are indexed in Appendix A
Recorded-response intervention (naturalistic)	`scripts/run_lebel_tune.py`	Final merged or full-parameter student evaluated immediately; temporary voxel head discarded	Recorded grids and result bundles remain under their owning configurations; verified paths are delegated to Appendix A
Synthetic-response intervention	`scripts/run_tribe_phase3.py`	Final student evaluated in memory; optional checkpoint written only when a save directory is supplied; temporary target map discarded	Target, saved-student, and analysis bundles are indexed in Appendix A

The current replay environment uses the Python requirement in `pyproject.toml` and the exact package and CUDA build resolved by `uv.lock`. These files are current replay pins; they are not complete historical environment manifests for every earlier run. Result JSONs retain model identifiers and executed training configurations, and later saved-student bundles bind selected weights, inputs, runners, and analyses by hash. Many early runs do not record immutable upstream Hugging Face revision identifiers, so exact historical replay additionally depends on a retained model cache or a verified saved-student artifact. This limitation must be preserved rather than filled with an inferred revision.

Smoke runs, logs, temporary caches, throughput probes, and model-loading checks are diagnostic outputs and do not receive manuscript-level path–hash rows. Appendix A owns the verified artifact ledger and any stated omissions. With the optimization and replay scope fixed, Appendix D specifies how the retained students are evaluated and how uncertainty is assigned.

D Estimators, Aggregation, and Inferential Scope

Section 3.5 states the design-level estimands and inference units. This appendix answers four narrower questions: how the predictive quantities are related, how fitting and aggregation are nested, how uncertainty and decision rules are assigned, and where information-theoretic analogies stop. Appendix B owns the inputs and preprocessing definitions, Appendix C owns the training objectives and retained model states, and Section 4 owns the empirical outcomes. The quantities below are assay-conditional predictive contrasts. They are neither causal effects nor direct estimates of information-theoretic objects.

D.1 Predictive estimands and ridge estimation

For response coordinate j, Section 3.2 defines the held-out score $R^2_j(M)$ for a readout design M in Equation 1. Let N denote the declared nuisance design and $F = [N, X]$ the full design after adding contextual LM representations X. The nuisance-only and full scores are $R^2_j(N)$ and $R^2_j(F)$, and the reported semipartial increment is

$$\Delta R^2_{\mathrm{semi},j}(X) = R^2_j(F) - R^2_j(N).$$

This difference asks how much held-out predictive value is gained by adding X to the nuisance design. Because the nuisance-only and full ridge readouts are separately regularized and evaluated out of sample, the difference can be negative. It is not a nonnegative variance decomposition.

Population partial R^2 answers a different question. Let $\mathcal{R}^2_{N,j}$ and $\mathcal{R}^2_{F,j}$ be the population coefficients of determination for nested, unregularized linear projections on N and F. When $\mathcal{R}^2_{N,j} < 1$,

$$R^2_{\mathrm{partial},j} = \frac{\mathcal{R}^2_{F,j} - \mathcal{R}^2_{N,j}}{1 - \mathcal{R}^2_{N,j}} = 1 - \frac{\mathrm{Var}(Y_j - \Pi_F Y_j)}{\mathrm{Var}(Y_j - \Pi_N Y_j)}, \tag{30}$$

where Π_N and Π_F are the corresponding population linear projections. Equation 30 measures the fraction of variance left after the nuisance projection that is removed by adding X. The semipartial quantity instead measures the gain relative to total outcome variance. The thesis reports the cross-validated semipartial increment, not a transformed cross-validated partial coefficient.

Aggregation across target or response coordinates defines another distinction. For J held-out coordinates, let SSE_j and SST_j denote the coordinate-specific residual and total sums of squares. The two aggregations used in the thesis are

$$\overline{R^2}_{\mathrm{equal}} = \frac{1}{J}\sum_{j=1}^{J}\left(1 - \frac{\mathrm{SSE}_j}{\mathrm{SST}_j}\right), \qquad R^2_{\mathrm{pooled}} = 1 - \frac{\sum_{j=1}^{J}\mathrm{SSE}_j}{\sum_{j=1}^{J}\mathrm{SST}_j}. \tag{31}$$

The first expression gives every coordinate equal weight. The second weights coordinates by their held-out variance, so a nearly constant coordinate cannot contribute as much as a variable coordinate. The WikiText synthetic-target recovery assay uses equal-coordinate averaging, whereas the saved-student target-retention analysis pools sums of squares across the fixed target coordinates. Neither calculation treats coordinates as independent repetitions.

Ridge regression supplies a regularized linear predictor. Let $y \in \mathbb{R}^n$ be a centered response vector, let $M \in \mathbb{R}^{n\times p}$ be the centered and standardized predictor matrix, and let $\beta \in \mathbb{R}^p$ be its coefficient vector. If

$$y = M\beta + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I), \qquad \beta \sim \mathcal{N}(0, \tau_\beta^2 I),$$

then the posterior mode minimizes

$$\|y - M\beta\|_2^2 + \lambda\|\beta\|_2^2, \qquad \lambda = \frac{\sigma^2}{\tau_\beta^2}.$$

This is a maximum a posteriori interpretation only under the stated Gaussian likelihood, isotropic zero-mean prior, and scaling convention. The fitted linear predictor represents the conditional mean only when that mean is linear and the model is correctly specified.

The sentence-level Tuckute and WikiText readouts use the candidate penalties

$$\{1, 10, 100, 1000, 10000\}.$$

Each multi-output fit selects one scalar penalty on its training partition by generalized cross-validation, the closed-form leave-one-out criterion, rather than by an explicit fold split; it does not select a separate penalty for every response or target coordinate. The naturalistic LeBel readouts use

$$\{10, 10^2, 10^3, 10^4, 10^5, 10^6, 10^7\},$$

again selecting one multi-output penalty from training stories only.

The comparator-adequacy audit uses

$$\{0.1, 1, 10, 100, 1000\}$$

and selects one penalty by aggregate held-out R^2 across both target families and both fixed coordinate samples within five contiguous inner folds. These selection rules are nested inside the corresponding training partition; the untouched outer block or held-out corpus is used once for evaluation. In the executed assays, λ is selected from a finite grid using the relevant training complement; operationally, it is predictive regularization rather than a fixed scientific prior. Neither this derivation nor a positive held-out score makes the fitted coefficients causal, identifies a biological mechanism, or shows that an LM and the brain implement the same computation.

D.2 Cross-validation, aggregation, and inference units

The governing leakage rule is simple: every data-adaptive operation that can use stimulus-domain information is fitted on the relevant training complement. Fixed protocol operations may be applied before the split, but learned centering, scaling, dimensionality reduction, penalty selection, and readout fitting remain inside it. Appendix B specifies the exact fixed preprocessing and split definitions.

The naturalistic assays require one explicit distinction. Feature resampling, edge slicing, within-story feature and BOLD z-scoring, and finite-impulse-response delay copies are fixed story-local operations. Because the held-out story's response normalization uses that story's own moments, the resulting estimand is conditional on this normalization protocol. Across-story PCA, assembled-design scaling, response centering, ridge selection, and readout fitting are learned only from the readout-training stories. The 50-component linear target-projection assay has one other declared exception: it reuses target-standardization moments fixed during target construction, while its target PCA and readouts remain fold-local.

Table 16 records the held-out boundary, learned operations, and final inference unit for each scientific role.

Table 16: Cross-validation nesting and inference units by scientific role.

Scientific role	Held-out boundary	Operations learned inside the boundary	Aggregation and final unit
Controlled regional predictivity assay	Five contiguous Tuckute folds, each with an 800-sentence training complement and a 200-sentence test block	Centering, scaling, PCA, one multi-output ridge penalty, and the response readout	Declared ROIs and folds are averaged; the sentence set is fixed, and untrained-model seeds are technical repetitions
Distillation-headroom analysis	The same five Tuckute folds for brain predictivity and a fixed WikiText probe for language quality	Brain-response transforms and readouts are fitted inside each fold; quality scores use the fixed scoring protocol	Model condition is the descriptive unit, seeds index stochastic arms, and model families are the resampling clusters for the cross-family quality association

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Table 16 (continued)

Scientific role	Held-out boundary	Operations learned inside the boundary	Aggregation and final unit
Recorded-response intervention	The student trains on each 800-sentence outer complement; the untouched 200-sentence block is divided into five inner readout folds	Inner-fold transforms, penalty selection, and the fresh response readout	Participant-averaged analyses end at five shared outer folds; participant-specific analyses aggregate seed–fold cells within each participant
Controlled naturalistic voxelwise predictivity assay	Five complete-story folds; voxel reliability uses a separate repeated story	Training-story PCA, assembled-design scaling, response centering, one multi-output ridge penalty, and the readout	Story-block signs and voxel summaries describe within-participant robustness; they do not create participant inference
Naturalistic recorded-response intervention	Complete stories remain held out from fresh evaluation readouts	Training-story transforms, ridge selection, and participant-specific voxel readouts	Seeds are technical repetitions within three descriptive participant probes
Synthetic-target recovery assay	The student trains on 95,999 WikiText sentences and is evaluated on 1,999 fixed held-out sentences	The fresh target-readout penalty is selected using training data	Equal-coordinate target R^2 is compared within six matched technical seeds
Saved-student biological-transfer assay	Frozen students enter five contiguous Tuckute 800/200 response-readout folds	All representation transforms, nuisance transforms, penalties, and fresh response readouts use the 800-sentence complement	Six seeds and five folds are aggregated within each of nine participants; participants are the biological units
50-component linear target-projection assay	Five contiguous Tuckute folds; aligned targets and the frozen row twin share the train-fold target-PCA basis	Each aligned or twin multi-output readout selects one scalar penalty on its own training complement	Folds are aggregated within each of nine participants; there is no student-seed axis
Saved-student target-retention analysis	Frozen students and targets use the same five outer folds	Representation and target transforms and the multi-output target readout remain train-fold-local	Sums of squares are pooled across target coordinates; paired contrasts end at six technical seeds
Composed-path diagnostic	Both linear paths are fitted on the same outer-training rows and scored once on the untouched test block	Target-to-response and student-to-target coefficient blocks are fitted separately inside each fold	Folds and six seeds are aggregated within each of nine participants
Practical-utility evaluation	Fixed in-domain and out-of-domain corpora are scored without fitting a biological readout	No evaluation-time model fitting; corpus-level losses are computed under the fixed scoring protocol	Paired arm contrasts end at eight technical seeds and remain conditional on the fixed models and corpora

The aggregation order can be written without treating technical repetitions as new scientific units. Let $\mathcal{C}_u$ be the set of technical cells contributing to final unit u, such as its seeds, folds, or seed–fold combinations, let $d_{u,c}$ be the matched contrast in cell c, and let $\mathcal{A}_u$ denote the declared within-unit aggregation:

$$\widehat{\theta}_u = \mathcal{A}_u(\{d_{u,c} : c \in \mathcal{C}_u\}), \qquad \widehat{\Theta} = \mathcal{I}\Big(\widehat{\theta}_1, \ldots, \widehat{\theta}_U\Big). \tag{32}$$

Here $\mathcal{A}_u$ may be a mean, a median, or pooled sums of squares, while $\mathcal{I}$ is the descriptive summary or inferential procedure applied to the U final units. Equation 32 says that seeds and folds are first reduced within a participant when the participant is the scientific unit. Analyzing every seed–fold cell as if it were an independent participant would change both the estimator and its uncertainty.

The final unit determines what can generalize. Seeds vary optimizer randomness under one fixed procedure. Stimulus folds share observations and have overlapping training complements. Voxels and ROIs share a participant, preprocessing, and stimuli. Target coordinates share one generated target and student. None can replace a participant when the claim concerns people [23]. Conversely, several participants evaluated on one fixed sentence set do not by themselves establish generalization to a population of stimuli. Shared-fold and story-block summaries are therefore robustness checks, not substitutes for participant or stimulus-population inference.

D.3 Uncertainty, multiplicity, and decision rules

Four choices must remain separate. The inference unit identifies the observations that enter uncertainty. The interval or test describes variation across those units. The multiplicity family identifies which simultaneous claims require adjustment. The decision rule states what evidence was required to continue to a later stage. Unless stated otherwise, the reported intervals are two-sided 95% intervals.

For U final-unit estimates $\widehat{\theta}_1, \ldots, \widehat{\theta}_U$, the ordinary t interval is

$$\overline{\theta} \;\pm\; t_{1-\alpha/2, U-1} \frac{s_\theta}{\sqrt{U}}, \qquad \overline{\theta} = \frac{1}{U} \sum_{u=1}^{U} \widehat{\theta}_u. \tag{33}$$

Here α is the two-sided error rate and s_θ is the sample standard deviation of the U final-unit estimates. Equation 33 uses the number of final units U, not the number of coordinates, folds, or technical runs. Where a table reports a descriptive interval over technical seeds, it applies the same formula to the K matched seed contrasts below and is read as a stability summary of one frozen procedure, not as a sampling interval over scientific units. Its sampling interpretation requires independent final-unit estimates and approximate normality when U is small. Participant intervals are conditional on the fixed stimuli, trained students, and analysis protocol used to construct each participant estimate.

For K matched technical contrasts $d_1, \ldots, d_K$, the exact two-sided magnitude-preserving sign-flip test used in the seed-level analyses is

$$p_{\text{flip}} = \frac{1}{2^K} \sum_{\epsilon \in \{-1,+1\}^K} \mathbf{1}\left[\left|\frac{1}{K}\sum_{k=1}^{K} \epsilon_k d_k\right| \geq \left|\frac{1}{K}\sum_{k=1}^{K} d_k\right|\right]. \tag{34}$$

Under the null hypothesis that each matched contrast is symmetrically distributed about zero, the contrast signs are exchangeable conditional on their magnitudes, which is what this test requires. It differs from the participant sign test, which uses only the number of positive and negative participant effects and requires independent participant signs. The Wilcoxon signed-rank sensitivity additionally assumes independent and approximately symmetric paired effects. The shared-fold cluster bootstrap resamples the five outer-fold summaries as clusters. Because the folds reuse observations through overlapping training complements, both that bootstrap and the fold t interval are fixed-block stability summaries rather than stimulus-population intervals. The cross-family quality bootstrap instead resamples model families rather than individual models.

Table 17 maps each scientific role to its uncertainty procedure and stage-specific decision scope.

Table 17: Uncertainty procedures and decision rules by scientific role.

Scientific role	Interval or test and its unit	Multiplicity and decision scope
Controlled regional predictivity assay	Held-out-fold dispersion and trained–untrained contrasts on the fixed sentence set	Prespecified layer and nuisance grids bound the assay; folds do not support participant or stimulus-population inference
Controlled naturalistic voxelwise predictivity assay	Within-participant trained–untrained point contrasts and five story-block signs; no voxel-resampling interval	Spatial voxels are not independent units, and overlapping story folds remain robustness checks
Distillation-headroom analysis	Fold–seed bootstrap conditional on convergence; cross-family quality association uses a family-cluster bootstrap, with pooled Fisher intervals only as a sensitivity	The declared retention bands classify opportunity within this assay and are not universal alignment thresholds
Participant-averaged recorded-response intervention	Seeds are averaged within each outer fold before a five-fold t interval	The interval describes stability across the five shared blocks of the fixed sentence set
Participant-specific recorded-response intervention	Participant medians over seed–fold cells receive a participant t interval and exact one-sided sign test; a shared-fold interval and cluster bootstrap assess block sensitivity	Requiring agreement between participant and shared-fold readings is a conservative conjunctive rule, not a joint crossed-population confidence interval
Naturalistic recorded-response intervention	Seed summaries are technical repetitions within three participant-specific probes	The small participant set supports descriptive mechanism probing, not population inference
Synthetic-target recovery assay and retained-student movement audit	Six matched-seed contrasts receive descriptive intervals and declared exact sign-flip tests	Manipulation and language-quality criteria qualify later interpretation but provide no biological replication
Comparator-adequacy audit and attribution assessment	Fixed-sample margins and paired arm contrasts are reported without coordinate-level p-values	Comparator adequacy is a pre-outcome or baseline design condition for attribution; realized movement is diagnostic, and statistical precision cannot repair a non-identifying comparator
Saved-student biological-transfer assay	Nine participant effects receive a t interval, a one-sided upper bound, an exact sign test, and a Wilcoxon sensitivity	Seeds and blocks remain descriptive; the upper bound is compared only with the improvement threshold fixed before the results were seen
50-component linear target-projection assay	Participant effects receive t intervals and exact sign tests	Its participant quantities belong to a separate Bonferroni family with a simultaneous one-sided bound
Saved-student target-retention analysis	Six paired technical-seed contrasts receive a descriptive interval and an exact 2^6 sign-flip test	Target retention has its own adjusted interval family and does not create biological evidence

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Table 17 (continued)

Scientific role	Interval or test and its unit	Multiplicity and decision scope
Composed-path diagnostic	Participant path-product quantities receive t intervals and exact sign tests	The composed path has its own adjusted interval family; it is predictive and is not a causal mediation effect
Practical-utility evaluation	Eight paired technical-seed endpoint observations were recorded, but the completed analysis reports only mean contrasts and an observed-variance sensitivity calculation	No paired endpoint interval or equivalence test was retained; the MDE supplies no hypothesis or equivalence decision

For the primary retained-student movement audit, let H_s be the layer-6 representation matrix of the retained synthetic-target student and H_s^{KD} that of its seed-matched ordinary-KD student, both on the same $n = 1{,}999$ WikiText sentences held out from every training arm in technical seed s, and let $C = I - \mathbf{1}\mathbf{1}^\top/n$ center examples. The centered relative-Frobenius distance and linear-CKA distance are

$$D_{F,s} = \frac{\|C(H_s - H_s^{\mathrm{KD}})\|_F}{\|CH_s^{\mathrm{KD}}\|_F}, \qquad D_{\mathrm{CKA},s} = 1 - \frac{\|(CH_s)^\top (CH_s^{\mathrm{KD}})\|_F^2}{\|(CH_s)^\top (CH_s)\|_F \, \|(CH_s^{\mathrm{KD}})^\top (CH_s^{\mathrm{KD}})\|_F}. \tag{35}$$

The first quantity measures centered activation change relative to the ordinary-KD activation norm. The second measures the change in centered linear representation geometry, with zero denoting identical linear-CKA geometry. Global parameter displacement is reported separately as

$$D_{\theta,s} = \frac{\|\theta_s - \theta_s^{\mathrm{KD}}\|_2}{\|\theta_s^{\mathrm{KD}}\|_2}. \tag{36}$$

These post-training quantities diagnose the realized intervention; they are not comparator-identification conditions.

An MDE describes the resolution of a planned contrast under its declared variance model, sample size, error rate, and power. It does not show that a smaller effect is zero or that a larger effect matters biologically. Likewise, language-quality criteria decide whether two arms are sufficiently comparable for a specified analysis; they do not establish a biological endpoint.

The $+0.002$ unique-R^2 rule was fixed before the transfer results were seen, to decide whether participant-level biological transfer justified a matched-control follow-up, and was then kept unchanged as the reference for the target-projection assay. It can support a statement about whether an assay-specific simultaneous bound reaches that continuation rule. It cannot define biological importance across another response scale, nuisance model, layer, cohort, or target. No retrospective choice of a more favorable analysis can change that scope.

D.4 Information-theoretic boundaries

Population partial R^2 has an information-theoretic identity only under restrictive assumptions. For scalar Y, jointly Gaussian (X, Y, Z), correctly specified population-linear conditional means, and unregularized population residual variances,

$$I(X; Y \mid Z) = -\frac{1}{2} \log\bigl(1 - R^2_{\mathrm{partial}}\bigr), \qquad \Delta\mathcal{R}^2_{\mathrm{semi}} = \bigl(1 - \mathcal{R}^2_N\bigr)\, R^2_{\mathrm{partial}}. \tag{37}$$

Here $\Delta\mathcal{R}^2_{\mathrm{semi}} = \mathcal{R}^2_F - \mathcal{R}^2_N$ is the population analogue of the cross-validated increment defined above. Equation 37 converts a population residual-variance ratio into conditional mutual information under the stated Gaussian model. The multivariate analogue uses a log-determinant ratio. The thesis instead estimates finite-sample, cross-validated semipartial differences from separately regularized ridge fits. Those differences may be negative and depend on the nuisance design, rank, penalty, and held-out substrate, so they are neither estimates nor lower bounds of $I(X; Y \mid Z)$ [32].

A data-processing claim requires a separate conditional-independence premise. Let S be stimulus text, B a recorded brain response, $T = g(S)$ the deterministic synthetic brain-response target, and H the retained student state. The chain

$$B \longrightarrow T \longrightarrow H \qquad \Longleftrightarrow \qquad B \perp H \mid T$$

would require all information from B that reaches H to pass through T, which would imply $I(B; H) \leq I(B; T)$. The executed design does not establish this chain because S directly supplies both T and H, and it also drives B. Conditioning on S changes the question: because T is deterministic given S, $I(T; B \mid S) = 0$. That identity does not show that the target is useless. It shows that conditioning on the exact stimulus cannot establish additional target information beyond that stimulus or recover the desired unconditioned mediation claim. Data processing therefore supplies a bound only after the relevant conditional independence is defended; it does not predict the sign of a held-out encoding contrast [32].

Output matching and auxiliary-target fitting impose still weaker constraints on retained representations. Let p^{τ}_{tch} and q^{τ}_{stu} be the teacher and student output distributions at softmax temperature τ, let H_{tch} and H_{stu} be their internal states, let d_H be any declared discrepancy between those states, and let $\Delta R^2_{\text{transfer}}$ be the held-out biological-transfer contrast on recorded responses. Then, without additional identifying assumptions,

$$D_{\text{KL}}(p^{\tau}_{\text{tch}} \| q^{\tau}_{\text{stu}}) = 0 \;\not\Rightarrow\; d_H(H_{\text{stu}}, H_{\text{tch}}) = 0, \qquad \mathcal{L}_{\text{target}} \downarrow \not\Rightarrow\; \Delta R^2_{\text{transfer}} > 0.$$

The first non-implication holds because different internal parameterizations can realize the same output distribution. The second holds because a temporary target map can absorb part of the auxiliary objective, and a predictable target can be learned without producing participant-general improvement on recorded responses. Low output or target loss can establish optimization success under its own objective. It cannot replace fresh tests of retained-student movement, comparator adequacy, brain-response attribution, biological transfer, or practical utility. Section 4 reports those empirical stages without treating any formal analogy as their outcome.

E Supporting Analyses and Stopped Routes

This appendix separates secondary evidence from work that did not reach an interpretable scientific test. The distinction matters because a working pipeline, a restricted local diagnostic, and a stopped route answer different questions. Only executed analyses can bear on the thesis claims, and even then only within the scope of the design that produced them. Table 18 states the status language used below. Appendix A provides the corresponding provenance records, while Appendix F reports the full numerical results and sensitivity analyses. The headline empirical verdicts remain in Section 4.

Table 18: Evidence-status categories used in this appendix.

Category	Reader question	Permitted contribution	Excluded claim
Pipeline validation	Did the pipeline behave correctly when the expected signal was known?	Supports implementation and analysis checks.	No claim about recorded brain responses or training benefit.
Supporting analysis	Does an alternative specification change a settled interpretation?	Tests a named objection within its recorded design.	No broader claim beyond the tested specification.
Restricted local diagnostic	Does a reduced local implementation recover a reported effect?	Reports whether the effect appears under the available local conditions.	No reproduction, refutation, or cross-study equivalence claim.
Failed instrument	Could the proposed measurement support the intended inference?	Identifies why the instrument was not interpretable.	No biological null from a measurement failure.
Precondition-stopped route	Did the design satisfy the prerequisite required before intervention?	Records why the route stopped before producing an attribution claim that could not have been interpreted.	No result for the intervention that was not run.
Analysis-only or prospective design	Was a future test screened or specified without opening its outcome?	Clarifies the intended identifying question and activation criterion.	No empirical evidence for the proposed mechanism.

E.1 Pipeline validation and scientific scope

The end-to-end pipeline was first tested in a setting where the expected signal was known. Real Pereira sentences [27] were paired with synthetic responses generated from teacher features, named nuisance variables, and noise. The test exercised ordinary and target-guided KD, the temporary target head, middle-layer extraction, the contiguous held-out split, fold-local ridge fitting, nuisance subtraction, and aggregation across optimization seeds. The expected teacher–student ordering appeared, the nuisance component remained small, and excessive target weight degraded the result rather than receiving an automatic advantage.

These observations rule out several basic implementation failures. They show that the held-out evaluation tail was not used for target-guided training, that the scoring pipeline could recover a planted feature–response relation, and that the analysis did not reward every target-guided run by construction. They do not establish that an LM predicts recorded brain responses, that the nuisance set is sufficient for real stimuli, or that a brain-response objective improves a student. The synthetic test therefore validates the end-to-end implementation, not the scientific claim.

E.2 Response averaging and recorded-response robustness

Two averaging analyses addressed different questions. The first changed how many participants contributed to the recorded-response intervention target. Because each target was rebuilt from a different participant subset, increasing the subset size also changed participant identity, response reliability, and target construction. A corrected analysis showed that the apparent gain in controlled brain predictivity did not rise steadily with subset size. The result therefore does not identify a causal benefit of adding more participants to the training target.

The second analysis held the frozen representation and encoding procedure fixed while averaging increasing numbers of recorded responses. Controlled brain predictivity increased as averaging suppressed idiosyncratic and measurement noise and emphasized the shared stimulus-evoked component. This is a valid signal-to-noise gain for a group-response estimand. It changes what is being predicted, however, and does not supply participant replication or rescue the individual-response intervention.

The remaining analyses tested whether specific design choices concealed a recorded-response benefit. Table 19 maps each change to its objection and scope.

Table 19: Recorded-response robustness analyses and their scope.

Objection	Design change	Supported disposition	Remaining open
Auxiliary weight was poorly chosen.	Repeated the recorded-response intervention over the tested weight range.	No stable target-specific gain appeared within that range.	Other objectives or schedules could behave differently.
Adapter capacity was too limited.	Increased the tested low-rank adaptation capacity.	The tested capacity increase did not recover the effect.	This does not test every stronger manipulation.
Pointwise regression was the wrong loss.	Replaced it with a contrastive response objective.	The alternative loss did not produce a reliable target-specific benefit.	Other response objectives remain possible.
Regional targets hid a voxelwise effect.	Tested the available naturalistic voxelwise training procedure.	Target-matched tuning lowered held-out predictivity below the untuned model at every tested target weight, so the intended change was never induced and the arms compare two degraded models.	The broader multi-participant full-parameter route was not built.
Parameter-efficient tuning was too restrictive.	Tested full-parameter tuning, including a gentler three-participant probe.	Aggressive tuning caused forgetting; the gentler probe showed no reliable target-specific gain.	Three participants do not support population inference.

These analyses weaken the listed specific alternative explanations. They do not establish that every recorded-response objective must fail. Section 4.3 owns the participant-level intervention conclusion, and Appendix F reports the numerical variants.

E.3 Reduced external and cross-modal probes

External protocols were used as local diagnostics only when their decisive ingredients could be approximated with the available data and models. The standard is therefore narrower than reproduction: the question is whether a reported effect appears under the local conditions, not whether a reduced and mismatched local setup can confirm or refute the source study.

The first diagnostic adapted two elements of Negi et al.: a differentiable head that aligns model features to response timing, and their controls for target specificity [9], to the available LeBel responses [25] and decoder-only LMs. Under the gentlest tested training schedule the student representations that remain after the temporary head is discarded changed little, and stronger schedules degraded language quality severely. The local runs therefore did not establish a target-specific manipulation over their controls. The comparison nevertheless differs from the source study in population, response substrate, model family, language, training scale, and control construction. It is therefore a restricted local failure to establish the target-specific manipulation, not a reproduction or refutation of the published result.

The second diagnostic adapted a speech-based protocol from Moussa et al. [8]. Starting from the same Wav2Vec2-base model family, the design required a positive phoneme-prediction gain before constructing a matched-quality specificity test. In the reduced single-participant setup, target loss decreased but the phoneme-gain criterion was not met, so the downstream specificity comparison was not built. The source study used a larger, multi-participant setting with different timing and model choices. The result is thus a reduced local non-reproduction of the prerequisite effect, not evidence against the published claim.

Neither diagnostic supplied an external positive control for the present intervention pipeline. They narrow what the local pipeline recovered while leaving the external literature open under its original conditions.

E.4 Failed instruments and precondition-stopped routes

Three routes stopped because an upstream measurement or attribution condition failed. In each case, continuing to intervention would have produced a number without a defensible interpretation.

The shared-response route asked whether the predictable group response could serve as a ceiling for participant-specific intervention effects. In principle, this reference estimates the expected response shared across participants for the same stimulus, $m(S) = \mathbb{E}[Y \mid S]$. In practice, the leave-one-participant reference was not reliable enough, while temporally adjacent samples preserved apparent residual predictivity through leakage and autocorrelation. Once a clean temporal gap was imposed, the controlled residual predictivity was small but too imprecise to fix a ceiling for participant-specific effects. The ceiling instrument therefore failed; the analysis does not show that participant-specific structure is absent or that a universal biological ceiling has been reached.

The reading-time route asked whether a cognitive target could provide a simpler intervention signal than fMRI. Probes trained on frozen model representations initially predicted raw and residualized reading-time targets better than simple control targets. A phase-randomized control preserved the target's marginal distribution and temporal structure, however, and matched those advantages. The design therefore did not isolate cognition-specific content from target shape. This attribution failure does not imply that reading time contains no useful signal; it means that the available contrast could not identify that signal.

The gaze route required three conditions before privileged-information training: a measurable target, evidence that it added information beyond text, and enough independent groups for leakage-resistant evaluation. Natural-reading gaze was measurable but did not add information beyond the text itself under the tested assay. Task-directed gaze was more predictable, but the signal could reflect task intent or label leakage, and the number of independent paragraph groups was too small for a reliable group split. The intervention therefore stopped at its preconditions. The route's other planned signal, electroencephalography (EEG), never reached testing, so no corresponding claim was tested for it.

Stopping at these failed prerequisites means no downstream number exists that could be misread as an identified intervention effect. They identify why the proposed instruments could not answer their questions, but they do not turn measurement failure into a biological null.

E.5 Prospective designs without empirical evidence

The remaining routes specify questions that would be useful only after their activation conditions are met. They have no weight in the empirical verdicts of this thesis.

The same-lineage alignment–quality design would compare models from one lineage along an alignment–quality frontier, then test paired ordinary fine-tuning where alignment changes while language quality remains approximately stable. A screen of the alignment-versus-quality curve, run on recorded values without new training, found no region where language quality stays high while alignment varies enough to justify the intervention. Because no such region was found, no paired training test was run, so the screen contributes no evidence for or against whether ordinary fine-tuning loses brain alignment because of its training objective.

The matched auxiliary-control design would ask whether any intervention benefit came from brain-response content rather than generic target geometry, learnability, optimization pressure, or retained-student movement. The planned control would match those properties while removing brain-response content. Building it required the preceding saved-student transfer result to clear its predeclared participant-level improvement threshold, and that threshold was not cleared. The text-derived auxiliary target already in hand could not stand in, because it differs from the synthetic target on properties that must match before outcomes are compared: how well ordinary KD already predicts it, how much room for improvement is left, its covariance geometry and effective rank, and how well low-level text features predict it. No target or student from this branch supports a scientific comparison.

The cross-modal falsification design would compare real training fMRI, a phase-randomized response control, and a geometry-matched control on an independent biological endpoint. The route remains on design hold because the exact external protocol depends on unavailable author artifacts; proceeding would require a deliberate decision to run a clearly labeled reimplementation instead. Only outcome-blind metadata and pipeline development were completed: no endpoint data were inspected, no student was trained, and no biological score was computed. The planned analysis was intended to predict the sign and ordering of those outcomes before inspection. It remained contingent on the cross-modal setting carrying usable signal, and only an internal check that its predictions are mutually consistent was completed, without independent review or execution.

These designs clarify what stronger tests would need to identify. They remain prospective and do not alter the thesis conclusions.

F Claim-Relevant Sensitivities and Verification Tables

This appendix reports the verification detail needed to judge whether the main conclusions depend on an analysis choice or an unstable aggregate. Section 4 owns the decisive observations and scientific verdicts; Appendix A owns the visible evidence-record crosswalk and the verified subset of path–hash bindings. Every value below traces to its owning evidence record, including values whose artifacts are not listed in that verified subset.

The tables preserve the distinction between inference units. Participant-level effects support inference across the sampled biological units under the assumptions in Appendix D. Held-out blocks diagnose dependence on the fixed stimulus partition, and optimization seeds diagnose the stability of a fixed training procedure. Neither blocks nor seeds create additional biological or model-population replicates.

F.1 Claim-relevant analysis choices

An analysis choice can affect a thesis claim in four ways. An interpretation-changing analysis replaces an earlier estimand or inference unit. A scope-bounding sensitivity narrows what the primary result supports without reversing its frozen decision rule. A confirmatory check leaves the scoped interpretation unchanged. A limitation or correction removes authority from a design element rather than generating a more favorable result. Tables 20 and 21 apply these labels to the analyses that matter for the thesis claims.

Table 20: Interpretation-changing and scope-bounding analyses.

Analysis choice	Scientific role	Status	Effect on interpretation
Representation rank	Distillation-headroom analysis	Scope-bounding	The ordering of the model arms survived, but absolute magnitudes and some signs changed. Headroom is conditional on the declared rank.
Auxiliary weight and language quality	Participant-averaged recorded-response intervention	Scope-bounding	Near-quality-matched settings remained uncertain across folds. The clearest fold-positive setting incurred a substantial language-quality cost.
Quality band and model family	Distillation-headroom analysis	Scope-bounding	The full-range Pearson correlation between bits per byte and unique R^2 was -0.783; within the capable-model band it was -0.501, with interval $[-0.859, +0.188]$. Language quality remains a required control, not a deterministic explanation.
Response construction and inference unit	Participant-specific recorded-response intervention	Interpretation-changing	Participant-specific targets and participant-first inference answer a different question from an averaged target and fold-level summary. The change removed support for a participant-general gain.
Participant and subgroup deletion	Saved-student biological-transfer assay and 50-component linear target-projection assay	Scope-bounding	Removing the predeclared highest-baseline participant reversed the relative transfer point estimate. Every leave-one-participant-out linear target-projection mean remained negative.
Held-out stimulus block	Controlled naturalistic voxelwise predictivity assay; recorded-response intervention; saved-student biological-transfer assay; 50-component linear target-projection assay	Scope-bounding	The naturalistic assay remains a one-participant result. No leave-one-block-out intervention analysis produced a positive participant-mean contrast that changed the participant-level conclusion, and the linear target-projection result held under every block deletion.
Controlled-predictivity contrast by seed	Practical-utility evaluation	Scope-bounding	Only 5/8 of the eight technical-seed signs were positive. The utility analysis therefore follows a seed-unstable prerequisite rather than a reliably induced target-specific change.

Table 21: Confirmatory checks and design limitations.

Analysis choice	Scientific role	Status	Effect on interpretation
Layer, nuisance set, and target aggregation	Saved-student biological-transfer assay and 50-component linear target-projection assay	Confirmatory	Prespecified nearby choices did not overturn the primary rules. The 50-component linear target projection did not meet the improvement threshold fixed for it in advance (Section 4.4); retention and composed-path quantities remain diagnostics of that pathway.
Saved-student optimization seed	Synthetic-target recovery assay; retained-student movement audit; saved-student target-retention analysis; saved-student biological-transfer assay; composed-path diagnostic. The saved student and the retained student are the same post-training model	Confirmatory	The seeds agreed where the manipulation itself was measured, with target extraction positive and the movement criterion met in all six, and disagreed in sign on the endpoint contrasts for incremental retention, biological transfer, and the composed path. Seed agreement can establish reproducible optimization or manipulation, but it cannot substitute for participant agreement.
Saved controls audited after training	Comparator-adequacy audit and attribution assessment	Limitation or correction	The saved block-permutation sensitivity is defective, and the text-derived auxiliary control fails the comparability checks made before outcomes were inspected, including comparability with the ordinary-KD baseline. Neither can identify an effect of brain-response content.

Table 22: Classification of the comparator-audit checks into identification checks and post-training diagnostics.

Role	Total	Pass or within tolerance	Fail or outside tolerance	Interpretation
Identification checks	37	20	16	One additional check is unresolved; failures include KD-baseline recovery and headroom, covariance geometry, effective rank, and nuisance predictability.
Post-training diagnostics	18	4	14	Parameter, Frobenius, and CKA margins describe realized intervention differences and do not determine comparator validity.

The original combined engineering count, 30/55, is retained only as provenance for the frozen audit.

The interpretation-changing analysis altered the question being answered; it was not a more favorable reanalysis of the earlier averaged-target result. The remaining sensitivities either narrowed the scope, confirmed stability near a frozen specification, or exposed a limitation. None licenses replacing a prospective primary estimator or decision rule with a more favorable variant.

F.2 Participant-level intervention contrasts

The participant-specific recorded-response estimator first forms a paired target-specific contrast for participant u, held-out block k, and optimization seed s, where $P = 5$ is the number of matched permuted-target draws.

$$d_{uks} = R^{2,\text{unique}}_{\text{recorded},uks} - \frac{1}{P}\sum_{p=1}^{P} R^{2,\text{unique}}_{\text{perm}(p),uks}, \qquad e_u = \text{median}_{k,s}\, d_{uks}, \qquad \widehat{\theta}_{\text{part}} = \frac{1}{U}\sum_{u=1}^{U} e_u. \tag{38}$$

Equation 38 subtracts the mean matched permuted-target score inside every seed–block cell, then aggregates the crossed cells within participant before averaging participants. The resulting U =9 participant effects, each based on 15 paired cells, are the biological inputs to the cohort estimate.

Table 23: Participant-specific recorded-response intervention contrasts.

UID	Participant effect e_u (unique R^2 difference)
797	+0.00002885
837	+0.00100412
841	−0.00018259
848	−0.00063374
856	−0.00037809
865	−0.00058994
875	+0.00070575
876	+0.00011067
880	+0.00083826
Mean	+0.00010

The signs are mixed around a small cohort mean. The table supports participant-first aggregation and heterogeneity inspection; it does not imply that every participant has exactly zero effect.

The saved-student biological-transfer assay uses a different target and within-participant aggregator. Let T_u, X_u, and K_u denote the participant-level scores for the synthetic-response target, text-derived auxiliary control, and ordinary KD. The three displayed contrasts satisfy

$$\Delta_u^{T-X} = T_u - X_u, \quad \Delta_u^{T-K} = T_u - K_u, \quad \Delta_u^{X-K} = X_u - K_u, \quad \Delta_u^{T-X} = \Delta_u^{T-K} - \Delta_u^{X-K}. \tag{39}$$

Equation 39 shows why a positive relative contrast against the auxiliary control need not imply improvement over ordinary KD: the auxiliary control may instead have moved downward. Seeds and outer folds are averaged within each participant before these contrasts are formed.

Table 24: Participant-level saved-student biological-transfer contrasts.

UID	Δ_u^{T-X}	Δ_u^{T-K}	Δ_u^{X-K}
797	−0.00023801	−0.00005311	+0.00018490
837	+0.00227168	+0.00226591	−0.00000577
841	+0.00000452	−0.00078813	−0.00079265
848	−0.00002346	−0.00002667	−0.00000320
856	−0.00022167	−0.00036255	−0.00014087
865	−0.00064994	−0.00108762	−0.00043767
875	+0.00009116	−0.00006339	−0.00015455
876	−0.00013023	−0.00014342	−0.00001319
880	+0.00012305	+0.00013517	+0.00001212
Mean	+0.000136	−0.000014	−0.000150

The transfer contrasts are also heterogeneous. The relative mean Δ^{T-X} is small and positive, while the direct mean Δ^{T-K} is -0.000014; removing the participant with the highest ordinary-KD brain-predictivity baseline, predeclared for deletion, changes the relative point estimate to -0.000131. These rows explain why the relative comparator cannot replace the direct contrast or participant-first inference.

The two participant tables must not be pooled or compared row by row. They use different targets, contrasts, and within-participant aggregators; the shared identifiers only show that they were evaluated in the same participant cohort.

F.3 Participant-level mechanism diagnostics

The target-projection diagnostic evaluates the declared 50-component linear pathway on the same sentence substrate; it is not a prerequisite for the independent direct-transfer contrast. For each participant, A_u measures the target's unique response predictivity above nuisance, and Q_u compares that score with the frozen row-twin control, which keeps the target rows and the readout procedure but breaks their pairing with the sentences (Section 4.4). These quantities compare target-above-nuisance predictivity with sentence-pairing specificity. They are read before the later target-retention analysis and composed-path diagnostic.

Table 25: Participant-level linear target-projection diagnostics.

UID	Target above nuisance A_u	Aligned minus frozen twin Q_u
797	−0.00266759	−0.00031777
837	−0.00448823	−0.00785204
841	−0.02042083	+0.00588154
848	−0.00001541	+0.00158126
856	−0.01394534	−0.00066703
865	+0.00208088	+0.00684375
875	+0.00266981	+0.01519279
876	−0.00660269	−0.00590127
880	+0.00293994	+0.00106256
Mean	−0.004494	+0.001758

The mean target-above-nuisance score is -0.004494, and every leave-one-participant-out mean remains negative. This first prerequisite misses the improvement threshold fixed for it in advance (Section 4.4). The aligned-minus-frozen-twin values have substantial effects of both signs, so the paired specificity question remains unresolved. It does not contradict the first result.

The later diagnostics ask whether the part of the target that a student predicts overlaps with the target direction that predicts participant responses. For participant u, O_u is the absolute composed student-to-target-to-response overlap, D_u is its brain-target-minus-KD increment, and J_u is the aligned-minus-frozen-twin increment. All three average optimization seeds and outer folds within participant before participant-level inspection.

Table 26: Participant-level composed-path quantities.

UID	Absolute overlap O_u	Synthetic response minus KD D_u	Aligned minus twin J_u
797	+0.00045651	+0.00018799	+0.00019106
837	+0.00484230	+0.00029143	+0.00026774
841	+0.00038073	+0.00004280	+0.00003954
848	+0.00010189	−0.00005708	−0.00005644
856	−0.00040275	−0.00005387	−0.00005261
865	+0.00216009	−0.00006632	−0.00005151
875	+0.00447180	−0.00018830	−0.00019123
876	−0.00126093	+0.00008691	+0.00008438
880	+0.00043030	+0.00003540	+0.00003336
Mean	+0.001242	+0.000031	+0.000029

Absolute composed overlap remains unresolved. The two increments are smaller and have mixed signs. Within the declared linear pathway, these downstream diagnostics cannot repair the target-projection result or identify its cause. They do not constrain the independent direct-transfer contrast.

F.4 Technical-seed diagnostics

Optimization seeds test whether a fixed training procedure behaves reproducibly. They do not support biological generalization, because the same participants and stimuli are reused within each seed.

The practical-utility evaluation has a prerequisite: a reliable contrast in controlled brain predictivity between the recorded-response arm and its matched permuted-response control. Its brain-specific interpretation depends on that contrast. Table 27 reports the complete seed vector used to judge that prerequisite contrast; it declared eight seeds, where the transfer and target-extraction procedures declared six.

Table 27: Technical-seed values for the practical-utility prerequisite.

Seed	Recorded-minus-permuted contrast in controlled brain predictivity
0	+0.063
1	+0.014
2	+0.004
3	+0.016
4	+0.005
5	−0.006
6	−0.002
7	−0.005
Mean	+0.011
Median	+0.0042

The signs are split, and the mean is strongly influenced by seed 0. The seed rows are rounded independently to three decimal places; the displayed mean and median are computed from artifact-precision values. The complete vector documents mixed signs and seed-0 sensitivity; it does not establish a null or equivalence.

For the saved-student biological-transfer assay, each seed row averages participants and outer folds. The three columns use the same contrasts as Equation 39.

Table 28: Technical-seed values for saved-student biological transfer.

Seed	Δ^{T-X}	Δ^{T-K}	Δ^{X-K}
0	-0.00015727	-0.00035769	-0.00020042
1	$+0.00032614$	$+0.00003556$	-0.00029058
2	$+0.00004025$	$+0.00000892$	-0.00003133
3	$+0.00025080$	$+0.00015307$	-0.00009773
4	$+0.00018344$	-0.00006376	-0.00024721
5	$+0.00017470$	$+0.00014137$	-0.00003333
Mean	$+0.000136$	-0.000014	-0.000150

Before participant responses were inspected, the synthetic-target recovery, retained-student movement, and language-quality criteria were satisfied. The largest saved-arm language-quality difference, 0.002579, remained within the frozen 0.005 margin. The seed table therefore checks an interpretable manipulation while leaving biological inference to the participant table.

The exact-target seed diagnostics separate absolute target extraction from improvement over seed-matched ordinary KD and from composed-path increments. Within each seed, target extraction and incremental retention aggregate held-out target recovery; the two overlap increments average participants and outer folds.

Table 29: Technical-seed target-extraction and retention values.

Seed	Target extraction M_s	Incremental retention B_s
0	$+0.03965524$	$+0.00013878$
1	$+0.04096723$	$+0.00103098$
2	$+0.03932328$	-0.00159873
3	$+0.04291468$	-0.00234707
4	$+0.03920675$	-0.00038085
5	$+0.04228561$	-0.00197014
Mean	$+0.040725$	-0.000855

Table 30: Technical-seed composed-path increments.

Seed	Synthetic response minus KD D_s	Aligned minus twin J_s
0	-0.00003781	-0.00004017
1	-0.00004152	-0.00004668
2	-0.00001666	-0.00001488
3	$+0.00006481$	$+0.00006414$
4	$+0.00014778$	$+0.00014581$
5	$+0.00006938$	$+0.00006797$
Mean	$+0.000031$	$+0.000029$

Absolute target extraction is positive in every seed. Incremental retention has mixed signs and an interval containing zero. Both composed-path increments show the same uncertainty. The separation shows that fitting the target in absolute terms does not establish improvement over ordinary KD or transfer along the composed path.

Table 31: Movement of the retained synthetic-target student away from its seed-matched ordinary-KD student, by technical seed.

Seed	Centered relative-Frobenius	Linear-CKA distance
0	0.186369	0.002633
1	0.216294	0.005457
2	0.298694	0.011778
3	0.195493	0.003585
4	0.197089	0.003925
5	0.196519	0.003423

The values use layer 6 and the same 1,999 WikiText sentences held out from every training arm; Equation 35 defines both distances. Before participant responses were opened, every seed was required to exceed 10^{-4} in centered relative-Frobenius change and to exceed ten times the duplicate-extraction noise in linear-CKA distance; all six seeds met both, and repeated extraction of the same checkpoint reproduced it exactly.

No seed-only table is reported for the participant-specific recorded-response intervention. Its primary participant statistic is a median over crossed seed–block cells, so a seed-only vector cannot reconstruct the nonlinear participant aggregate in Equation 38. Seed variability remains inside each participant value, while the shared-block summary diagnoses the other crossed axis.

Table 32: Models in the cross-family language-quality analysis.

Model	Family	Parameters	Mid layer	Bits/byte	Unique R^2
Mistral-7B-v0.1	Mistral	7.24B	16	1.135	+0.0438
Qwen2.5-7B	Qwen	7.62B	14	1.152	+0.0235
Llama-3.2-3B	Llama	3.21B	14	1.163	+0.0464
Qwen2.5-3B	Qwen	3.09B	18	1.200	+0.0371
OPT-6.7B	OPT	6.66B	16	1.222	+0.0233
Pythia-2.8B	Pythia	2.78B	16	1.240	+0.0174
Qwen2.5-1.5B	Qwen	1.54B	14	1.242	+0.0297
Llama-3.2-1B	Llama	1.24B	8	1.249	+0.0360
OPT-2.7B	OPT	2.65B	16	1.276	+0.0324
Pythia-1.4B	Pythia	1.41B	12	1.283	+0.0213
Pythia-1B	Pythia	1.01B	8	1.315	+0.0226
OPT-1.3B	OPT	1.32B	12	1.315	+0.0281
Qwen2.5-0.5B	Qwen	0.49B	12	1.342	+0.0356
GPT-2 Large	GPT-2	0.77B	18	1.351	+0.0173
Pythia-410M	Pythia	0.41B	12	1.381	+0.0233
GPT-2 Medium	GPT-2	0.35B	12	1.394	+0.0177
OPT-350M	OPT	0.33B	12	1.453	+0.0259
GPT-2 Small	GPT-2	0.12B	6	1.486	+0.0170
OPT-125M	OPT	0.13B	6	1.514	+0.0147
Pythia-160M	Pythia	0.16B	6	1.523	+0.0209
DistilGPT-2	GPT-2	0.08B	3	1.628	+0.0071
Pythia-70M	Pythia	0.07B	3	1.699	-0.0012

Unique R^2 is the variance explained above the declared nuisance set, and all 22 models enter the primary middle-layer correlation. The capable-model sensitivity retains the ten models below 1.30 bits per byte. The family-factor analysis excludes Llama and Mistral because each has fewer than three model sizes; those exclusions do not apply to the primary cross-family correlation.

These tables distinguish choices that changed or bounded an interpretation from checks of local stability. Participant heterogeneity and technical-seed variation answer different questions; neither can substitute for the other or expand the claims reported in Section 4.

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